

# Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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# Chapter 1

## Low Dimensional Asymptotic Normality

### 1.1 Basic Setup

Given a "high dimensional" data set  $\{\mathbf{x}_i\}_{i=1}^n$  where  $\mathbf{x}_i \in \mathbb{R}^p$  for all  $i = 1, \dots, n$ , the statistician fixes a positive integer  $q < p$  and wishes to find the  $q$  principal components. Assuming that the data is centred (that is, with mean 0), the probabilistic principal component analysis model is a statistical model that treats  $\mathbf{x}_i$ 's as realizations of independent random variables  $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$ , where  $\mathbf{W}$  is an unknown  $p \times q$  loading matrix of full rank,  $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$  is the latent normal variable, and  $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$  is the Gaussian noise with unknown standard deviation  $\sigma > 0$ , and  $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$  are all independent (for now). Put  $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$  for the unknown parameter to estimate and  $\Theta$  for all possible choices of  $\theta$ .  $\Theta$  is thus the collection of all tuples  $(\mathbf{W}, \sigma^2)$  such that  $\mathbf{W}$  is a real  $p \times q$  matrix with  $\text{rank}(\mathbf{W}) = q$  and  $\sigma^2 \in (0, \infty)$ . Given that the collection of all real  $p \times q$  matrix with rank  $q$  is an open subset of the Euclidean space  $\mathbb{R}^{pq}$ , we consider  $\Theta$  as an open subset of the Euclidean space  $\mathbb{R}^{pq+1}$  and our statistical model is

$$\mathcal{P} = \{N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)\}_{\theta \in \Theta}. \quad (1.1)$$

To resolve the non-identifiability of  $\mathcal{P}$ , we introduce the quotient manifold structure. For  $(\mathbf{A}, s) \in \Theta$  and  $(\mathbf{B}, t) \in \Theta$ , we define  $(\mathbf{A}, s) \sim (\mathbf{B}, t)$  to mean that they give rise to the same covariance matrix, namely when  $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$ . The equivalence classes are denoted by  $[(\mathbf{A}, s)]$  or more briefly  $[\mathbf{A}, s]$ . Unless otherwise mentioned,  $(\mathbf{A}, s) \in \Theta$  is the default representative we pick when we write an equivalence class as  $[\mathbf{A}, s]$ . Writing  $O(q)$  for the collection of all  $q \times q$  real orthogonal matrices, it was shown in Datta's thesis that  $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$ . Therefore,  $M := \Theta / \sim$  can be identified as the orbits of the right action of  $O(q)$  on  $\Theta$  and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee, 2013](#), Theorem 21.10) shows that there is a unique smooth structure making  $M$  a manifold of dimension

$$d = pq + 1 - \frac{q(q-1)}{2} \quad (1.2)$$

and the map  $\pi : \Theta \rightarrow M$  defined by

$$\pi((\mathbf{A}, s)) = [\mathbf{A}, s] \quad (1.3)$$

is a smooth submersion. Moreover, this action is vertical, transitive on fibers and is isometric with respect to the Euclidean metric. Thus, there exists a unique metric  $g$  on  $M$  making  $(M, g)$  a Riemannian manifold and  $\pi$  a Riemannian submersion (see [Lee, 2018](#), Theorem 2.28). In more concrete terms, if  $g^\Theta$  is the standard Euclidean metric given by

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = \text{tr}(\mathbf{V}\mathbf{U}^T) + cb, \quad (1.4)$$

whenever  $(\mathbf{A}, s) \in \Theta$ ,  $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$ , then

$$g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{h}) = g_{[\mathbf{A}, s]}(d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)), d\pi_{(\mathbf{A}, s)}((\mathbf{U}, b))) = g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)), \quad (1.5)$$

whenever  $\mathbf{v}, \mathbf{h} \in T_{[\mathbf{A}, s]}M$  lifting to  $(\mathbf{V}, c)$  and  $(\mathbf{U}, b)$  respectively. A standard fact is that  $\Theta$  is connected. Consequently,  $M$  is also connected. For later convenience, let us define two functions  $\Phi : \Theta \rightarrow \mathbb{R}^{p \times p}$  and  $\Sigma : M \rightarrow \mathbb{R}^{p \times p}$  by setting, for every  $[\mathbf{A}, s] \in M$ :

$$\Sigma([\mathbf{A}, s]) := \Phi([\mathbf{A}, s]) := \mathbf{A}\mathbf{A}^T + s\mathbf{I}_p. \quad (1.6)$$

In other words,  $\Sigma \circ \pi = \Phi$  gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\mathbf{0}, \Sigma([\mathbf{A}, s]))\}_{[\mathbf{A}, s] \in M} \quad (1.7)$$

and this is the identifiable model we shall work with.

Finally, throughout the entire document, our default norm for a matrix  $\mathbf{A}$  will be the spectral norm, denoted by  $\|\mathbf{A}\|$ . We will also make use of the Frobenius norm  $\|\mathbf{A}\|_F$  as well. The eigenvalues and the singular values of a matrix will always be labelled in decreasing order. In symbols, if  $\mathbf{A}$  is a matrix of size  $p \times q$ , then we denote its  $q$  singular values as

$$\|\mathbf{A}\| = \sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_q(\mathbf{A}) = \sigma_{\min}(\mathbf{A}) \geq 0$$

and, if  $\mathbf{S}$  is a symmetric matrix of size  $p \times p$ , then we denote its  $p$  eigenvalues as

$$\lambda_1(\mathbf{S}) \geq \lambda_2(\mathbf{S}) \geq \cdots \geq \lambda_p(\mathbf{S}).$$

## 1.2 Basic Facts about $(M, g)$

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and vertical tangent spaces.

### Proposition 1.

1. The map  $\Sigma$  defined in (1.6) is a smooth injective immersion.
2. At every  $(\mathbf{A}, s) \in \Theta$ , the vertical tangent space

$$V_{(\mathbf{A}, s)} := \ker(d\pi_{(\mathbf{A}, s)}) = \ker(d\Phi_{(\mathbf{A}, s)}) \quad (1.8)$$

equals to the collection of all tangent vectors of the form  $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A}, s)}\Theta$ , with  $\mathbf{\Omega}$  ranged over all  $q \times q$  skew symmetric matrices.

3. At every  $(\mathbf{A}, s) \in \Theta$ , the horizontal tangent space  $H_{(\mathbf{A}, s)} := V_{(\mathbf{A}, s)}^\perp$  equals to the collection of all tangent vectors of the form  $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$  ranged over all  $c \in \mathbb{R}$  and all  $p \times q$  matrices  $\mathbf{V}$  satisfying  $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$ .

*Proof.* Fix a point  $(\mathbf{A}, s) \in \Theta$  throughout this proof. Before we start the proof, let us first study the set  $\ker(d\Phi_{(\mathbf{A}, s)})$ , where  $\Phi$  is given in (1.6). For  $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ , define a curve  $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$  by setting  $\gamma(t) = (\mathbf{A} + t\mathbf{V}, s + tc)$ . Then:

$$d\Phi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \left. \frac{d}{dt} \right|_{t=0} \Phi(\gamma(t)) = \mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p.$$

If  $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$ , then

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p = \mathbf{0}. \quad (1.9)$$

If  $\mathbf{u} \in \mathbb{R}^p$  is any nonzero vector orthogonal to all column vectors of  $\mathbf{A}$ , then  $\mathbf{A}^T \mathbf{u} = \mathbf{0}$  and

$$0 = \mathbf{u}^T (\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p) \mathbf{u} = c \mathbf{u}^T \mathbf{u}.$$

Given that  $\mathbf{u} \neq \mathbf{0}$ , we must have  $c = 0$  and (1.9) reduces to

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T = \mathbf{0}. \quad (1.10)$$

Apply  $\mathbf{u}$  to the right to get:

$$\mathbf{A}\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Given that  $\mathbf{A}$  has full rank, rank nullity theorem forces

$$\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Therefore, we have proven that every vector  $\mathbf{u}$  orthogonal to columns vectors of  $\mathbf{A}$  will also be orthogonal to column vectors of  $\mathbf{V}$ . This means  $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$  for some  $q \times q$  matrix  $\mathbf{\Omega}$ . Substitute this back to (1.10) gives

$$\mathbf{A}(\mathbf{\Omega}^T + \mathbf{\Omega})\mathbf{A}^T = \mathbf{0}.$$

Applying  $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$  to the left and  $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$  to the right yields  $\mathbf{\Omega}^T + \mathbf{\Omega} = \mathbf{0}$ . Therefore,  $\mathbf{\Omega}$  is skew-symmetric. Conversely, (1.10) is satisfied by  $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$  with any skew-symmetric matrix  $\mathbf{\Omega}$ . We have proven that  $\ker(d\Phi_{(\mathbf{A},s)})$  equals to the collection of all tangent vectors of the form  $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A},s)}\Theta$ , with  $\mathbf{\Omega}$  ranged over all  $q \times q$  skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.8). Clearly,  $\Sigma$  is well-defined and is injective. The smoothness of  $\Sigma$  is guaranteed by relation  $\Sigma \circ \pi = \Phi$  (see for example Lee, 2013, Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\mathbf{A},s)} = d\Sigma_{[\mathbf{A},s]} \circ d\pi_{(\mathbf{A},s)}. \quad (1.11)$$

It follows immediately that  $\ker(d\pi_{(\mathbf{A},s)}) \subseteq \ker(d\Phi_{(\mathbf{A},s)})$ . The other direction will follow if we can prove that  $d\Sigma_{[\mathbf{A},s]}$  is injective.

Suppose that  $\mathbf{v} \in T_{[\mathbf{A},s]}M$  is such that  $d\Sigma_{[\mathbf{A},s]}(\mathbf{v}) = \mathbf{0}$ . We prove that  $\mathbf{v} = \mathbf{0}$ . Let  $(\mathbf{V}, c) \in T_{(\mathbf{A},s)}\Theta$  be the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ , meaning that  $(\mathbf{V}, c)$  is the unique horizontal tangent vector at  $(\mathbf{A}, s)$  such that  $\mathbf{v} = d\pi_{(\mathbf{A},s)}((\mathbf{V}, c))$ . (1.11) then implies that  $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A},s)})$  and the preceding discussion shows that  $c = 0$  and  $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$  for some skew-symmetric  $\mathbf{\Omega}$ . To proceed, we look at the curve  $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$  defined by  $\gamma(t) = (\mathbf{A}e^{t\mathbf{\Omega}}, s)$ . Clearly  $\gamma(0) = (\mathbf{A}, s)$  and  $\gamma'(0) = (\mathbf{A}\mathbf{\Omega}, 0)$ . Moreover, the matrix  $e^{t\mathbf{\Omega}}$  is always orthogonal for any  $t$  and any skew-symmetric  $\mathbf{\Omega}$  so that  $\pi(\gamma(t)) = [\mathbf{A}, s]$  for all  $t$ . We have

$$\mathbf{v} = d\pi_{(\mathbf{A},s)}((\mathbf{A}\mathbf{\Omega}, 0)) = d\pi_{(\mathbf{A},s)}(\gamma'(0)) = \left. \frac{d}{dt} \right|_{t=0} \pi(\gamma(t)) = \mathbf{0} \in T_{[\mathbf{A},s]}M.$$

Therefore,  $d\Sigma_{[\mathbf{A},s]}$  is injective and the vertical tangent space is the same as  $\ker(d\Phi_{(\mathbf{A},s)})$ . Since  $[\mathbf{A}, s]$  is arbitrary,  $\Sigma$  is an immersion.

To compute the horizontal tangent space, any  $(\mathbf{V}, c) \in V_{(\mathbf{A},s)}^\perp$  will satisfy

$$0 = g_{(\mathbf{A},s)}^\Theta((\mathbf{V}, c), (\mathbf{A}\mathbf{\Omega}, 0)) = \text{tr}(\mathbf{A}^T \mathbf{V} \mathbf{\Omega}) \quad (1.12)$$

for all skew-symmetric  $\mathbf{\Omega}$ . Decomposing  $\mathbf{A}^T \mathbf{V} = \mathbf{S} + \mathbf{K}$ , where  $\mathbf{S}$  is the symmetric part and  $\mathbf{K}$  is the skew-symmetric part, we find out that

$$0 = \text{tr}(\mathbf{S}\mathbf{\Omega}) + \text{tr}(\mathbf{K}\mathbf{\Omega}) = 0 + \text{tr}(\mathbf{K}\mathbf{\Omega}) = \text{tr}(\mathbf{K}\mathbf{\Omega}).$$

Setting  $\mathbf{\Omega} = \mathbf{K}^T$  will yield that  $\mathbf{K} = \mathbf{0}$ . Therefore,  $\mathbf{A}^T \mathbf{V}$  is a symmetric matrix. Conversely, any  $\mathbf{V}$  such that  $\mathbf{A}^T \mathbf{V}$  is symmetric will satisfy (1.12). The proof is complete.  $\square$

We record a proposition about lifting curves.

**Proposition 2.** Given a smooth curve  $\gamma : (a, b) \rightarrow M$ , a point  $t_0 \in (a, b)$ , and a representative  $(\mathbf{A}, s)$  of  $\gamma(t_0)$ , there exists a unique smooth curve  $c : (a, b) \rightarrow \Theta$ , which we refer to as the horizontal lift of  $\gamma$  at  $t_0$  and representative  $(\mathbf{A}, s)$ , such that  $c(t_0) = (\mathbf{A}, s)$  and, for all  $t \in (a, b)$ ,  $\pi(c(t)) = \gamma(t)$ ,  $c'(t) \in H_{c(t)}$ , and  $d\pi_{c(t)}(c'(t)) = \gamma'(t)$ .

*Proof.* This result is immediate from Theorem 17.8 and 17.9 from [Michor, 2008](#).

Given that the notations and language are slightly different in [Michor, 2008](#), we provide a bit more explanation for convenience. We apply the two theorems for the fiber bundle  $(E, p, M, S)$  with  $E = \Theta$ ,  $p = \pi$ ,  $S = O(q)$ , and  $M$  is just our  $M$ . The map  $\Phi$  mentioned in their text (called the Ehresmann connection), merely assigns each  $\theta \in \Theta$  the linear projection map  $\Phi_\theta : T_\theta\Theta \rightarrow T_\theta\Theta$  onto the subspace  $V_\theta$ . Given the smooth curve  $\gamma$  in the statement of our proposition, Theorem 17.8 implemented with  $x = \gamma(t_0)$  and  $u_x = (\mathbf{A}, s)$  produces a map  $\text{Pt}_\gamma$ . We set  $c(t) = \text{Pt}_\gamma((\mathbf{A}, s), t)$ , which is a smooth curve. Property (1) of Theorem 17.8 shows that  $c(t_0) = (\mathbf{A}, s)$  and  $\pi(c(t)) = \gamma(t)$  for all  $t$  such that  $c$  is well-defined. Property (2) of Theorem 17.8 shows that  $c'(t) \in H_{c(t)}$  for all  $t$  such that  $c$  is well-defined. Given that  $S = O(q)$  is compact, Theorem 17.9 shows that  $c$  is defined on the entire  $(a, b)$ . An application of chain rule from  $\pi \circ c = \gamma$  shows that  $d\pi_{c(t)}(c'(t)) = \gamma'(t)$  for all  $t \in (a, b)$ . Finally, as pointed out in the first proof of Theorem 17.8,  $\text{Pt}_\gamma$  is unique, or equivalently  $c$  is unique.  $\square$

Here is a basic geodesic classification result.

**Proposition 3.** Given  $[\mathbf{A}, s] \in M$  and  $\mathbf{v} \in T_{[\mathbf{A}, s]}M$ , all geodesics in  $M$  starting at  $[\mathbf{A}, s]$  with initial velocity  $\mathbf{v}$  takes the format of curves  $\gamma : I \rightarrow M$ , where  $I$  is an interval containing 0 in its interior, defined by

$$\gamma(t) = [\mathbf{A} + t\mathbf{V}, s + tc], \quad (1.13)$$

where  $(\mathbf{V}, c) \in H_{(\mathbf{A}, s)}$  is the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ .

*Proof.* First of all, the curve  $\gamma$  given by (1.13) is well defined. If  $(\mathbf{A}\mathbf{Q}, s)$ , where  $\mathbf{Q} \in O(q)$ , is another representative of  $[\mathbf{A}, s]$ , let  $(\mathbf{V}', c')$  be the horizontal lift of  $\mathbf{v}$  to  $(\mathbf{A}\mathbf{Q}, s)$ . With the function  $R_{\mathbf{Q}} : \Theta \rightarrow \Theta$  defined by  $R_{\mathbf{Q}}((\mathbf{A}, s)) = (\mathbf{A}\mathbf{Q}, s)$ , we have

$$\mathbf{v} = d\pi_{(\mathbf{A}, s)}(\mathbf{V}, c) = d(\pi \circ R_{\mathbf{Q}})_{(\mathbf{A}, s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q}, s)}(dR_{\mathbf{Q}})_{(\mathbf{A}, s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q}, s)}(\mathbf{V}\mathbf{Q}, c).$$

Thus,  $\mathbf{V}' = \mathbf{V}\mathbf{Q}$  and  $c' = c$ . Substitute this into the definition of  $\gamma$  will yield the exact same curve.

Given that  $\gamma$  is well-defined, it must be that  $(\mathbf{A} + t\mathbf{V}, s + tc) \in \Theta$  for all  $t \in I$ . The curve  $c : I \rightarrow \Theta$  with

$$c(t) = (\mathbf{A} + t\mathbf{V}, s + tc) \quad (1.14)$$

for all  $t \in I$  is then well-defined. Being a straight line in Euclidean space,  $c$  is a geodesic. Moreover, it has horizontal initial velocity and satisfies  $\pi \circ c = \gamma$ . By Corollary 26.12 from [Michor, 2008](#),  $\gamma$  must be a geodesic. Conversely, suppose that  $\gamma : I \rightarrow M$  is a geodesic in  $M$  starting at  $[\mathbf{A}, s]$  with initial velocity  $\mathbf{v}$ . Let  $c$  be the horizontal lift of  $\gamma$  at 0 and representative  $(\mathbf{A}, s)$  given by Proposition 2. Then  $c$  starts at  $(\mathbf{A}, s)$  and has its initial velocity  $(\mathbf{V}, c)$  being the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ . Lemma 26.11 in [Michor, 2008](#) shows that  $c$  is a geodesic in  $\Theta$ . Given that straight lines are the only geodesics in Euclidean spaces,  $c$  must be defined by (1.14) for all  $t \in I$  and  $\gamma = \pi \circ c$  must take the format (1.13).  $\square$

We now move to compute the exponential map and the logarithmic map. The following two lemmas are crucial for computing the orthogonal matrix based on given points in  $M$ .

**Lemma 4.** If  $\mathbf{L}$  is an invertible  $q \times q$  matrix and  $\mathbf{Q} \in O(q)$  are such that  $\mathbf{L}\mathbf{Q}$  is symmetric, then there exists matrices  $\mathbf{R} \in O(q)$ ,  $\mathbf{P} \in O(q)$ , a  $q \times q$  diagonal matrix  $\mathbf{D}$  whose diagonal entries are 1 or  $-1$ , and a  $q \times q$  diagonal matrix  $\mathbf{S}$  with diagonal entries  $\sigma_1(\mathbf{L}) \geq \dots \geq \sigma_q(\mathbf{L}) > 0$  such that  $\mathbf{L} = \mathbf{R}\mathbf{S}\mathbf{P}^T$  and  $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$ .

If we add furthermore the assumption that  $\mathbf{L}$  is symmetric positive definite, then we can achieve the results mentioned above with  $\mathbf{P} = \mathbf{R}$ . In particular, this implies  $\mathbf{Q} = \mathbf{Q}^T$  and  $\|\mathbf{Q} - \mathbf{I}_q\| = \|\mathbf{D} - \mathbf{I}_q\|$  can take only 0 or 2 as values.

*Proof.* Let us start by applying the singular value decomposition of  $\mathbf{L}$  to get matrices  $\mathbf{R}' \in O(q)$ ,  $\mathbf{P}' \in O(q)$  and a diagonal matrix  $\mathbf{S}$  with diagonal entries  $\sigma_1(\mathbf{L}) \geq \dots \geq \sigma_q(\mathbf{L}) > 0$  such that  $\mathbf{L} = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T$ . Note that all singular values are positive because  $\mathbf{L}$  is invertible. Substituting the given condition that  $\mathbf{L}\mathbf{Q}$  is symmetric will yield that  $\mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$ , where  $\mathbf{J} := (\mathbf{P}')^T\mathbf{Q}\mathbf{R}'$ .

Clearly  $\mathbf{J} \in O(q)$ . The equation

$$\mathbf{S}^2 = \mathbf{S}\mathbf{J}\mathbf{J}^T\mathbf{S} = (\mathbf{S}\mathbf{J})(\mathbf{S}\mathbf{J})^T = (\mathbf{J}^T\mathbf{S})(\mathbf{J}^T\mathbf{S})^T = \mathbf{J}^T\mathbf{S}^2\mathbf{J}$$

shows that

$$\mathbf{J}\mathbf{S}^2 = \mathbf{S}^2\mathbf{J}.$$

Looking at each entry of both side, we end up with

$$\mathbf{J}_{ij}(\sigma_j(\mathbf{L})^2 - \sigma_i(\mathbf{L})^2) = 0$$

for  $i, j = 1, \dots, q$ . Given that all singular values are positive, we in fact have

$$\mathbf{J}_{ij}(\sigma_j(\mathbf{L}) - \sigma_i(\mathbf{L})) = 0 \quad (1.15)$$

for all  $i$  and all  $j$  so that  $\mathbf{J}\mathbf{S} = \mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$ . Inverting  $\mathbf{S}$  yields that  $\mathbf{J}$  is actually symmetric. Being a symmetric orthogonal matrix,  $\mathbf{J}^2 = \mathbf{I}_q$  and the eigenvalues of  $\mathbf{J}$  must be 1 or  $-1$ . Another look at (1.15) shows that  $\mathbf{J}$  is a block-diagonal matrix, diagonal if all singular values are distinct. Furthermore, each block of  $\mathbf{J}$  must still be orthogonal and symmetric.

We now diagonalize  $\mathbf{J}$  based on the above information. Suppose that  $\mathbf{S} = \bigoplus_{i=1}^k s_i \mathbf{I}_{m_i}$ , where  $s_1 > \dots > s_k$  are the unique diagonal entries of  $\mathbf{S}$  and each repeats for  $m_i$  times. Thus,  $\mathbf{J}$  has  $k$  blocks. Write the  $i$ th block by  $\mathbf{J}_i$  so that  $\mathbf{J} = \bigoplus_{i=1}^k \mathbf{J}_i$ . The spectral theorem then allows us to orthogonally diagonalize  $\mathbf{J}_i = \mathbf{K}_i \mathbf{D}_i \mathbf{K}_i^T$ , where  $\mathbf{K}_i \in O(m_i)$  and  $\mathbf{D}_i$  is an  $m_i \times m_i$  diagonal matrix with all diagonal entries being 1 or  $-1$ . Put  $\mathbf{K} = \bigoplus_{i=1}^k \mathbf{K}_i$  and  $\mathbf{D} = \bigoplus_{i=1}^k \mathbf{D}_i$ . Then  $\mathbf{K} \in O(q)$ ,  $\mathbf{D}$  is a  $q \times q$  diagonal matrix with 1 or  $-1$  diagonal entry, and  $\mathbf{J} = \mathbf{K}\mathbf{D}\mathbf{K}^T$ . This involved construction leads us to the outcome that  $\mathbf{K}$  in fact commutes with  $\mathbf{S}$ :

$$\mathbf{K}\mathbf{S} = \bigoplus_{i=1}^k s_i \mathbf{K}_i = \mathbf{S}\mathbf{K}.$$

Put  $\mathbf{P} = \mathbf{P}'\mathbf{K}$ ,  $\mathbf{R} = \mathbf{R}'\mathbf{K}$ . We have

$$\mathbf{Q} = \mathbf{P}'\mathbf{J}(\mathbf{R}')^T = \mathbf{P}'\mathbf{K}\mathbf{D}\mathbf{K}^T(\mathbf{R}')^T = \mathbf{P}\mathbf{D}\mathbf{R}^T$$

and

$$\mathbf{R}\mathbf{S}\mathbf{P}^T = \mathbf{R}'\mathbf{K}\mathbf{S}\mathbf{K}^T(\mathbf{P}')^T = \mathbf{R}'\mathbf{S}\mathbf{K}\mathbf{K}^T(\mathbf{P}')^T = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T = \mathbf{L}.$$

This completes the proof.

Finally, let us just note that, in case  $\mathbf{L}$  is symmetric positive definite, the spectral theorem allows us to take  $\mathbf{R}' = \mathbf{P}'$  from the start and it will yield in the end  $\mathbf{L} = \mathbf{P}\mathbf{S}\mathbf{P}^T$  and  $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{P}^T$ . It then follows easily that  $\mathbf{Q}^2 = \mathbf{I}_q$  and  $\mathbf{Q}^T = \mathbf{Q}$ . If  $\mathbf{D} = \mathbf{I}_q$ , then  $\|\mathbf{Q} - \mathbf{I}_q\| = \|\mathbf{D} - \mathbf{I}_q\| = 0$ . If any entry of  $\mathbf{D}$  is  $-1$ ,  $\|\mathbf{D} - \mathbf{I}_q\| = 2$ .  $\square$

**Lemma 5.** If  $\mathbf{L}$  is an invertible  $q \times q$  matrix with singular value decomposition  $\mathbf{L} = \mathbf{R}\mathbf{S}\mathbf{P}^T$ , then

$$\mathbf{P}\mathbf{R}^T = (\mathbf{L}^T\mathbf{L})^{-\frac{1}{2}}\mathbf{L}^T.$$

*Proof.* Given that

$$\mathbf{L}^T\mathbf{L} = \mathbf{P}\mathbf{S}^2\mathbf{P}^T,$$

we have

$$(\mathbf{L}^T\mathbf{L})^{\frac{1}{2}} = \mathbf{P}\mathbf{S}\mathbf{P}^T.$$

Therefore:

$$\mathbf{L} = \mathbf{R}\mathbf{P}^T \mathbf{P}\mathbf{S}\mathbf{P}^T = \mathbf{R}\mathbf{P}^T (\mathbf{L}^T \mathbf{L})^{\frac{1}{2}}.$$

The result follows from inverting  $(\mathbf{L}^T \mathbf{L})^{\frac{1}{2}}$  and take transpose.  $\square$

We now state and prove all the results about the exponential and the logarithmic map in one single proposition.

**Proposition 6.** The exponential map is computed as

$$\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}) = [\mathbf{A} + \mathbf{V}, s + c], \quad (1.16)$$

where  $[\mathbf{A}, s] \in M$ ,  $\mathbf{v} \in T_{[\mathbf{A}, s]}M$ , and  $(\mathbf{V}, c)$  is the horizontal lift of  $\mathbf{v}$  at  $(\mathbf{A}, s)$ .

Let us use  $\mathcal{E}_{[\mathbf{A}, s]}$  to denote the open ball in  $T_{[\mathbf{A}, s]}M \cong \mathbb{R}^d$  centred at  $\mathbf{0}$  of radius  $\min(\sigma_{\min}(\mathbf{A}), s)$ . Given that (1.16) is well-defined at least over all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ , we from now on treat  $\mathcal{E}_{[\mathbf{A}, s]}$  as the domain of the exponential map. That is, the function  $\text{Exp}_{[\mathbf{A}, s]} : \mathcal{E}_{[\mathbf{A}, s]} \rightarrow U_{[\mathbf{A}, s]}$ , where  $U_{[\mathbf{A}, s]} := \text{Exp}_{[\mathbf{A}, s]}(\mathcal{E}_{[\mathbf{A}, s]})$  is an open subset of  $M$  known as a geodesic ball, is defined by (1.16). Note the following facts:

1. For every  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ ,  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$  is symmetric positive definite.
2.  $\text{Exp}_{[\mathbf{A}, s]}$  is a diffeomorphism.
3. If  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ , then for every representative  $(\mathbf{A}, s)$  of  $[\mathbf{A}, s]$  and every representative  $(\mathbf{B}, t)$  of  $[\mathbf{B}, t]$ , the matrix  $\mathbf{A}^T \mathbf{B}$  will always be invertible. Given representative  $(\mathbf{A}, s)$  of  $[\mathbf{A}, s]$ , some representative  $(\mathbf{B}, t)$  of  $[\mathbf{B}, t]$  will furthermore make  $\mathbf{A}^T \mathbf{B}$  symmetric positive definite.
4. The logarithmic map  $\text{Exp}_{[\mathbf{A}, s]}^{-1} : U_{[\mathbf{A}, s]} \rightarrow \mathcal{E}_{[\mathbf{A}, s]}$  is computed as follows: for every  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ ,

$$\text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t]) = d\pi_{(\mathbf{A}, s)}((\mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B}) - \mathbf{A}, t - s)), \quad (1.17)$$

where

$$\mathbf{Q}(\mathbf{A}, \mathbf{B}) = (\mathbf{B}^T \mathbf{A} \mathbf{A}^T \mathbf{B})^{-\frac{1}{2}} \mathbf{B}^T \mathbf{A} \in O(q). \quad (1.18)$$

5. The geodesic ball  $U_{[\mathbf{A}, s]}$  equals to the set of all  $[\mathbf{B}, t] \in M$  such that  $\mathbf{A}^T \mathbf{B}$  is invertible and

$$\|\mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B}) - \mathbf{A}\|_F^2 + (t - s)^2 < \min(\sigma_{\min}(\mathbf{A}), s)^2. \quad (1.19)$$

6. If  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ , then the Riemannian distance between  $[\mathbf{A}, s]$  and  $[\mathbf{B}, t]$  equals to

$$\left| \text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t]) \right| = \sqrt{\|\mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B}) - \mathbf{A}\|_F^2 + (t - s)^2}, \quad (1.20)$$

which is realized by the geodesic  $\gamma : [0, 1] \rightarrow M$  defined by, for all  $t \in [0, 1]$ :

$$\gamma(t) = \text{Exp}_{[\mathbf{A}, s]}(t \text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t])).$$

*Proof.* Before we start the proof, let us first check that all the statements are indeed well-defined. If  $\mathbf{R} \in O(q)$  and  $\mathbf{P} \in O(q)$ , then

$$\mathbf{Q}(\mathbf{A}\mathbf{P}, \mathbf{B}\mathbf{R}) = \mathbf{R}^T \mathbf{Q}(\mathbf{A}, \mathbf{B}) \mathbf{P}. \quad (1.21)$$

Therefore, the same computation as in the proof of Proposition 3 will show that (1.17) is indeed well-defined. Also, the contents of the inequality (1.19) and the expression (1.20) are independent of representatives and the description of the set  $U_{[\mathbf{A}, s]}$  makes sense.

Proposition 3 already shows that the computation of the exponential map is valid as long as  $(\mathbf{A} + \mathbf{V}, s + c) \in \Theta$ . Therefore, the exponential map is certainly well-defined over all  $\mathbf{v}$  with small enough norm. To have a quantified

upper bound of the norm, we use the classical result of Weyl's perturbation inequality (see for example Chapter 7.3 Problem 16 equation (7.3.15) of [Horn and Johnson, 2013](#)) to get

$$\sigma_{\min}(\mathbf{A} + \mathbf{V}) \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\| \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\|_F \geq \sigma_{\min}(\mathbf{A}) - |\mathbf{v}| > 0,$$

whenever  $|\mathbf{v}| < \min(\sigma_{\min}(\mathbf{A}), s)$  or equivalently  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ . We have made use of the two facts that  $\|\mathbf{V}\| \leq \|\mathbf{V}\|_F$  for any matrix  $\mathbf{V}$  and that  $|\mathbf{v}| = \sqrt{\|\mathbf{V}\|_F^2 + c^2}$  by isometry. Thus, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ ,  $\text{rank}(\mathbf{A} + \mathbf{V}) = q$  and the exponential map is well-defined in  $\mathcal{E}_{[\mathbf{A}, s]}$ .

We now start the proof of the five facts of the function  $\text{Exp}_{[\mathbf{A}, s]} : \mathcal{E}_{[\mathbf{A}, s]} \rightarrow U_{[\mathbf{A}, s]}$ .

The first fact can be quickly settled. Let  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$  with horizontal lift  $(\mathbf{V}, c)$  to  $(\mathbf{A}, s)$  be given. As proved in Proposition 1,  $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$  and thus  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$  is symmetric. If  $\mathbf{x} \in \mathbb{R}^q$  is any unit vector, then

$$\mathbf{x}^T \mathbf{A}^T(\mathbf{A} + \mathbf{V})\mathbf{x} = |\mathbf{A}\mathbf{x}|^2 + (\mathbf{A}\mathbf{x})^T \mathbf{V}\mathbf{x} \geq |\mathbf{A}\mathbf{x}|(|\mathbf{A}\mathbf{x}| - |\mathbf{V}\mathbf{x}|) \geq \sigma_{\min}(\mathbf{A})(\sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\|_F) > 0$$

because  $\sigma_{\min}(\mathbf{A}) > |\mathbf{v}| \geq \|\mathbf{V}\|_F$ .

The second fact has a longer proof. Being a well-defined exponential map,  $\text{Exp}_{[\mathbf{A}, s]}$  is automatically a smooth map. To prove that it is actually a diffeomorphism in  $\mathcal{E}_{[\mathbf{A}, s]}$ , we must first show that it is injective in  $\mathcal{E}_{[\mathbf{A}, s]}$  so that the inverse function, the logarithmic map  $\text{Exp}_{[\mathbf{A}, s]}^{-1}$ , exists. Once this is done, we prove that  $U_{[\mathbf{A}, s]}$  is open and that  $\text{Exp}_{[\mathbf{A}, s]}^{-1}$  is smooth in one shot by proving that its differential  $d(\text{Exp}_{[\mathbf{A}, s]})_{\mathbf{v}}$  is an isomorphism for every given  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ .

We thus move to show that  $\text{Exp}_{[\mathbf{A}, s]}$  is injective in  $\mathcal{E}_{[\mathbf{A}, s]}$ . Suppose that  $\mathbf{v}_1$  and  $\mathbf{v}_2$  are two vectors in  $\mathcal{E}_{[\mathbf{A}, s]}$  who lifts to  $(\mathbf{V}_1, c_1)$  and  $(\mathbf{V}_2, c_2)$  respectively at  $(\mathbf{A}, s)$  such that  $\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}_1) = \text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}_2)$ . We prove that  $\mathbf{v}_1 = \mathbf{v}_2$ . (1.16) immediately shows that  $c_1 = c_2$  and

$$\mathbf{A} + \mathbf{V}_1 = (\mathbf{A} + \mathbf{V}_2)\mathbf{Q}$$

for some  $\mathbf{Q} \in O(q)$ . Lemma 4 with the symmetric positive definite matrix  $\mathbf{L} = \mathbf{A}^T(\mathbf{A} + \mathbf{V}_2)$  shows that  $\|\mathbf{Q} - \mathbf{I}_q\|$  can only equal to 0 or 2. Given that

$$\sigma_{\min}(\mathbf{A})\|\mathbf{Q} - \mathbf{I}_q\| \leq \|\mathbf{A}\mathbf{Q} - \mathbf{A}\| = \|\mathbf{V}_1 - \mathbf{V}_2\mathbf{Q}\| \leq \|\mathbf{V}_1\| + \|\mathbf{V}_2\| < 2\sigma_{\min}(\mathbf{A}),$$

we have proved that  $\|\mathbf{Q} - \mathbf{I}_q\| < 2$  and  $\mathbf{Q} = \mathbf{I}_q$ . Therefore,  $\mathbf{V}_1 = \mathbf{V}_2$  and  $\mathbf{v}_1 = \mathbf{v}_2$ . The exponential map is indeed injective in  $\mathcal{E}_{[\mathbf{A}, s]}$ .

Next, let  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$  be given. We prove that  $d(\text{Exp}_{[\mathbf{A}, s]})_{\mathbf{v}}$  is an isomorphism because  $\ker(d(\text{Exp}_{[\mathbf{A}, s]})_{\mathbf{v}})$  consists only of the zero vector. Suppose that  $\mathbf{v}$  lifts to  $(\mathbf{V}, c)$  at  $(\mathbf{A}, s)$ . To make the task notationally feasible, we implicitly canonically identify the Euclidean spaces together. Let us write  $l : T_{[\mathbf{A}, s]}M \rightarrow H_{(\mathbf{A}, s)}$  for the horizontal lift map:  $l(\mathbf{v}) = (\mathbf{V}, c)$ , which is a linear isometry. Define a map  $F : l(\mathcal{E}_{[\mathbf{A}, s]}) \rightarrow M$  by setting, for any  $(\mathbf{U}, b) \in l(\mathcal{E}_{[\mathbf{A}, s]}) \subseteq H_{(\mathbf{A}, s)}$ , that  $F((\mathbf{U}, b)) = \pi((\mathbf{A} + \mathbf{U}, s + b))$ . In this notation,  $\text{Exp}_{[\mathbf{A}, s]} = F \circ l$  and the chain rule shows that

$$d(\text{Exp}_{[\mathbf{A}, s]})_{\mathbf{v}} = dF_{(\mathbf{V}, c)} \circ dl_{\mathbf{v}}.$$

Therefore, the only thing left to show is that  $\ker(dF_{(\mathbf{V}, c)})$  contains only the zero vector. For any  $(\mathbf{U}, b) \in l(\mathcal{E}_{[\mathbf{A}, s]})$ , we always have

$$dF_{(\mathbf{V}, c)}((\mathbf{U}, b)) = \left. \frac{d}{dt} \right|_{t=0} \pi((\mathbf{A} + \mathbf{V} + t\mathbf{U}, s + c + tb)) = d\pi_{(\mathbf{A} + \mathbf{V}, s + c)}((\mathbf{U}, b)).$$

Therefore, if  $(\mathbf{U}, b) \in \ker(dF_{(\mathbf{V}, c)})$ , then  $(\mathbf{U}, b) \in V_{(\mathbf{A} + \mathbf{V}, s + c)}$  so that  $b = 0$  and  $\mathbf{U} = (\mathbf{A} + \mathbf{V})\mathbf{\Omega}$  for some  $q \times q$  skew-symmetric matrix  $\mathbf{\Omega}$ . But  $(\mathbf{U}, b) \in H_{(\mathbf{A}, s)}$  as well. Thus,  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})\mathbf{\Omega} = \mathbf{\Omega}^T(\mathbf{A} + \mathbf{V})^T \mathbf{A}$ . Given that  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$  is also symmetric, we arrive at

$$\mathbf{A}^T(\mathbf{A} + \mathbf{V})\boldsymbol{\Omega} + \boldsymbol{\Omega}\mathbf{A}^T(\mathbf{A} + \mathbf{V}) = \mathbf{0}.$$

By the spectral theorem, choose  $\mathbf{P} \in O(q)$  such that  $\mathbf{A}^T(\mathbf{A} + \mathbf{V}) = \mathbf{P}\mathbf{S}\mathbf{P}^T$  where  $\mathbf{S}$  is the  $q \times q$  diagonal matrix with diagonal entries  $\lambda_1(\mathbf{A}^T(\mathbf{A} + \mathbf{V})), \dots, \lambda_q(\mathbf{A}^T(\mathbf{A} + \mathbf{V}))$ . Substitute this yields

$$\mathbf{S}(\mathbf{P}^T\boldsymbol{\Omega}\mathbf{P}) + (\mathbf{P}^T\boldsymbol{\Omega}\mathbf{P})\mathbf{S} = \mathbf{0}.$$

Looking at each entry to get

$$(\lambda_i(\mathbf{A}^T(\mathbf{A} + \mathbf{V})) + \lambda_j(\mathbf{A}^T(\mathbf{A} + \mathbf{V}))) (\mathbf{P}^T\boldsymbol{\Omega}\mathbf{P})_{ij} = 0,$$

for all  $i$  and all  $j \in \{1, \dots, q\}$ . Given that  $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$  is positive definite, this forces  $\mathbf{P}^T\boldsymbol{\Omega}\mathbf{P} = \mathbf{0}$  and hence  $\boldsymbol{\Omega} = \mathbf{0}$ . We have proven that  $\text{Exp}_{[\mathbf{A}, s]}$  is a diffeomorphism in  $\mathcal{E}_{[\mathbf{A}, s]}$  and that  $U_{[\mathbf{A}, s]}$  is an open subset of  $M$ .

Now that the logarithmic map  $\text{Exp}_{[\mathbf{A}, s]}^{-1} : U_{[\mathbf{A}, s]} \rightarrow \mathcal{E}_{[\mathbf{A}, s]}$  is well-defined and is smooth, we will compute it and show the next three facts in one shot. Let  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$  be given. By (1.16),  $[\mathbf{B}, t] = [\mathbf{A} + \mathbf{V}, s + c]$ , where  $(\mathbf{V}, c)$  is the horizontal lift of  $\mathbf{v} = \text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t])$  to  $(\mathbf{A}, s)$ . It immediately follows that  $c = t - s$ . The first fact shows that, with  $\mathbf{B}_0 := \mathbf{A} + \mathbf{V}$ ,  $\mathbf{A}^T\mathbf{B}_0$  is symmetric positive definite. Given that the value of  $|\det(\mathbf{A}^T\mathbf{B})|$  is independent of representatives,  $\mathbf{A}^T\mathbf{B}$  is always invertible for any representative. The third fact is therefore established. Given that  $\mathbf{Q}(\mathbf{A}, \mathbf{B}_0) = \mathbf{I}_q$ , we immediately see that (1.17) holds with  $\mathbf{B}_0$  in place of  $\mathbf{B}$ . However, (1.17) is independent of representatives so it is valid for every representative  $\mathbf{B}$ . Finally, let us write  $C$  for the set of all points  $[\mathbf{B}, t] \in M$  such that  $\mathbf{A}^T\mathbf{B}$  is invertible and satisfies (1.19). By using  $\mathbf{B}_0$  in place of  $\mathbf{B}$  in (1.19), it is clear that  $U_{[\mathbf{A}, s]} \subseteq C$ . Conversely, let  $[\mathbf{B}, t] \in C$  be given. Put  $\mathbf{B}_0 := \mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B})$ . Then  $\mathbf{A}^T\mathbf{B}_0$  is again symmetric positive definite. In particular,  $\mathbf{A}^T(\mathbf{B}_0 - \mathbf{A})$  is symmetric and  $(\mathbf{V}, c) := (\mathbf{B}_0 - \mathbf{A}, t - s)$  is a horizontal vector at  $(\mathbf{A}, s)$ . If  $\mathbf{v} := d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c))$ , then (1.19) shows that  $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$  and  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ . We conclude that  $U_{[\mathbf{A}, s]} = C$ .

Finally, let us calculate the Riemannian distance between  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$  and  $[\mathbf{A}, s]$ . Proposition 6.11 in Lee (2018) shows that the Riemannian distance between points in a geodesic ball is indeed attained by exactly one geodesic. It suffices to find a geodesic minimizing the Riemannian distance among all geodesics. By Proposition 3, all geodesics connecting  $[\mathbf{A}, s]$  to  $[\mathbf{B}, t]$  takes the form  $\gamma(t) = [\mathbf{A} + t\mathbf{V}, s + tc]$ , where  $(\mathbf{V}, c)$  is the horizontal lift of some vector  $\mathbf{v} \in T_{[\mathbf{A}, s]}M$ . It is clear that  $c = t - s$  and  $\mathbf{V} = \mathbf{B}\mathbf{Q} - \mathbf{A}$  for some  $\mathbf{Q} \in O(q)$ . Given that the endpoints are fixed, the time is fixed, and  $\gamma$  has constant speed  $|\mathbf{v}|$ , the choice of  $\mathbf{v}$  with minimal speed will give the smallest arclength and hence the smallest Riemannian distance. To carry out this minimization, Lemma 4 shows that there exists  $\mathbf{R} \in O(q)$ ,  $\mathbf{P} \in O(q)$  such that  $\mathbf{A}^T\mathbf{B} = \mathbf{R}\mathbf{S}\mathbf{P}^T$  and  $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$ . Let us compute the norm of  $\mathbf{v}$ :

$$\begin{aligned} g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v}) &= g_{(\mathbf{A}, s)}^\Theta((\mathbf{B}\mathbf{Q} - \mathbf{A}, t - s), (\mathbf{B}\mathbf{Q} - \mathbf{A}, t - s)) \\ &= \text{tr}((\mathbf{B}\mathbf{Q} - \mathbf{A})(\mathbf{B}\mathbf{Q} - \mathbf{A})^T) + (t - s)^2 = \text{tr}(\mathbf{B}\mathbf{B}^T) + \text{tr}(\mathbf{A}\mathbf{A}^T) - 2\text{tr}(\mathbf{S}\mathbf{D}) + (t - s)^2, \end{aligned}$$

where the last equality substituted  $\mathbf{A}^T\mathbf{B}\mathbf{Q}' = \mathbf{R}\mathbf{S}\mathbf{P}^T\mathbf{P}\mathbf{D}\mathbf{R}^T = \mathbf{R}\mathbf{S}\mathbf{D}\mathbf{R}^T$ . The fact that the diagonal entries of  $\mathbf{S}$  are strictly positive show that  $g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v})$  is minimized precisely when  $\mathbf{D} = \mathbf{I}_q$ . By Lemma 5, it must be that  $\mathbf{Q} = \mathbf{P}\mathbf{R}^T = \mathbf{Q}(\mathbf{A}, \mathbf{B})$  and the last fact holds. The proof is complete.  $\square$

**Remark 7.** In the above proof, it might seem that the proof of (1.17) is trivialized while the amazing matrix (1.18) seems to come from nowhere. This is not at all the case. Originally, the matrix  $\mathbf{Q}(\mathbf{A}, \mathbf{B})$  comes from considering the cases for a generic representative whose matrix  $\mathbf{A}^T\mathbf{B}$  is only invertible, but not necessarily symmetric or positive definite. Given  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ , (1.16) shows that

$$\text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t]) = d\pi_{(\mathbf{A}, s)}((\mathbf{B}\mathbf{Q} - \mathbf{A}, t - s))$$

for some  $\mathbf{Q} \in O(q)$ . Given that  $\mathbf{A}^T(\mathbf{B}\mathbf{Q} - \mathbf{A})$  is symmetric, it follows that  $\mathbf{A}^T\mathbf{B}\mathbf{Q}$  is symmetric and Lemma 4 shows that there exists  $\mathbf{R} \in O(q)$ ,  $\mathbf{P} \in O(q)$  such that  $\mathbf{A}^T\mathbf{B} = \mathbf{R}\mathbf{S}\mathbf{P}^T$  and  $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$ . Thus, by continuity

of the logarithmic map, if  $[B, t] \rightarrow [A, s]$  and for some representative  $(B, t) \rightarrow (A, s)$ , then it must be the case that  $BQ - A \rightarrow 0$  or equivalently  $Q \rightarrow I_q$ . Therefore, heuristically, the only reasonable choice of  $D$  is  $I_q$ . So  $Q = PR^T$ . This value is actually uniquely determined by  $A$  and  $B$  and gives exactly  $Q = Q(A, B)$ , as shown in Lemma 5, despite that neither  $P$  nor  $R$  is unique. This argument is not discarded. In the proof of injectivity of the exponential map, the argument is present but simplified by symmetric positive definiteness. The problem of calculating the Riemannian distance actually makes us show a stronger result that minimizing the norm of  $BQ - A$  with fixed  $A$  and  $B$  will already lead to  $Q = Q(A, B)$ .

We end this section with a result concerning the limit of a sequence.

**Proposition 8.** Given a sequence  $\{A_n\}_{n=1}^\infty$  of real  $p \times q$  matrices with the property  $A_n A_n^T \rightarrow A A^T$  as  $n \rightarrow \infty$  for a  $p \times q$  matrix  $A$  with  $\text{rank}(A) = q$ , then

$$\lim_{n \rightarrow \infty} \min_{Q \in O(q)} \|A_n Q - A\|_F = 0.$$

In particular, if  $\{(A_n, s_n)\}_{n=1}^\infty$  is a sequence in  $\Theta$  such that, for some  $(A, s) \in \Theta$ , we have both  $s_n \rightarrow s$  and  $\Phi((A_n, s_n)) \rightarrow \Phi((A, s))$  as  $n \rightarrow \infty$ , then  $[A_n, s_n] \rightarrow [A, s]$  in  $M$  as  $n \rightarrow \infty$ .

*Proof.* Assume, for the sake of contradiction, that there exists an  $\epsilon > 0$  and a subsequence  $\{n_k\}_{k=1}^\infty$  such that, for all  $k$ ,

$$\min_{Q \in O(q)} \|A_{n_k} Q - A\|_F \geq \epsilon.$$

Given that  $\|A_n\|_F^2 = \text{tr}(A_n A_n^T)$  is convergent, in particular bounded, choose subsubsequence  $\{n_{k_l}\}_{l=1}^\infty$  such that  $A_{n_{k_l}} \rightarrow B$  as  $l \rightarrow \infty$  for some  $p \times q$  matrix  $B$ . It follows that  $BB^T = AA^T$ . Therefore,  $\text{rank}(A) = \text{rank}(B) = q$  and  $B = AP$  for some  $P \in O(q)$ . We have

$$\min_{Q \in O(q)} \|A_{n_{k_l}} Q - A\|_F \leq \|A_{n_{k_l}} P^T - A\|_F = \|A_{n_{k_l}} P^T - B P^T\|_F = \|A_{n_{k_l}} - B\|_F \rightarrow 0$$

as  $l \rightarrow \infty$ , a contradiction.

Now suppose that  $\{(A_n, s_n)\}_{n=1}^\infty$  is a sequence in  $\Theta$  such that, for some  $(A, s) \in \Theta$ , we have both  $s_n \rightarrow s$  and  $\Phi((A_n, s_n)) \rightarrow \Phi((A, s))$  as  $n \rightarrow \infty$ . It follows that  $A_n A_n^T \rightarrow A A^T$  and the previous paragraph shows that there exists a sequence  $\{Q_n\}_{n=1}^\infty$  of  $q \times q$  orthogonal matrices such that  $A_n Q_n \rightarrow A$  as  $n \rightarrow \infty$ . Thus,

$$\lim_{n \rightarrow \infty} [A_n, s_n] = \lim_{n \rightarrow \infty} \pi((A_n Q_n, s_n)) = \pi((A, s)) = [A, s].$$

The proof is complete. □

### 1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model  $\mathcal{P}$  given by (1.7) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in Sun et al. (2026), definition III.1 and III.2.

Let us introduce a few notations. For each  $[W, \sigma^2] \in M$ , let

$$p(\mathbf{x}; [W, \sigma^2]) = (2\pi)^{-\frac{p}{2}} \det^{-\frac{1}{2}}(\Sigma([W, \sigma^2])) \exp\left(-\frac{1}{2} \mathbf{x}^T (\Sigma([W, \sigma^2]))^{-1} \mathbf{x}\right) \quad (1.22)$$

be the density of  $N_p(\mathbf{0}, \Sigma([W, \sigma^2]))$ . For  $\mathbf{v} \in \mathcal{E}_{[W, \sigma^2]} \subset T_{[W, \sigma^2]} M \cong \mathbb{R}^d$ , write  $l_{[W, \sigma^2]}(\mathbf{v})$  for the log-likelihood function of the local experiment

$$l_{[W, \sigma^2]}(\mathbf{v})(\mathbf{x}) = \log\left(p(\mathbf{x}; \text{Exp}_{[W, \sigma^2]}(\mathbf{v}))\right). \quad (1.23)$$

Define the square root likelihood function  $r_{[\mathbf{W}, \sigma^2]} : \mathcal{E}_{[\mathbf{W}, \sigma^2]} \rightarrow L^2(\mathbb{R}^p)$  by setting, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ , that

$$r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) = \sqrt{p(\mathbf{x}; \text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))}. \quad (1.24)$$

**Theorem 9.**  $r_{[\mathbf{W}, \sigma^2]}$  is Fréchet differentiable at any  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ . In particular,  $\mathcal{P}$  is differentiable in quadratic mean with score

$$s(\mathbf{x}; [\mathbf{W}, \sigma^2])(\mathbf{h}) = \frac{1}{2} \text{tr} \left\{ \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} [\mathbf{x}\mathbf{x}^T - \left( \Sigma([\mathbf{W}, \sigma^2]) \right)] \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} (D_{\mathbf{h}}\Sigma([\mathbf{W}, \sigma^2])) \right\}, \quad (1.25)$$

where

$$D_{\mathbf{h}}\Sigma([\mathbf{W}, \sigma^2]) := \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(t\mathbf{h})) = \mathbf{U}\mathbf{W}^T + \mathbf{W}\mathbf{U}^T + b\mathbf{I}_p, \quad (1.26)$$

if  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  lifts to  $(\mathbf{U}, b)$  at  $(\mathbf{W}, \sigma^2)$ . Furthermore,  $\mathcal{P}$  is a regular model in the sense that the Fisher information operator

$$G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}_1, \mathbf{h}_2) = \frac{1}{2} \text{tr} \left\{ \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} D_{\mathbf{h}_1}\Sigma([\mathbf{W}, \sigma^2]) \left( \Sigma([\mathbf{W}, \sigma^2]) \right)^{-1} D_{\mathbf{h}_2}\Sigma([\mathbf{W}, \sigma^2]) \right\} \quad (1.27)$$

is positive definite.

*Proof.* The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of  $r_{[\mathbf{W}, \sigma^2]}$  at  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$  in a given direction  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to  $\mathbf{v} = \mathbf{0}$  to prove (1.25) and (1.27).

Given  $\mathbf{v}$  and  $\mathbf{h}$ , there exists an  $\epsilon > 0$  such that  $\mathbf{v} + t\mathbf{h} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$  whenever  $t \in (-\epsilon, \epsilon)$ . If they respectively lift to  $(\mathbf{V}, c)$  and  $(\mathbf{U}, b)$  at  $(\mathbf{W}, \sigma^2)$ , then  $\mathbf{v} + t\mathbf{h}$  lifts to  $(\mathbf{V} + t\mathbf{U}, c + tb)$  at  $(\mathbf{W}, \sigma^2)$ . First of all:

$$\begin{aligned} D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) &:= \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ (\mathbf{W} + \mathbf{V} + t\mathbf{U})(\mathbf{W} + \mathbf{V} + t\mathbf{U})^T + (\sigma^2 + c + tb)\mathbf{I}_p \right\} \\ &= \mathbf{U}\mathbf{V}^T + \mathbf{V}\mathbf{U}^T + \mathbf{U}\mathbf{W}^T + \mathbf{W}\mathbf{U}^T + b\mathbf{I}_p, \end{aligned} \quad (1.28)$$

proving (1.26) with special case  $\mathbf{v} = \mathbf{0}$ . Secondly, for all  $\mathbf{x} \in \mathbb{R}^p$ :

$$\begin{aligned} D_{\mathbf{h}}l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &:= \left. \frac{d}{dt} \right|_{t=0} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) - \frac{1}{2} \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right)^{-1} \mathbf{x} \right\} \\ &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= \frac{1}{2} \text{tr} \left\{ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} [\mathbf{x}\mathbf{x}^T - \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)] \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\}. \end{aligned} \quad (1.29)$$

Finally:

$$\begin{aligned} D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &= D_{\mathbf{h}} \exp \left( \frac{1}{2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \right) = \frac{1}{2} \exp \left( \frac{1}{2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \right) D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \\ &= \frac{1}{2} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}). \end{aligned} \quad (1.30)$$

To prove that (1.30) indeed defines a valid Gâteaux derivative, we must prove that  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$ . Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in [Gupta and Nagar \(2000\)](#). For  $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma)$ , Theorem 7.7.1(i) in [Gupta and Nagar \(2000\)](#) states that

$$\mathbb{E} [\mathbf{X}^T \mathbf{Z} \mathbf{X}] = \text{tr}(\mathbf{Z} \Sigma)$$

for any  $p \times p$  matrix  $\mathbf{Z}$  and Theorem 7.7.2 in [Gupta and Nagar \(2000\)](#) states that

$$\text{Cov}(\mathbf{X}^T \mathbf{Z} \mathbf{X}, \mathbf{X}^T \mathbf{Y} \mathbf{X}) = \text{tr}(\mathbf{Z} \Sigma \mathbf{Y}^T \Sigma) + \text{tr}(\mathbf{Z}^T \Sigma \mathbf{Y}^T \Sigma)$$

for any two  $p \times p$  matrices  $\mathbf{Y}$  and  $\mathbf{Z}$ . Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned} &\int_{\mathbb{R}^p} D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right] \\ &+ \frac{1}{2} \int_{\mathbb{R}^p} \left[ \mathbf{x}^T \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \right] p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} = 0 \end{aligned}$$

and, whenever  $\mathbf{h}_1$  and  $\mathbf{h}_2 \in T_{[\mathbf{W}, \sigma^2]}M$ :

$$\begin{aligned} &\int_{\mathbb{R}^p} D_{\mathbf{h}_1} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}_2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} \\ &= \frac{1}{4} \text{Cov}(\mathbf{X}^T \mathbf{Z}_1 \mathbf{X}, \mathbf{X}^T \mathbf{Z}_2 \mathbf{X}) = \frac{1}{4} \cdot 2 \text{tr} \left( \mathbf{Z}_1 \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \mathbf{Z}_2 \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_1} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_2} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right\}, \end{aligned} \quad (1.31)$$

where  $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})))$  and

$$\mathbf{Z}_j := \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}_j} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \quad (1.32)$$

is symmetric for  $j = 1, 2$ , so that the two terms in Theorem 7.7.2 of [Gupta and Nagar \(2000\)](#) coincide. Thus, in the particular case  $\mathbf{h} = \mathbf{h}_1 = \mathbf{h}_2$ :

$$\begin{aligned} \|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|^2 &= \int_{\mathbb{R}^p} [D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})]^2 d\mathbf{x} \\ &= \frac{1}{4} \int_{\mathbb{R}^p} (D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}))^2 p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} = \frac{1}{8} \text{tr} \left\{ \left[ \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right]^2 \right\}. \end{aligned} \quad (1.33)$$

Therefore,  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$  and (1.30) is indeed a valid Gâteaux derivative at  $\mathbf{v}$  in the direction  $\mathbf{h}$ .

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$R(\mathbf{v}, \mathbf{h})(\mathbf{x}) := r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})$$

and prove that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 = 0.$$

To start, the fundamental theorem of calculus shows that

$$\begin{aligned} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &= \int_0^1 \frac{d}{dt} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt \\ &= \int_0^1 \lim_{s \rightarrow 0} \frac{r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + (t+s)\mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x})}{s} dt = \int_0^1 D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt. \end{aligned}$$

Therefore, by Minkowski's integral inequality and the linearity of  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$  in  $\mathbf{h}$ , we have

$$\begin{aligned} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\| &= \left\| \int_0^1 (D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})) dt; L^2(\mathbb{R}^p) \right\| \\ &\leq \int_0^1 \|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = |\mathbf{h}| \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \end{aligned}$$

and

$$\begin{aligned} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 &\leq \left\{ \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2 \\ &\leq \left\{ \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2. \end{aligned}$$

The proof will be complete once we can show that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = 0. \quad (1.34)$$

We must therefore examine the continuity of  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ , which is easy to see if we revisit the formulas. (1.28) shows that  $D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))$  is smooth in both  $\mathbf{h}$  and  $\mathbf{v}$ . Thus, (1.30) shows that, for each fixed  $\mathbf{x} \in \mathbb{R}^p$ ,  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})$  is smooth in both  $\mathbf{h}$  and  $\mathbf{v}$  and (1.33) shows that  $\|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|$  is smooth in both  $\mathbf{h}$  and  $\mathbf{v}$ . Scheffé's lemma shows that  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ , as a map to  $L^2(\mathbb{R}^p)$ , is continuous in  $\mathbf{h}$  and in  $\mathbf{v}$ . The integrand inside (1.34) is thus bounded uniformly over all  $t \in [0, 1]$  and all  $\mathbf{h}$  in a compact neighborhood of  $\mathbf{0}$  and it suffices to show that, for every  $t \in [0, 1]$ , that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| = 0, \quad (1.35)$$

thanks to the bounded convergence theorem. To prove (1.35), let  $\mathbf{e}_1, \dots, \mathbf{e}_d$  be a  $g_{[\mathbf{W}, \sigma^2]}$  orthonormal basis. Every unit vector  $\mathbf{u}$  can be expressed as  $\mathbf{u} = \sum_{i=1}^d u_i \mathbf{e}_i$  with  $\sum_{i=1}^d u_i^2 = 1$ . By linearity of  $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$  in  $\mathbf{h}$ , we have

$$\begin{aligned} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| &\leq \sum_{i=1}^d |u_i| \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \\ &\leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|. \end{aligned}$$

Taking supremum yields

$$\sup_{|u|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|.$$

Given that  $D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) \rightarrow D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$  in  $L^2(\mathbb{R}^p)$  as  $\mathbf{h} \rightarrow \mathbf{0}$  for every  $i$ , every  $t$ , and every  $\mathbf{v}$ , the proof of Fréchet differentiability is complete.

Finally, the formulas (1.28), (1.29), and (1.30) simplifies when  $\mathbf{v} = \mathbf{0}$ :

$$\begin{aligned} D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) &= \frac{1}{2} r_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) \\ &= \frac{1}{4} \sqrt{p(\mathbf{x}; [\mathbf{W}, \sigma^2])} \operatorname{tr} \left\{ (\Sigma([\mathbf{W}, \sigma^2]))^{-1} [\mathbf{x}\mathbf{x}^T - (\Sigma([\mathbf{W}, \sigma^2]))] (\Sigma([\mathbf{W}, \sigma^2]))^{-1} (D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2])) \right\}. \end{aligned}$$

The score  $s(\mathbf{x}; [\mathbf{W}, \sigma^2])(\mathbf{h})$  therefore equals to  $D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x})$  and is indeed given by (1.25). The Fisher information operator (1.27) is obtained by taking  $\mathbf{v} = \mathbf{0}$  in (1.31). To show that  $G_{[\mathbf{W}, \sigma^2]}$  is positive definite, let  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  be given. We show that  $G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) \geq 0$  and equality holds only when  $\mathbf{h} = \mathbf{0}$ . The non-negative part is clear from (1.27):

$$G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr} \left\{ \left[ (\Sigma([\mathbf{W}, \sigma^2]))^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) \right]^2 \right\} \geq 0.$$

Assume from now on that the equality holds. A slight annoyance here is that the product of two symmetric matrices is not necessarily symmetric and we cannot immediately conclude that  $D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) = \mathbf{0}$ . We easily overcome this difficulty by using the cyclic property of the trace to get a symmetric matrix

$$\mathbf{N} := (\Sigma([\mathbf{W}, \sigma^2]))^{-\frac{1}{2}} D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) (\Sigma([\mathbf{W}, \sigma^2]))^{-\frac{1}{2}}$$

such that

$$0 = G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr}(\mathbf{N}^2).$$

Therefore,  $\mathbf{N} = \mathbf{0}$  and  $D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) = \mathbf{0}$ . Now (1.26) reads the same as (1.9) in the proof of Proposition 1. We therefore get  $(\mathbf{U}, b) \in \ker(d\Phi_{(\mathbf{W}, \sigma^2)}) = V_{(\mathbf{W}, \sigma^2)}$ . However, being lifted from  $\mathbf{h}$ ,  $(\mathbf{U}, b)$  must also be a horizontal vector. Therefore,  $\mathbf{h} = \mathbf{0}$ . We have proven that  $G_{[\mathbf{W}, \sigma^2]}$  is positive definite.  $\square$

**Proposition 10.** Suppose that  $[\mathbf{W}, \sigma^2] \in M$  is the true parameter so that each  $\mathbf{X}_i$  has density  $p(\cdot; [\mathbf{W}, \sigma^2])$  as in (1.22). We have the local asymptotic normality expansion

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \operatorname{Exp}_{[\mathbf{W}, \sigma^2]}\left(\frac{\mathbf{h}}{\sqrt{n}}\right)\right)}{p(\mathbf{X}_i; [\mathbf{W}, \sigma^2])} = \sum_{i=1}^n s(\mathbf{X}_i; [\mathbf{W}, \sigma^2]) \left(\frac{\mathbf{h}}{\sqrt{n}}\right) - \frac{1}{2} G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) + o_{\mathbb{P}}(1) \quad (1.36)$$

whenever  $n$  large enough that  $\frac{\mathbf{h}}{\sqrt{n}} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ . In particular,

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \operatorname{Exp}_{[\mathbf{W}, \sigma^2]}\left(\frac{\mathbf{h}}{\sqrt{n}}\right)\right)}{p(\mathbf{X}_i; [\mathbf{W}, \sigma^2])} \rightarrow N\left(-\frac{1}{2} G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}), G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h})\right) \quad (1.37)$$

in distribution as  $n \rightarrow \infty$ .

*Proof.* Given that  $\mathcal{P}$  is differentiable in quadratic mean and has positive definite Fisher operator, this follows from Proposition III.1 in Sun et al. (2026).  $\square$

## 1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

In this section, we assume that there is a true parameter  $[\mathbf{W}_0, \sigma_0^2] \in M$  for the model  $\mathcal{P}$  given in (1.7) and we investigate the consistency and asymptotic normality of the maximum likelihood estimator towards this true parameter.

More concretely, given data points  $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ , we consider them as realizations of independent identically distributed random variables  $\mathbf{X}_1, \dots, \mathbf{X}_n$  whose true density is  $p(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])$ , where  $p$  is given in (1.22). In what follows, we denote the true covariance matrix by

$$\Sigma_0 := \Sigma([\mathbf{W}_0, \sigma_0^2]) = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 \mathbf{I}_p. \quad (1.38)$$

The eigenvalues of the true covariance matrix are

$$\lambda_j(\Sigma_0) = \begin{cases} \sigma_j(\mathbf{W}_0)^2 + \sigma_0^2 & j = 1, \dots, q \\ \sigma_0^2 & j = q + 1, \dots, p. \end{cases} \quad (1.39)$$

The sample covariance matrix is defined by

$$\mathbf{S}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T. \quad (1.40)$$

Given that we will discuss  $\mathbf{S}_n$  a lot, we shall write  $\lambda_j$  in place of  $\lambda_j(\mathbf{S}_n)$ . We need also to fix one choice of orthonormal basis of eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_p$  where  $\mathbf{v}_j = \mathbf{v}_j(\mathbf{S}_n)$  corresponds to the eigenvalue  $\lambda_j$ . The facts we will use repeatedly is that, when  $p$  and  $q$  are both fixed,  $\mathbf{S}_n \rightarrow \Sigma_0$  almost surely as  $n \rightarrow \infty$  and, due to the fact that eigenvalues depends continuously on the matrices,  $\lambda_j \rightarrow \lambda_j(\Sigma_0)$  almost surely as  $n \rightarrow \infty$ .

We now formulate the maximum likelihood estimator as computed in [Tipping and Bishop \(1999\)](#). Put  $\mathbf{U} = [\mathbf{v}_1, \dots, \mathbf{v}_q]$ , a  $p \times q$  matrix, and  $\Delta_q$  to be a  $q \times q$  diagonal matrix with diagonal entries  $\lambda_1, \dots, \lambda_q$ . Then, the collection of all choices of  $(\hat{\mathbf{W}}_n, \hat{\sigma}_n^2)$  maximizing the likelihood is given by

$$\hat{\mathbf{W}}_n = \mathbf{U} (\Delta_q - \hat{\sigma}_n^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R} \quad (1.41)$$

and

$$\hat{\sigma}_n^2 = \frac{1}{p - q} \sum_{j=q+1}^p \lambda_j, \quad (1.42)$$

ranged over all rotation matrices  $\mathbf{R} \in O(q)$ .

It is immediately clear from (1.39) that, almost surely, for all sufficiently large  $n$ ,  $\lambda_j > 0$  for all  $j$  and  $\hat{\sigma}_n^2 > 0$  so that  $\hat{\sigma}_n^2$  is well-defined. Also,  $\hat{\sigma}_n^2 \rightarrow \sigma_0^2$  almost surely as  $n \rightarrow \infty$ . Another look at (1.39) shows that, almost surely, for all sufficiently large  $n$ ,  $\lambda_j - \hat{\sigma}_n^2 > 0$  for all  $j \in \{1, \dots, q\}$ ,  $\lambda_q > \lambda_{q+1}$ , and  $\text{rank}(\hat{\mathbf{W}}_n) = q$  so that  $\hat{\mathbf{W}}_n$  is well-defined. Thus, almost surely, for all sufficiently large  $n$ , the same model parametrized by  $M$  has a unique maximum likelihood estimator  $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$ .

First of all, I would like to argue that this quotient manifold parameter space allows us to settle consistency in an (arguably more straightforward) way that does not use [Datta and Chakrabarty \(2023\)](#) or [Redner \(1981\)](#).

**Theorem 11.** *The maximum likelihood estimator is consistent. That is,  $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \rightarrow [\mathbf{W}_0, \sigma_0^2]$  almost surely as  $n \rightarrow \infty$ .*

*Proof.* Put

$$\hat{\Sigma}_n := \Sigma([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]) = \hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T + \hat{\sigma}_n^2 \mathbf{I}_p. \quad (1.43)$$

In light of Proposition 8 and the fact that  $\hat{\sigma}_n^2$  is consistent, it suffices to prove that  $\hat{\Sigma}_n \rightarrow \Sigma_0$  almost surely as  $n \rightarrow \infty$ . The fact that  $\sum_{j=1}^p \mathbf{v}_j \mathbf{v}_j^T = \mathbf{I}_p$  allows us to write

$$\hat{\Sigma}_n = U (\Delta_q - \hat{\sigma}_n^2 \mathbf{I}_q) U^T + \hat{\sigma}_n^2 \mathbf{I}_p = \sum_{j=1}^q (\lambda_j - \hat{\sigma}_n^2) \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \mathbf{I}_p = \sum_{j=1}^q \lambda_j \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \sum_{j=q+1}^p \mathbf{v}_j \mathbf{v}_j^T$$

and

$$\mathbf{S}_n = \mathbf{S}_n \sum_{j=1}^p \mathbf{v}_j \mathbf{v}_j^T = \sum_{j=1}^p \lambda_j \mathbf{v}_j \mathbf{v}_j^T.$$

Thus,

$$\hat{\Sigma}_n - \mathbf{S}_n = \sum_{j=q+1}^p (\hat{\sigma}_n^2 - \lambda_j) \mathbf{v}_j \mathbf{v}_j^T$$

and

$$\|\hat{\Sigma}_n - \mathbf{S}_n\| \leq \sum_{j=q+1}^p |\hat{\sigma}_n^2 - \lambda_j| \|\mathbf{v}_j \mathbf{v}_j^T\| = \sum_{j=q+1}^p |\hat{\sigma}_n^2 - \lambda_j|.$$

Therefore, given that both  $\hat{\sigma}_n^2$  and  $\lambda_j$  will converge almost surely to  $\sigma_0^2$  when  $j \in \{q+1, \dots, p\}$ ,  $\|\hat{\Sigma}_n - \mathbf{S}_n\| \rightarrow 0$  almost surely as  $n \rightarrow \infty$ . The proof is complete.  $\square$

We now move to prove that, locally at the true parameter, the maximum likelihood estimator is asymptotically normal in the tangent space of the true parameter by local linearity. Given that the manifold is locally Euclidean, this result can be attacked with the classical approach, namely by verifying conditions of Theorem 5.23 in [van der Vaart \(1998\)](#), without too much use of geometric analysis. We start with computational lemmas.

**Lemma 12.** Given  $[\mathbf{W}, \sigma^2] \in M$  and any convex neighborhood  $U$  of  $\mathbf{0}$  in  $T_{[\mathbf{W}, \sigma^2]}M$  such that  $\bar{U} \subset \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ , there exists a function  $m : \mathbb{R}^p \rightarrow \mathbb{R}$  such that, whenever  $\mathbf{v}_1, \mathbf{v}_2 \in U$  and  $\mathbf{x} \in \mathbb{R}^p$ , we always have

$$|l_{[\mathbf{W}, \sigma^2]}(\mathbf{v}_1)(\mathbf{x}) - l_{[\mathbf{W}, \sigma^2]}(\mathbf{v}_2)(\mathbf{x})| \leq m(\mathbf{x}) |\mathbf{v}_1 - \mathbf{v}_2|.$$

This  $m$  has its second moment under any distribution  $\mu$  on  $\mathbb{R}^p$  that has the fourth moment. That is,

$$\int_{\mathbb{R}^p} (m(\mathbf{x}))^2 d\mu(\mathbf{x}) < \infty$$

whenever

$$\int_{\mathbb{R}^p} |\mathbf{x}|^4 d\mu(\mathbf{x}) < \infty.$$

*Proof.* Given  $[\mathbf{W}, \sigma^2] \in M$  and a convex neighborhood  $U$  of  $\mathbf{0}$  compactly contained in  $\mathcal{E}_{[\mathbf{W}, \sigma^2]}$ , the result will follow by the mean value inequality if we can find a function  $m$  such that

$$|D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})| \leq m(\mathbf{x})$$

for all unit vector  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$ , all  $\mathbf{v} \in \bar{U}$ , and all  $\mathbf{x} \in \mathbb{R}^p$ . Given that we have proved in Proposition 9 that  $(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})))^{-1}$  is continuous in  $\mathbf{v}$  and that  $D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))$  is continuous in both  $\mathbf{h}$  and  $\mathbf{v}$ , we choose constants  $C_1 \in (0, \infty)$  and  $C_2 \in (0, \infty)$  such that

$$\frac{1}{2} \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \right\|_F \leq C_1$$

and

$$\frac{1}{2} \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\|_F \leq C_2$$

for all unit vector  $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$  and all  $\mathbf{v} \in \bar{U}$ , thanks to the fact that  $\bar{U}$  is compact. from Proposition 6. By (1.29), we have

$$\begin{aligned} |D_{\mathbf{h}}l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})| &\leq \frac{1}{2} \|\mathbf{x}\mathbf{x}^T\|_F \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \right\|_F \\ &\quad + \frac{1}{2} \left\| \left( \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left( D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\|_F \leq C_1 \mathbf{x}^T \mathbf{x} + C_2 \end{aligned}$$

Setting  $m(\mathbf{x}) = C_1 \mathbf{x}^T \mathbf{x} + C_2$  will finish the proof.  $\square$

**Lemma 13.** Define function  $L : \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]} \rightarrow \mathbb{R}$  by setting, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$ ,

$$L(\mathbf{v}) = \int_{\mathbb{R}^p} l_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma_0) d\mathbf{x}.$$

Then  $\mathbf{0}$  is the unique maximizer of  $L$  and, for all  $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$ :

$$L(\mathbf{v}) = L(\mathbf{0}) - \frac{1}{2} G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{v}) + R(\mathbf{v})$$

with the remainder of the second order Taylor expansion satisfying

$$\lim_{|\mathbf{v}| \rightarrow 0} \frac{R(\mathbf{v})}{|\mathbf{v}|^2} = 0.$$

*Proof.* One should immediately recognize that  $L(\mathbf{0}) - L(\mathbf{v})$  equals to the Kullback-Leibler divergence of the true model  $N_p(\mathbf{0}, \Sigma_0)$  with respect to the model  $N_p(\mathbf{0}, \Sigma(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})))$ . Therefore,  $L(\mathbf{v})$  is maximized at all  $\mathbf{v}$  with  $\Sigma(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})) = \Sigma_0$ . Given that  $\Sigma$  is injective and that we have proven in Proposition 6 that  $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}$  is injective in  $\mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$ , this only happens when  $\mathbf{v} = \mathbf{0}$  and  $\mathbf{0}$  is the unique maximizer of  $L$ .

To obtain the Taylor expansion, we basically have to repeat the computation part in the proof of Proposition 9 on a slightly different problem with the second derivative at  $\mathbf{0}$  being computed as well. The first order derivative at  $\mathbf{0}$  is automatically 0 because  $\mathbf{0}$  is the maximizer.

Let  $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$  be given. We can easily compute that

$$L(t\mathbf{h}) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) - \frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}.$$

Suppose that  $\mathbf{h}$  lifts to  $(U, b)$  at  $(\mathbf{W}_0, \sigma_0^2)$ . (1.28) immediately shows that

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) = 2UU^T.$$

We now compute the second derivative  $L$  term by term. For the log determinant term, we have

$$\begin{aligned} \frac{d}{dt} \left( -\frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) &= -\frac{1}{2} \text{tr} \left[ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right], \\ \frac{d^2}{dt^2} \left( -\frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) &= -\frac{1}{2} \frac{d}{dt} \text{tr} \left[ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right] \\ &= -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d^2}{dt^2} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right\} \\ &\quad + \frac{1}{2} \text{tr} \left\{ \left[ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right]^2 \right\}, \end{aligned}$$

and therefore

$$\frac{d^2}{dt^2} \Big|_{t=0} \left( -\frac{1}{2} \log \left( \det \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) = -\text{tr} \{ \Sigma_0^{-1} \mathbf{U} \mathbf{U}^T \} + \frac{1}{2} \text{tr} \left\{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \right\}.$$

Next, for the trace term, we have

$$\begin{aligned} & \frac{d}{dt} \left( -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}, \\ & \frac{d^2}{dt^2} \left( -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) \\ &= \frac{1}{2} \text{tr} \left\{ -2 \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \right. \right. \\ & \quad \left. \frac{d}{dt} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \left. \right\} \\ & \quad + \frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d^2}{dt^2} \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}, \end{aligned}$$

and therefore

$$\frac{d^2}{dt^2} \Big|_{t=0} \left( -\frac{1}{2} \text{tr} \left\{ \left( \Sigma \left( \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) = -\text{tr} \left\{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \right\} + \text{tr} (\Sigma_0^{-1} \mathbf{U} \mathbf{U}^T).$$

We end up the negative of the Fisher information operator, as computed in (1.27).

$$\frac{d^2}{dt^2} \Big|_{t=0} L(t\mathbf{h}) = -\frac{1}{2} \text{tr} \left\{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \right\} = -G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{h}, \mathbf{h}).$$

This completes the proof.  $\square$

**Definition 14.** We define the **inverse Fisher information operator**  $G_{[\mathbf{W}, \sigma^2]}^{-1}$  as the inverse of the flat version of the Fisher information bilinear form. More concretely, instead of treating  $G_{[\mathbf{W}, \sigma^2]}$  as a bilinear form on  $T_{[\mathbf{W}, \sigma^2]}M$ , we treat it as a map  $G_{[\mathbf{W}, \sigma^2]} : T_{[\mathbf{W}, \sigma^2]}M \rightarrow T_{[\mathbf{W}, \sigma^2]}^*M$  by setting  $G_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) = G_{[\mathbf{W}, \sigma^2]}(\mathbf{v}, \cdot)$  and we define  $G_{[\mathbf{W}, \sigma^2]}^{-1} : T_{[\mathbf{W}, \sigma^2]}^*M \rightarrow T_{[\mathbf{W}, \sigma^2]}M$  to be the inverse.

**Proposition 15.** The maximum likelihood estimator is asymptotically linear at  $[\mathbf{W}_0, \sigma_0^2]$  with the influence function  $\text{IF} : \mathbb{R}^p \rightarrow T_{[\mathbf{W}_0, \sigma_0^2]}M$  given by

$$\text{IF}(\mathbf{x}) = G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}(s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])). \quad (1.44)$$

As described in Sun et al. (2026) Lemma III.3, this means

$$\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( \left[ \hat{\mathbf{W}}_n, \hat{\sigma}_n^2 \right] \right) - \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}(\mathbf{X}_i) \rightarrow 0 \quad (1.45)$$

in probability as  $n \rightarrow \infty$ . Also,  $\text{IF}(\mathbf{X})$  has mean  $\mathbf{0}$  and is square integrable if  $\mathbf{X}$  follows the true distribution  $N_p(\mathbf{0}, \Sigma_0)$ .

In particular, once we fix a  $g_{[\mathbf{W}_0, \sigma_0^2]}$  orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  and we identify  $T_{[\mathbf{W}_0, \sigma_0^2]}M$  canonically as  $\mathbb{R}^d$  (in concrete terms, identify  $\sum_{j=1}^d a_j \mathbf{e}_j$  to the point  $(a_1, \dots, a_d) \in \mathbb{R}^d$ ), then  $\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left( \left[ \hat{\mathbf{W}}_n, \hat{\sigma}_n^2 \right] \right)$  converges in distribution to  $N_d(\mathbf{0}, \mathbf{G}^{-1})$  as  $n \rightarrow \infty$ , where  $\mathbf{G}$  is the  $d \times d$  matrix defined by  $\mathbf{G}_{ij} = G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{e}_i, \mathbf{e}_j)$ .

*Proof.* Fix a  $g_{[\mathbf{W}_0, \sigma_0^2]}$  orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  of  $T_{[\mathbf{W}_0, \sigma_0^2]}M$ . Define the matrix  $\mathbf{G} \in \mathbb{R}^{d \times d}$  as in the statement of lemma and the vector  $\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]) \in \mathbb{R}^d$  by setting  $(\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))_j = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{e}_j)$ , for all  $j = 1, \dots, d$ . Identifying  $\sqrt{n}\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}\left(\left[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2\right]\right) \in T_{[\mathbf{W}_0, \sigma_0^2]}M$  to a point in  $\mathbb{R}^d$  through this basis, we immediately have, by combining the two previous lemmas and Theorem 5.23 in [van der Vaart \(1998\)](#), that

$$\sqrt{n}\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}\left(\left[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2\right]\right) - \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{G}^{-1} \nabla l(\mathbf{X}_i; [\mathbf{W}_0, \sigma_0^2]) \rightarrow 0 \quad (1.46)$$

in probability as  $n \rightarrow \infty$ . In particular,  $\sqrt{n}\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}\left(\left[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2\right]\right)$  converges in distribution to  $N_d(\mathbf{0}, \mathbf{G}^{-1})$  as  $n \rightarrow \infty$ . Also, if  $\mathbf{X}$  follows the true distribution  $N_p(\mathbf{0}, \Sigma_0)$ , then  $\mathbb{E}[\nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])] = \mathbf{0}$  and thus  $\mathbb{E}[\mathbf{G}^{-1} \nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])] = \mathbf{0}$ . Given that  $\nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])$  has its second moment, so does  $\mathbf{G}^{-1} \nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])$ .

We remark that (1.45) is simply (1.46) rephrased in a way using only vectors in  $T_{[\mathbf{W}_0, \sigma_0^2]}M$  defined without  $\mathbf{e}_1, \dots, \mathbf{e}_d$  and hence without the need of identifying  $T_{[\mathbf{W}_0, \sigma_0^2]}M$  with  $\mathbb{R}^d$ . Let us for simplicity define  $\mathbf{y} := \mathbf{G}^{-1} \nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]) \in \mathbb{R}^d$ . Put  $\mathbf{v} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$  by setting  $\mathbf{v} = \sum_{j=1}^d \mathbf{y}_j \mathbf{e}_j \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ . We claim that  $\mathbf{v}$  is the unique vector such that, for all  $\mathbf{z} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ :

$$G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{z}) = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{z}). \quad (1.47)$$

The uniqueness of  $\mathbf{v}$  is immediate by the positive definiteness of  $G_{[\mathbf{W}_0, \sigma_0^2]}$ . Given that  $(\mathbf{G}\mathbf{y})_j = (\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))_j$  for all  $j = 1, \dots, d$ , we have, for all  $j = 1, \dots, d$ :

$$G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{e}_j) = \sum_{i=1}^d \mathbf{y}_i G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{e}_i, \mathbf{e}_j) = \sum_{i=1}^d \mathbf{y}_i \mathbf{G}_{ij} = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{e}_j).$$

The claim is therefore proved. By looking at Definition 14, (1.47) reads  $G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}) = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])$  and hence  $\mathbf{v} = G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}(s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))$ , giving (1.45). The results about the integral of  $\text{IF}(\mathbf{X})$  translates to here as well. This completes the proof.  $\square$

# Chapter 2

## Draft

The first chapter produced three things: the quotient Riemannian manifold  $(M, g)$  together with its exponential and logarithmic maps; the verification that the probabilistic principal component analysis model is differentiable in quadratic mean with positive definite Fisher information operator, hence locally asymptotically normal; and, for the maximum likelihood estimator, consistency (Theorem 11) together with asymptotic linearity and asymptotic normality in the tangent space at the true parameter (Proposition 15).

That last result is a statement made under a single law and about a single estimator: it says nothing about what happens when the truth is perturbed. Section 2.1 computes the parallel transport of  $(M, g)$  and uses it to supply the missing statement, namely that the asymptotics do not degrade when the truth is moved at the scale  $n^{-\frac{1}{2}}$ , which is *local regularity*. Section 2.2 draws the inferential consequences, confidence sets with locally uniform coverage and the limit law of the fitted covariance matrix. Section 2.3 computes the Fisher information operator and its inverse in closed form, which is what makes any of the preceding usable in practice. Nothing from the first chapter is restated below; it is cited, and the complements it does not contain are proved where they are first needed.

### Notation

Throughout this chapter we abbreviate

$$\theta_0 := [\mathbf{W}_0, \sigma_0^2] \in M, \quad \hat{\theta}_n := [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \in M, \quad G := G_{\theta_0}, \quad (2.1)$$

and we keep  $d = pq + 1 - \frac{q(q-1)}{2}$  of (1.2) for the dimension of  $M$ , together with  $\Sigma_0$  of (1.38),  $\mathbf{S}_n$  of (1.40), its eigenvalues  $\lambda_1 \geq \dots \geq \lambda_p$  and eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_p$ , the estimators (1.41) and (1.42) of Tipping and Bishop (1999), and the fitted covariance matrix  $\hat{\Sigma}_n$  of (1.43), all as in Section 1.4. The letter  $\theta$ , which in Section 1.1 denoted a point of  $\Theta$ , was retired at the beginning of Section 1.4 when the coordinates on  $\Theta$  were renamed  $(\mathbf{W}, \sigma^2)$ ; from here on  $\theta$  always denotes a point of  $M$ . We write  $\mathbb{P}_\theta^n$  for the law of  $(\mathbf{X}_1, \dots, \mathbf{X}_n)$  when the true parameter is  $\theta$ , that is the  $n$ -fold product of  $p(\cdot; \theta) d\mathbf{x}$  with  $p$  as in (1.22), and  $\rightsquigarrow$  for convergence in distribution; unless another law is named,  $o_{\mathbb{P}}(1)$  and  $O_{\mathbb{P}}(1)$  refer to  $\mathbb{P}_{\theta_0}^n$ . Finally  $\text{Sym}(m)$  denotes the space of real symmetric  $m \times m$  matrices and  $\preceq$  the Loewner order on it.

The horizontal lift was used repeatedly in the first chapter but never named; we name it now. For  $(\mathbf{A}, s) \in \Theta$  let

$$\iota_{(\mathbf{A}, s)} : T_{[\mathbf{A}, s]}M \longrightarrow H_{(\mathbf{A}, s)} \quad (2.2)$$

be the inverse of the isomorphism  $d\pi_{(\mathbf{A}, s)}|_{H_{(\mathbf{A}, s)}}$ , so that  $\iota_{(\mathbf{A}, s)}(\mathbf{v})$  is the unique horizontal vector at  $(\mathbf{A}, s)$  projecting to  $\mathbf{v}$ ; by the very definition (1.5) of  $g$ , it is a linear isometry onto  $H_{(\mathbf{A}, s)}$ . We abbreviate

$$\iota := \iota_{(\mathbf{W}_0, \sigma_0^2)}, \quad \iota(\mathbf{h}) =: (\mathbf{U}_h, b_h) \quad \text{for } \mathbf{h} \in T_{\theta_0}M, \quad (2.3)$$

so that, by part 3 of Proposition 1 and by (1.4),

$$U_h^T W_0 = W_0^T U_h \quad \text{and} \quad |h|^2 = \|U_h\|_F^2 + b_h^2. \quad (2.4)$$

Finally, put

$$\rho := \min(\sigma_{\min}(W_0), \sigma_0^2) > 0, \quad B_\epsilon := \{u \in T_{\theta_0} M : |u| < \epsilon\} \quad \text{for } 0 < \epsilon \leq \rho, \quad (2.5)$$

so that  $B_\rho = \mathcal{E}_{\theta_0}$  is the domain of the exponential map supplied by Proposition 6, and put  $\mathcal{U} := \text{Exp}_{\theta_0}(B_\rho)$ . That proposition makes  $\text{Exp}_{\theta_0} : B_\rho \rightarrow \mathcal{U}$  a diffeomorphism onto an open subset of  $M$ , and (1.16) makes it *affine* in the horizontal lift:  $\text{Exp}_{\theta_0}(u) = [W_0 + U_u, \sigma_0^2 + b_u]$ , in the notation (2.3).

## 2.1 Parallel Transport and Local Regularity

Local regularity is the assertion that the conclusion of Proposition 15 survives a perturbation of the true parameter at the estimation scale  $n^{-\frac{1}{2}}$ . Stating it requires comparing the estimation error measured at  $\theta_0$  with the error measured at a nearby point  $\theta_{n,h}$ , and these two errors live in *different* tangent spaces. Something is therefore needed to identify  $T_{\theta_{n,h}} M$  with  $T_{\theta_0} M$ , and the canonical choice is the parallel transport of  $(M, g)$  along the minimizing geodesic joining the two points, which we compute first.

### The minimizing geodesics of $M$ and their parallel transport

The metric  $g$  was constructed in Section 1.1 so that  $\pi$  is a Riemannian submersion from the *flat* manifold  $\Theta \subseteq \mathbb{R}^{pq+1}$ , and Proposition 6 exploited this to make the exponential map affine in the horizontal lift. The same structure turns the parallel transport into an explicit linear ordinary differential equation.

Parallel transport is an operation attached to a *curve*, whereas the object we need — the map written  $\Pi_{x_1}^{x_2}$  in Section II of Sun et al. (2026) — is attached to a *pair of points*. The two are reconciled in the way that paper prescribes: between two points close enough to each other, transport is taken along the unique distance minimizing geodesic joining them. That geodesic has already been supplied. Writing  $d_M$  for the Riemannian distance of  $(M, g)$ , the infimum of the  $g$ -lengths of the piecewise smooth curves joining two points (Lee, 2018, Chapter 2), Proposition 3 classifies the geodesics of  $M$ , Proposition 2 lifts them horizontally, Proposition 6 exhibits  $U_{[A,s]}$  as a geodesic ball, and part 6 of that proposition computes  $d_M$  inside the ball together with the geodesic realizing it. Proposition 19 below has therefore only to restate that geodesic in the notation used here, and then to transport along it, the transport itself being solved beforehand, in Proposition 18, for an arbitrary curve.

Two preliminaries are needed. The first makes the orthogonal splitting  $T_{(A,s)}\Theta = V_{(A,s)} \oplus H_{(A,s)}$  of Proposition 1 explicit, through a Lyapunov equation that will turn out to govern the transport as well; it also shows that the horizontal lift (2.2) depends smoothly on its base point, which is used repeatedly below. The second is the submersion identity, in the form we need.

**Lemma 16.** Let  $A \in \mathbb{R}^{p \times q}$  have rank  $q$  and put  $P_A := A^T A$ , a positive definite  $q \times q$  matrix.

1. For every  $K \in \mathbb{R}^{q \times q}$  the Lyapunov equation

$$P_A \Omega + \Omega P_A = K \quad (2.6)$$

has a unique solution  $\Omega =: \mathcal{L}_A(K)$ . The map  $\mathcal{L}_A$  is linear, the map  $(A, K) \mapsto \mathcal{L}_A(K)$  is smooth, and  $\mathcal{L}_A(K)$  is skew-symmetric whenever  $K$  is.

2. For every  $(V, c) \in T_{(A,s)}\Theta$  the orthogonal projection  $\mathcal{H}_{(A,s)}$  onto the horizontal space is

$$\mathcal{H}_{(A,s)}((V, c)) = (V - A \mathcal{L}_A(A^T V - V^T A), c), \quad (2.7)$$

the subtracted term being the orthogonal projection of  $(V, c)$  onto  $V_{(A,s)}$ .

3. Consequently  $\iota_{(A,s)}(\mathrm{d}\pi_{(A,s)}((V, c))) = \mathcal{H}_{(A,s)}((V, c))$ , and this expression is smooth in  $(A, s, V, c)$ .

*Proof.* **1.** Diagonalize  $P_A = O \operatorname{diag}(p_1, \dots, p_q) O^T$  with  $O \in O(q)$  and every  $p_i > 0$ . Conjugating (2.6) by  $O$  turns the linear operator  $\Omega \mapsto P_A \Omega + \Omega P_A$  into the entrywise multiplication by  $p_i + p_j > 0$ , which is invertible; existence, uniqueness and linearity follow. Smoothness holds because the operator depends polynomially on  $A$  and the inverse of an invertible linear map depends smoothly on it. If  $K^T = -K$ , transposing (2.6) gives  $P_A(-\Omega^T) + (-\Omega^T)P_A = -K^T = K$ , so  $-\Omega^T$  solves the same equation and uniqueness forces  $\Omega^T = -\Omega$ .

**2.** By Proposition 1 the vertical space is  $\{(A\Omega, 0) : \Omega^T = -\Omega\}$ , so the orthogonal projection of  $(V, c)$  onto it is  $(A\Omega_*, 0)$ , where  $\Omega_*$  minimizes  $\|V - A\Omega\|_F^2$  over skew-symmetric  $\Omega$ ; the scalar component contributes the constant  $c^2$  and plays no role. Differentiating this strictly convex quadratic in a skew-symmetric direction  $K$  gives the stationarity condition  $\operatorname{tr}(K^T A^T (V - A\Omega_*)) = 0$  for every skew-symmetric  $K$ , which says exactly that the skew-symmetric part of  $A^T V - P_A \Omega_*$  vanishes:

$$A^T V - V^T A = P_A \Omega_* - \Omega_*^T P_A = P_A \Omega_* + \Omega_* P_A.$$

This is (2.6) with the skew-symmetric right hand side  $K = A^T V - V^T A$ , and part 1 identifies  $\Omega_* = \mathcal{L}_A(K)$  and confirms that it is skew-symmetric, as it must be; subtracting  $(A\Omega_*, 0)$  from  $(V, c)$  gives (2.7).

**3.** The vector  $\mathcal{H}_{(A,s)}((V, c))$  is horizontal and differs from  $(V, c)$  by a vertical vector, so it has the same image under  $\mathrm{d}\pi_{(A,s)}$ , whose kernel is  $V_{(A,s)}$  by (1.8). Smoothness is that of (2.7).  $\square$

**Lemma 17.** Let  $c : I \rightarrow \Theta$  be a smooth curve with horizontal velocity at every time, let  $\gamma := \pi \circ c$ , and let  $x$  be a smooth vector field along  $\gamma$ , with horizontal lift  $\tilde{x}(t) := \iota_{c(t)}(x(t))$ . Then the horizontal lift of the covariant derivative  $D_t x$  of  $(M, g)$  is the horizontal part of the ordinary derivative of  $\tilde{x}$ :

$$\iota_{c(t)}(D_t x(t)) = \mathcal{H}_{c(t)}(\dot{\tilde{x}}(t)). \quad (2.8)$$

In particular  $x$  is parallel along  $\gamma$  if and only if  $\dot{\tilde{x}}(t) \in V_{c(t)}$  for every  $t$ .

*Proof.* Write  $A(x) := D_t x$  and  $B(x) := \mathrm{d}\pi_{c(t)}(\mathcal{H}_{c(t)}(\dot{\tilde{x}}))$ ; both are  $\mathbb{R}$ -linear operators on smooth vector fields along  $\gamma$ . Both satisfy the Leibniz rule  $L(fx) = f'x + fL(x)$  for  $f \in C^\infty(I)$ : for  $A$  this is part of the definition of the covariant derivative along a curve, and for  $B$  it follows from  $\iota_c(fx) = f\tilde{x}$ , from  $\frac{d}{dt}(f\tilde{x}) = f'\tilde{x} + f\dot{\tilde{x}}$  and from  $\mathcal{H}_{c(t)}(\tilde{x}) = \tilde{x}$ . They agree on extendible fields. Indeed, let  $Y$  be a smooth vector field on an open subset of  $M$  and  $x = Y \circ \gamma$ ; then  $\tilde{x} = \tilde{Y} \circ c$  with  $\tilde{Y}$  the basic field lifting  $Y$ , so  $\dot{\tilde{x}}$  is the Euclidean covariant derivative  $\tilde{\nabla}_{\dot{c}} \tilde{Y}$  of  $\Theta$ , whose horizontal part is the lift of  $\nabla_{\mathrm{d}\pi(\dot{c})} Y = \nabla_{\dot{\gamma}} Y = D_t x$  by the fundamental identity for Riemannian submersions (O'Neill, 1967, Section 2; Valencia, 2021, Section 3), which applies because  $\dot{c}$  is horizontal. A covariant derivative along a curve is uniquely determined by  $\mathbb{R}$ -linearity, the Leibniz rule and its values on extendible fields (see Lee, 2018, Chapter 4, the uniqueness clause of the existence theorem for covariant derivatives along curves), so  $A = B$ , which is (2.8). The last sentence follows because  $\iota_{c(t)}$  is injective and  $\ker \mathcal{H}_{c(t)} = V_{c(t)}$ .  $\square$

Both lemmas are statements about a single time. Parallel transport is what they combine into, and it is solved here directly, in the form in which the question is posed. Given a smooth curve  $\gamma : I \rightarrow M$ , a time  $t_1 \in I$  and a vector  $w \in T_{\gamma(t_1)} M$ , find the vector field  $x$  along  $\gamma$  which is parallel and takes the value  $w$  at  $t_1$ :

$$D_t x(t) = 0 \quad \text{for every } t \in I, \quad x(t_1) = w. \quad (2.9)$$

The curve is arbitrary at this stage; the geodesic (2.24) is substituted only afterwards, in Proposition 19, where  $I = [0, 1]$  and the vector is prescribed at the far end  $t_1 = 1$ , the transport being read from  $[B, t]$  back to  $[A, s]$ . Keeping the curve arbitrary costs nothing and makes plain which features of the answer come from the submersion alone and which from the geodesic.

The method is the one the two lemmas prepare. Lift  $\gamma$  to a curve  $c$  in  $\Theta$  with  $\pi \circ c = \gamma$ , lift  $x$  to a vector field  $X$  along  $c$ , and observe that  $\mathrm{d}\pi$  carries the ordinary derivative of  $X$  to the covariant derivative of  $x$ , so that

$d\pi_{c(t)}(\dot{\mathbf{X}}(t)) = D_t \mathbf{x}(t) = 0$  is what (2.9) becomes upstairs. Now  $\Theta$  is flat, and one might hope that this forces  $\dot{\mathbf{X}}(t) = \mathbf{0}$ , a constant lift, hence a constant field downstairs. It does not: the kernel of  $d\pi_{c(t)}$  is the vertical space, not the origin, and all that is forced is that  $\dot{\mathbf{X}}(t)$  be vertical. That residue is exactly the failure of parallel transport in  $M$  to be the identity, and computing it is the content of the next proposition: the residue is  $\dot{\mathbf{X}} = \dot{P}\mathbf{X}$ , where  $P(t)$  denotes the orthogonal projection onto the horizontal space at  $c(t)$ . The lift is dragged by the rotation of the horizontal space, and by nothing else.

**Proposition 18** (The transport equation along a curve). Let  $\gamma : I \rightarrow M$  be a smooth curve and let  $c : I \rightarrow \Theta$  be a horizontal lift of it, as supplied by Proposition 2: a smooth curve with

$$\pi \circ c = \gamma, \quad \dot{c}(t) \in H_{c(t)}, \quad c(t) =: (\mathbf{A}(t), s(t)) \in \Theta, \quad t \in I. \quad (2.10)$$

Write  $P(t) := \mathcal{H}_{c(t)}$  for the orthogonal projection of  $T_{c(t)}\Theta$  onto the horizontal space  $H_{c(t)}$ , an operator which by (2.7) is given on  $(\mathbf{Z}, \beta) \in \mathbb{R}^{p \times q} \times \mathbb{R}$  by

$$P(t)((\mathbf{Z}, \beta)) = (\mathbf{Z} - \mathbf{A}(t)\Omega_{\mathbf{Z}}(t), \beta), \quad \Omega_{\mathbf{Z}}(t) := \mathcal{L}_{\mathbf{A}(t)}(\mathbf{A}(t)^T \mathbf{Z} - \mathbf{Z}^T \mathbf{A}(t)); \quad (2.11)$$

the letter  $P$  is reserved for this operator throughout, and is not to be confused with the  $q \times q$  matrix  $\mathbf{P}_{\mathbf{A}} = \mathbf{A}^T \mathbf{A}$  of Lemma 16. Let  $\mathbf{x}$  be a smooth vector field along  $\gamma$  and let

$$(\mathbf{X}(t), \beta(t)) := \iota_{c(t)}(\mathbf{x}(t)) \in H_{c(t)} \quad (2.12)$$

be its horizontal lift, in the notation (2.2). Then the following hold.

1. The derivative of the lift projects to the covariant derivative of the field:

$$d\pi_{c(t)}\left(\frac{d}{dt}(\mathbf{X}(t), \beta(t))\right) = D_t \mathbf{x}(t), \quad t \in I. \quad (2.13)$$

Consequently  $\mathbf{x}$  is parallel along  $\gamma$  if and only if  $\frac{d}{dt}(\mathbf{X}, \beta)$  is *vertical* at every time, which does not make it zero.

2.  $\mathbf{x}$  is parallel along  $\gamma$  if and only if its lift solves the linear ordinary differential equation

$$\dot{\mathbf{X}}(t) = \mathbf{A}(t)\Omega(t), \quad \Omega(t) := \mathcal{L}_{\mathbf{A}(t)}(\mathbf{X}(t)^T \dot{\mathbf{A}}(t) - \dot{\mathbf{A}}(t)^T \mathbf{X}(t)), \quad \dot{\beta}(t) = 0. \quad (2.14)$$

In particular the scalar component is transported trivially along any curve whatsoever, and  $\mathbf{X}$  is constant precisely when  $\Omega$  vanishes identically, that is when  $\mathbf{X}(t)^T \dot{\mathbf{A}}(t)$  is symmetric at every time.

3. Equation (2.14) is the equation

$$\frac{d}{dt}(\mathbf{X}(t), \beta(t)) = \dot{P}(t)((\mathbf{X}(t), \beta(t))), \quad (2.15)$$

in which  $\dot{P}(t)$  is the derivative of the operator valued map  $t \mapsto P(t)$ , given at a fixed  $(\mathbf{Z}, \beta) \in \mathbb{R}^{p \times q} \times \mathbb{R}$  by

$$\dot{P}(t)((\mathbf{Z}, \beta)) = -(\dot{\mathbf{A}}\Omega_{\mathbf{Z}} + \mathbf{A}\dot{\Omega}_{\mathbf{Z}}, 0), \quad (2.16)$$

$$\dot{\Omega}_{\mathbf{Z}} = \mathcal{L}_{\mathbf{A}}(\dot{\mathbf{A}}^T \mathbf{Z} - \mathbf{Z}^T \dot{\mathbf{A}} - \dot{\mathbf{P}}_{\mathbf{A}}\Omega_{\mathbf{Z}} - \Omega_{\mathbf{Z}}\dot{\mathbf{P}}_{\mathbf{A}}), \quad \dot{\mathbf{P}}_{\mathbf{A}} := \frac{d}{dt}(\mathbf{A}^T \mathbf{A}) = \dot{\mathbf{A}}^T \mathbf{A} + \mathbf{A}^T \dot{\mathbf{A}}, \quad (2.17)$$

the argument  $t$  being omitted from  $\mathbf{A}$ ,  $\dot{\mathbf{A}}$  and  $\Omega_{\mathbf{Z}}$ ; and on a vector  $(\mathbf{Z}, \beta)$  which is horizontal at  $c(t)$  the two displays collapse to

$$\dot{P}(t)((\mathbf{Z}, \beta)) = \left( \mathbf{A}(t) \mathcal{L}_{\mathbf{A}(t)} \left( \mathbf{Z}^T \dot{\mathbf{A}}(t) - \dot{\mathbf{A}}(t)^T \mathbf{Z} \right), 0 \right). \quad (2.18)$$

4. For every  $t_1 \in I$  and every  $\mathbf{w} \in T_{\gamma(t_1)}M$  the problem (2.9) has exactly one solution; it is defined on all of  $I$  and depends linearly on  $\mathbf{w}$ . A solution of (2.15) whose value at one time is horizontal is horizontal at every time, and its Euclidean norm is constant; hence  $|\mathbf{x}(t)| = |\mathbf{w}|$  for every  $t \in I$ .

*Proof. The curve is lifted.* Proposition 2 supplies  $c$ , and every  $\mathbf{A}(t)$  has rank  $q$  because  $c(t)$  lies in  $\Theta$ ; so Lemma 16 applies at every time and, by part 1 of it,  $t \mapsto \mathcal{L}_{\mathbf{A}(t)}$  is smooth. The lift (2.12) is smooth in  $t$  as well, the horizontal lift  $\iota$  depending smoothly on its base point by part 3 of the same lemma.

**The derivative of the lift is pushed down.** The curve  $c$  has horizontal velocity at every time by (2.10), so Lemma 17 applies and gives  $\iota_{c(t)}(D_t \mathbf{x}) = \mathcal{H}_{c(t)}\left(\frac{d}{dt}(\mathbf{X}, \beta)\right)$ . Apply  $d\pi_{c(t)}$  to both sides. On the left it undoes  $\iota_{c(t)}$ , which is by definition (2.2) the inverse of  $d\pi_{c(t)}|_{H_{c(t)}}$ , and returns  $D_t \mathbf{x}$ ; on the right it does not see  $\mathcal{H}_{c(t)}$  at all, a vector and its horizontal part differing by an element of  $V_{c(t)} = \ker d\pi_{c(t)}$ . This is (2.13). In particular  $D_t \mathbf{x} = 0$  if and only if  $\frac{d}{dt}(\mathbf{X}, \beta) \in \ker d\pi_{c(t)} = V_{c(t)}$ , by (1.8), which is strictly weaker than  $\frac{d}{dt}(\mathbf{X}, \beta) = 0$ ; the two agree only when the vertical vector computed in the next step happens to vanish.

**The verticality condition is solved.** By part 2 of Proposition 1 the vertical space at  $c(t) = (\mathbf{A}(t), s(t))$  consists of the pairs  $(\mathbf{A}(t)\Omega, 0)$  with  $\Omega$  skew-symmetric, and  $\Omega$  is determined by such a pair because  $\mathbf{A}(t)$  has rank  $q$ . The conclusion of the previous step therefore reads

$$\dot{\mathbf{X}}(t) = \mathbf{A}(t)\Omega(t) \quad \text{with } \Omega(t)^T = -\Omega(t), \quad \dot{\beta}(t) = 0, \quad (2.19)$$

and it remains to identify  $\Omega(t)$ . That is done by the one piece of information not yet used: the lift (2.12) is horizontal at *every* time, so that, by part 3 of Proposition 1, the matrix  $\mathbf{S}(t) := \mathbf{A}(t)^T \mathbf{X}(t) - \mathbf{X}(t)^T \mathbf{A}(t)$  vanishes identically, and so does its derivative. Substituting (2.19) and  $\Omega^T = -\Omega$  into that derivative, and omitting the argument  $t$ ,

$$0 = \dot{\mathbf{S}} = \dot{\mathbf{A}}^T \mathbf{X} + \mathbf{A}^T \mathbf{A} \Omega - \Omega^T \mathbf{A}^T \mathbf{A} - \mathbf{X}^T \dot{\mathbf{A}} = (\mathbf{P}_A \Omega + \Omega \mathbf{P}_A) - (\mathbf{X}^T \dot{\mathbf{A}} - \dot{\mathbf{A}}^T \mathbf{X}). \quad (2.20)$$

This is the Lyapunov equation (2.6) with the skew-symmetric right hand side  $\mathbf{K} = \mathbf{X}^T \dot{\mathbf{A}} - \dot{\mathbf{A}}^T \mathbf{X}$ , so part 1 of Lemma 16 identifies  $\Omega = \mathcal{L}_A(\mathbf{K})$  and confirms that it is skew-symmetric, as (2.19) requires. A parallel  $\mathbf{x}$  has therefore a lift solving (2.14). Conversely, if the lift solves (2.14), then  $\frac{d}{dt}(\mathbf{X}, \beta) = (\mathbf{A}\Omega, 0)$  is vertical,  $\Omega$  being skew-symmetric by the same lemma, and the second step gives  $D_t \mathbf{x} = 0$ . This is part 2, and part 1 was the second step.

**The equation is the derivative of the projection.** Fix  $(\mathbf{Z}, \beta)$ , not depending on  $t$ , and read (2.11): the vertical part subtracted there is  $\mathbf{A}(t)\Omega_Z(t)$ , with  $\Omega_Z$  the solution of

$$\mathbf{P}_A \Omega_Z + \Omega_Z \mathbf{P}_A = \mathbf{A}^T \mathbf{Z} - \mathbf{Z}^T \mathbf{A}. \quad (2.21)$$

Both  $\mathbf{A}$  and  $\mathcal{L}_A$  are smooth in  $t$  by the first step, so  $t \mapsto P(t)$  is a smooth curve of linear operators; differentiating (2.11) gives (2.16), and differentiating (2.21) and solving the resulting Lyapunov equation for  $\dot{\Omega}_Z$  with part 1 of Lemma 16 gives (2.17). Suppose now that  $(\mathbf{Z}, \beta)$  is horizontal at  $c(t)$ , that is that  $\mathbf{A}(t)^T \mathbf{Z} = \mathbf{Z}^T \mathbf{A}(t)$ . Then the right hand side of (2.21) vanishes at time  $t$ , so  $\Omega_Z(t) = \mathcal{L}_{\mathbf{A}(t)}(0) = 0$ ; the first term of (2.16) and the last two terms of (2.17) drop out, and what survives is

$$\dot{P}(t)((\mathbf{Z}, \beta)) = - \left( \mathbf{A} \mathcal{L}_A \left( \dot{\mathbf{A}}^T \mathbf{Z} - \mathbf{Z}^T \dot{\mathbf{A}} \right), 0 \right) = \left( \mathbf{A} \mathcal{L}_A \left( \mathbf{Z}^T \dot{\mathbf{A}} - \dot{\mathbf{A}}^T \mathbf{Z} \right), 0 \right),$$

which is (2.18). The lift (2.12) is horizontal at  $c(t)$  for every  $t$ , so  $(\mathbf{Z}, \beta) = (\mathbf{X}(t), \beta(t))$  is an admissible choice at each time, and it turns the right hand side of (2.18) into the right hand side of (2.14). The two equations (2.15) and (2.14) therefore say the same thing about a horizontal lift, which is part 3.

**Existence, horizontality and the norm.** Equation (2.15) is linear in  $(\mathbf{X}, \beta)$  with coefficients smooth in  $t$  on  $I$ , so for data prescribed at any  $t_1 \in I$  it has a unique solution, defined on all of  $I$  and linear in those data (Lee, 2018, Theorem 4.31). The two remaining properties are formal consequences of  $P(t)$  being an orthogonal projection and use nothing about  $\gamma$ . Differentiating  $P^2 = P$  gives  $\dot{P}P + P\dot{P} = \dot{P}$ ; multiplying that identity by  $P$  on both sides, and using  $P^2 = P$  once more, gives

$$P(t) \dot{P}(t) P(t) = 0. \quad (2.22)$$

Let  $\zeta$  solve (2.15) with  $\zeta(t_1) \in H_{c(t_1)}$ , that is with  $P(t_1)\zeta(t_1) = \zeta(t_1)$ , and put  $\mathbf{y} := \zeta - P\zeta$ , which vanishes exactly where  $\zeta$  is horizontal. Then, by (2.15), and then by  $\zeta = P\zeta + \mathbf{y}$  and (2.22),

$$\dot{\mathbf{y}} = \dot{\zeta} - \dot{P}\zeta - P\dot{\zeta} = \dot{P}\zeta - \dot{P}\zeta - P\dot{P}\zeta = -P\dot{P}(P\zeta + \mathbf{y}) = -P\dot{P}\mathbf{y},$$

a linear equation with  $\mathbf{y}(t_1) = 0$ , whence  $\mathbf{y} \equiv 0$ : the solution stays horizontal. For the norm, write  $g^\Theta$  for the Euclidean metric (1.4), which does not depend on the base point and for which each  $P(t)$  is self-adjoint, being an orthogonal projection. Using  $\zeta = P\zeta$  and (2.22) again,

$$\frac{d}{dt} g^\Theta(\zeta, \zeta) = 2 g^\Theta(\zeta, \dot{P}\zeta) = 2 g^\Theta(P\zeta, \dot{P}P\zeta) = 2 g^\Theta(\zeta, P\dot{P}P\zeta) = 0.$$

Finally, let  $\mathbf{w} \in T_{\gamma(t_1)}M$  and let  $\zeta$  be the solution of (2.15) with  $\zeta(t_1) = \iota_{c(t_1)}(\mathbf{w})$ , which is horizontal. By what precedes  $\zeta(t)$  is horizontal at  $c(t)$  for every  $t$ , hence is the horizontal lift of  $\mathbf{x}(t) := d\pi_{c(t)}(\zeta(t))$ ; parts 2 and 3 make  $\mathbf{x}$  parallel along  $\gamma$ , and  $\mathbf{x}(t_1) = \mathbf{w}$ . Any solution of (2.9) has, by parts 2 and 3, a lift solving (2.15) with the same value at  $t_1$ , so it coincides with  $\mathbf{x}$ ; and the linear dependence on  $\mathbf{w}$  is that of (2.15), together with the linearity of  $\iota_{c(t_1)}$  and of  $d\pi_{c(t)}$ . Since each  $\iota_{c(t)}$  is a linear isometry,  $|\mathbf{x}(t)|$  is the  $g^\Theta$  norm of  $\zeta(t)$ , just shown to be constant.  $\square$

Two comments on the answer, before it is applied. The first is that the hope raised above, that the flatness of  $\Theta$  should make the lifted field constant, is realized in exactly one circumstance: by part 2,  $\dot{\mathbf{X}} \equiv 0$  if and only if  $\mathbf{X}(t)^T \dot{\mathbf{A}}(t)$  is symmetric at every time. This is automatic when  $q = 1$ , all  $1 \times 1$  skew-symmetric matrices being zero, and it holds for the velocity field of a geodesic, for which  $\mathbf{X} = \dot{\mathbf{A}}$ ; both cases are taken up again after Proposition 19. Otherwise the lifted field turns, and (2.14) measures by how much. The second is that (2.15) is the whole answer in one line, and an intelligible one: a parallel field is the field whose lift is dragged by the moving horizontal space  $H_{c(t)}$  and by nothing else,  $P$  being the record of that motion. Only the tilting of  $H_{c(t)}$  inside the fixed ambient space  $\mathbb{R}^{p \times q} \times \mathbb{R}$  enters, which is why the scalar direction, along which nothing tilts, is transported trivially.

**Proposition 19** (Parallel transport along the minimizing geodesic). Let  $[\mathbf{A}, s] \in M$  and let  $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$ . Put

$$\mathbf{Q} := \mathbf{Q}(\mathbf{A}, \mathbf{B}), \quad \mathbf{V} := \mathbf{B}\mathbf{Q} - \mathbf{A}, \quad c := t - s, \quad \mathbf{A}_\tau := \mathbf{A} + \tau\mathbf{V}, \quad (2.23)$$

with  $\mathbf{Q}(\cdot, \cdot)$  the orthogonal matrix (1.18), and let

$$\gamma(\tau) := [\mathbf{A}_\tau, s + \tau c] \in M, \quad \tilde{\gamma}(\tau) := (\mathbf{A}_\tau, s + \tau c) \in \Theta, \quad \tau \in [0, 1]. \quad (2.24)$$

Write  $\mathbf{v} := \text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t]) \in \mathcal{E}_{[\mathbf{A}, s]}$ . Then  $\gamma(\tau) = \text{Exp}_{[\mathbf{A}, s]}(\tau\mathbf{v})$  is the geodesic of Proposition 3 from  $[\mathbf{A}, s]$  to  $[\mathbf{B}, t]$ , and  $\tilde{\gamma}$  is its horizontal lift at the representative  $(\mathbf{A}, s)$ , with constant velocity  $\iota_{(\mathbf{A}, s)}(\mathbf{v}) = (\mathbf{V}, c) \in H_{\tilde{\gamma}(\tau)}$ ; moreover  $\text{rank}(\mathbf{A}_\tau) = q$  for every  $\tau \in [0, 1]$ , so that Lemma 16 applies to  $\mathbf{A}_\tau$  and we may abbreviate  $\mathcal{L}_\tau := \mathcal{L}_{\mathbf{A}_\tau}$ .

For  $\tau_0, \tau_1 \in [0, 1]$  write

$$\Pi(\gamma)_{\tau_0}^{\tau_1} : T_{\gamma(\tau_0)}M \longrightarrow T_{\gamma(\tau_1)}M \quad (2.25)$$

for the parallel transport of  $(M, g)$  along  $\gamma$  from time  $\tau_0$  to time  $\tau_1$ . It exists, it is a linear isometry, its inverse is  $\Pi(\gamma)_{\tau_1}^{\tau_0}$ , and, read through the horizontal lifts  $\iota_{\tilde{\gamma}(\tau)}$ , it is the flow of the linear ordinary differential equation

$$\dot{\mathbf{X}}(\tau) = \mathbf{A}_\tau \Omega(\tau), \quad \Omega(\tau) := \mathcal{L}_\tau (\mathbf{X}(\tau)^T \mathbf{V} - \mathbf{V}^T \mathbf{X}(\tau)), \quad \dot{\beta}(\tau) = 0 : \quad (2.26)$$

if  $(\mathbf{X}, \beta)$  is the solution of (2.26) on  $[0, 1]$  with  $(\mathbf{X}(\tau_0), \beta(\tau_0)) = (\mathbf{X}_0, \beta_0) \in H_{\tilde{\gamma}(\tau_0)}$ , then  $(\mathbf{X}(\tau_1), \beta(\tau_1)) \in H_{\tilde{\gamma}(\tau_1)}$  and

$$\Pi(\gamma)_{\tau_0}^{\tau_1} (d\pi_{\tilde{\gamma}(\tau_0)} ((\mathbf{X}_0, \beta_0))) = d\pi_{\tilde{\gamma}(\tau_1)} ((\mathbf{X}(\tau_1), \beta(\tau_1))) . \quad (2.27)$$

The curve  $\gamma$  is determined by its two endpoints alone — if  $[\mathbf{B}, t] \neq [\mathbf{A}, s]$ , it is, up to reparametrization, the *only* curve in  $M$  from  $[\mathbf{A}, s]$  to  $[\mathbf{B}, t]$  of length  $d_M([\mathbf{A}, s], [\mathbf{B}, t])$  — and parallel transport is unchanged by a reparametrization; the notation

$$\Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]} := \Pi(\gamma)_1^0 : T_{[\mathbf{B}, t]}M \longrightarrow T_{[\mathbf{A}, s]}M \quad (2.28)$$

is therefore unambiguous, and this is the map  $\Pi_{x_1}^{x_2}$  of Section II of [Sun et al. \(2026\)](#). Explicitly, let  $\mathbf{z} \in T_{[\mathbf{B}, t]}M$ , let

$$(\mathbf{Y}, \beta) := \iota_{(\mathbf{BQ}, t)}(\mathbf{z}), \quad (\mathbf{BQ}, t) = (\mathbf{A} + \mathbf{V}, s + c) = \tilde{\gamma}(1), \quad (2.29)$$

be the horizontal lift of  $\mathbf{z}$  at the aligned representative  $(\mathbf{BQ}, t)$  of  $[\mathbf{B}, t]$ , and let  $\mathbf{X}$  be the unique solution on  $[0, 1]$  of the terminal value problem

$$\dot{\mathbf{X}}(\tau) = \mathbf{A}_\tau \mathcal{L}_\tau (\mathbf{X}(\tau)^T \mathbf{V} - \mathbf{V}^T \mathbf{X}(\tau)), \quad \mathbf{X}(1) = \mathbf{Y}. \quad (2.30)$$

Then

$$\Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\mathbf{z}) = d\pi_{(\mathbf{A}, s)} ((\mathbf{X}(0), \beta)), \quad \left| \Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\mathbf{z}) \right| = |\mathbf{z}|, \quad (2.31)$$

and  $(\mathbf{A}, \mathbf{V}, \mathbf{Y}, \beta) \mapsto (\mathbf{X}(0), \beta)$  is smooth, and linear in  $(\mathbf{Y}, \beta)$ . In particular  $\Pi_{[\mathbf{A}, s]}^{[\mathbf{A}, s]} = \text{id}$ ; and  $\mathbf{X} \equiv \mathbf{V}$  solves (2.30) for  $\mathbf{Y} = \mathbf{V}$ , so that  $\Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\dot{\gamma}(1)) = \dot{\gamma}(0) = \mathbf{v}$ , as a geodesic requires.

Finally, and this is the form in which the transport will be used, the two base points and the transported vector may be varied together. Write

$$\mathcal{D} := \{([\mathbf{A}, s], [\mathbf{B}, t], \mathbf{z}) : [\mathbf{A}, s] \in M, [\mathbf{B}, t] \in U_{[\mathbf{A}, s]}, \mathbf{z} \in T_{[\mathbf{B}, t]}M\}, \quad (2.32)$$

which is an open subset of  $M \times TM$  once the pair  $([\mathbf{B}, t], \mathbf{z})$  is read as the point  $\mathbf{z}$  of the tangent bundle  $TM$ . Then the map

$$\mathcal{D} \longrightarrow TM, \quad ([\mathbf{A}, s], [\mathbf{B}, t], \mathbf{z}) \longmapsto \Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\mathbf{z}), \quad (2.33)$$

is *smooth*.

**Proof. The geodesic.** Part 4 of Proposition 6 gives  $\mathbf{v} = d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c))$  with  $(\mathbf{V}, c)$  as in (2.23), and part 1 of that proposition makes  $\mathbf{A}^T(\mathbf{A} + \mathbf{V}) = \mathbf{A}^T \mathbf{BQ}$  symmetric, so  $\mathbf{A}^T \mathbf{V}$  is symmetric and  $(\mathbf{V}, c)$  is horizontal at  $(\mathbf{A}, s)$  by part 3 of Proposition 1; hence  $\iota_{(\mathbf{A}, s)}(\mathbf{v}) = (\mathbf{V}, c)$ , and (1.16) applied to  $\tau \mathbf{v}$  gives  $\gamma(\tau) = \text{Exp}_{[\mathbf{A}, s]}(\tau \mathbf{v})$ , with  $\gamma(1) = [\mathbf{BQ}, t] = [\mathbf{B}, t]$  because  $\mathbf{Q} \in O(q)$ . That  $\gamma$  is a geodesic, and that  $\tilde{\gamma}$ , the straight line of constant horizontal velocity  $(\mathbf{V}, c)$ , is the horizontal lift supplied by Proposition 2, is Proposition 3 and its proof. Finally  $|\tau \mathbf{v}| \leq |\mathbf{v}|$  puts  $\tau \mathbf{v}$  in  $\mathcal{E}_{[\mathbf{A}, s]}$  for every  $\tau \in [0, 1]$ , and part 1 of Proposition 6, applied to  $\tau \mathbf{v}$ , makes  $\mathbf{A}^T \mathbf{A}_\tau$  positive definite, hence invertible; therefore  $\mathbf{A}_\tau$  has rank  $q$ .

The uniqueness of  $\gamma$  is the only further feature of the geodesic that the transport map requires, and it is read off from the same proposition. Geodesics have constant speed, so the  $g$ -length of  $\gamma$  is  $|\dot{\gamma}(0)| = |\mathbf{v}|$ , which is  $d_M([\mathbf{A}, s], [\mathbf{B}, t])$  by part 6 of Proposition 6; and  $U_{[\mathbf{A}, s]}$  is a geodesic ball centred at  $[\mathbf{A}, s]$  of which  $\gamma$  is the radial geodesic reaching  $[\mathbf{B}, t]$ , again by that proposition, so Proposition 6.11 of [Lee \(2018\)](#) applies and makes  $\gamma$  the unique curve of that length joining the two points, up to reparametrization.

**The transport equation.** We next record what is needed about (2.26). Let  $\tau_0 \in [0, 1]$  and  $(\mathbf{X}_0, \beta_0) \in \mathbb{R}^{p \times q} \times \mathbb{R}$ . The equation has a unique solution on all of  $[0, 1]$  subject to  $(\mathbf{X}(\tau_0), \beta(\tau_0)) = (\mathbf{X}_0, \beta_0)$ ; that solution is linear in  $(\mathbf{X}_0, \beta_0)$  and smooth in  $(\tau, \tau_0, \mathbf{A}, \mathbf{V}, \mathbf{X}_0, \beta_0)$ ; and if moreover  $(\mathbf{X}_0, \beta_0) \in H_{\tilde{\gamma}(\tau_0)}$ , then it stays horizontal, that is  $(\mathbf{X}(\tau), \beta(\tau)) \in H_{\tilde{\gamma}(\tau)}$  for every  $\tau \in [0, 1]$ , and it preserves the Euclidean norm:  $\|\mathbf{X}(\tau)\|_F^2 + \beta(\tau)^2$  is constant.

Indeed, by part 1 of Lemma 16 the right hand side of (2.26) is linear in  $(\mathbf{X}, \beta)$ , with coefficients that are smooth in  $\tau$  and in  $(\mathbf{A}, \mathbf{V})$  on the open set where  $\mathbf{A}_\tau$  has rank  $q$ , a set which by the first step contains  $[0, 1]$ . Linear systems have solutions that exist and are unique on the whole interval, whatever the time  $\tau_0$  at which the data are prescribed, and that depend smoothly on the time and linearly on the data (Lee, 2018, Theorem 4.31); adjoining  $(\mathbf{A}, \mathbf{V})$  to the unknown as extra components with vanishing derivative turns the dependence on those parameters into a dependence on initial conditions, which is smooth as well (Lee, 2013, Theorem D.1 and Theorem D.6). By the same lemma each  $\Omega(\tau)$  is skew-symmetric, because  $\mathbf{X}^T \mathbf{V} - \mathbf{V}^T \mathbf{X}$  is.

Horizontality and the norm are already proved. Indeed  $\tilde{\gamma}$  is by the first step a horizontal lift of  $\gamma$ , and  $\dot{\mathbf{A}}_\tau = \mathbf{V}$ , so that (2.26) is verbatim the transport equation (2.14) of Proposition 18 written for the curve  $\gamma$ , and equivalently, by part 3 of that proposition, the equation  $\frac{d}{d\tau}(\mathbf{X}, \beta) = \dot{P}(\tau)((\mathbf{X}, \beta))$  driven by the orthogonal projection  $P(\tau) = \mathcal{H}_{\tilde{\gamma}(\tau)}$  onto  $H_{\tilde{\gamma}(\tau)}$ . Part 4 of that proposition asserts both properties.

**The transport map.** Let  $(\mathbf{X}_0, \beta_0) \in H_{\tilde{\gamma}(\tau_0)}$  and let  $(\mathbf{X}, \beta)$  be the solution of (2.26) it determines. By the second step the pair  $(\mathbf{X}(\tau), \beta(\tau))$  is horizontal at  $\tilde{\gamma}(\tau)$ , hence is the horizontal lift of

$$\mathbf{x}(\tau) := d\pi_{\tilde{\gamma}(\tau)}((\mathbf{X}(\tau), \beta(\tau))),$$

and part 2 of Proposition 18, applied to  $\gamma$  and its horizontal lift  $\tilde{\gamma}$  as in the second step, gives  $D_\tau \mathbf{x} = 0$ : the field  $\mathbf{x}$  is parallel along  $\gamma$ . Existence and uniqueness of the parallel field with a prescribed value at  $\tau_0$  (Lee, 2018, Theorem 4.32) identify  $\mathbf{x}$  with the parallel transport of  $\mathbf{x}(\tau_0)$ , which is (2.27). The map  $\Pi(\gamma)_{\tau_0}^{\tau_1}$  is linear because (2.26) is; it is an isometry by the conservation of the norm in the second step together with the fact that each  $\iota_{\tilde{\gamma}(\tau)}$  is one; and  $\Pi(\gamma)_{\tau_1}^{\tau_0}$  inverts it, by uniqueness.

Parallel transport is unchanged if  $\gamma$  is reparametrized. Indeed, let  $\phi : [a, b] \rightarrow [0, 1]$  be a diffeomorphism. The curve  $\tilde{\gamma} \circ \phi$  lies over  $\gamma \circ \phi$  and has velocity  $\phi(\varsigma)(\mathbf{V}, c)$ , horizontal at every time; and if  $\tilde{\mathbf{x}}$  is the horizontal lift of a field  $\mathbf{x}$  along  $\gamma$ , then  $\tilde{\mathbf{x}} \circ \phi$  is the horizontal lift of  $\mathbf{x} \circ \phi$  along  $\gamma \circ \phi$ , with derivative  $\dot{\phi} \cdot (\tilde{\mathbf{x}} \circ \phi)$ , which is vertical exactly when  $\tilde{\mathbf{x}} \circ \phi$  is. By the criterion closing Lemma 17, the field  $\mathbf{x} \circ \phi$  is parallel along  $\gamma \circ \phi$  if and only if  $\mathbf{x}$  is parallel along  $\gamma$ . Together with the uniqueness statement of the first step, this makes  $\Pi(\gamma)_1^0$  a function of the pair  $([\mathbf{A}, s], [\mathbf{B}, t])$  alone, so (2.28) is well posed.

For (2.31), apply the above with  $\tau_0 = 1$ ,  $\tau_1 = 0$  and  $(\mathbf{X}_0, \beta_0) = (\mathbf{Y}, \beta)$ , which is horizontal at  $\tilde{\gamma}(1)$  by (2.29): the second component of (2.26) stays equal to  $\beta$ , the first is (2.30), and the second step supplies existence, uniqueness, linearity in  $(\mathbf{Y}, \beta)$ , smoothness in  $(\mathbf{A}, \mathbf{V})$  and the preservation of the norm. If  $[\mathbf{B}, t] = [\mathbf{A}, s]$ , then  $\mathbf{V} = 0$  and  $c = 0$ , so (2.30) reads  $\dot{\mathbf{X}} = 0$  and  $\mathbf{X}(0) = \mathbf{Y}$ , whence  $\Pi_{[\mathbf{A}, s]}^{[\mathbf{A}, s]} = d\pi_{(\mathbf{A}, s)} \circ \iota_{(\mathbf{A}, s)} = \text{id}$ . Lastly  $\mathbf{V}^T \mathbf{V} - \mathbf{V}^T \mathbf{V} = 0$ , so the constant field  $\mathbf{X} \equiv \mathbf{V}$  solves (2.30) with  $\mathbf{Y} = \mathbf{V}$ ; since  $\iota_{\tilde{\gamma}(1)}(\dot{\gamma}(1)) = (\mathbf{V}, c)$ , this gives  $\Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\dot{\gamma}(1)) = d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \mathbf{v}$ .

**Joint smoothness.** Both the openness of  $\mathcal{D}$  and the smoothness of (2.33) are proved upstairs, on  $\Theta$ , and pushed down through  $\pi$ . Since  $\Theta$  is an open subset of  $\mathbb{R}^{pq+1}$  we identify  $T\Theta = \Theta \times (\mathbb{R}^{p \times q} \times \mathbb{R})$  and consider

$$\mathcal{N} := \{((\mathbf{A}, s), (\mathbf{B}, t), (\mathbf{Y}, \beta)) \in \Theta \times T\Theta : [\mathbf{B}, t] \in U_{[\mathbf{A}, s]}\}, \quad (2.34)$$

the preimage of  $\mathcal{D}$  under  $\pi \times d\pi$ . The latter is a surjective submersion:  $\pi$  is one by (1.3), and in local coordinates in which  $\pi$  is a projection so is  $d\pi$ . Consequently a subset of  $M \times TM$  is open, and a map out of it is smooth, as soon as its preimage under  $\pi \times d\pi$  is (Lee, 2013, Corollary 4.27 and Theorem 4.29).

For the openness, part 5 of Proposition 6 describes  $\mathcal{N}$  by the two conditions that  $\mathbf{A}^T \mathbf{B}$  be invertible and that the strict inequality (1.19) hold. The first is an open condition; on the open set it defines both sides of (1.19) are continuous, the right one because  $\sigma_{\min}$  is, and the left one because  $\mathbf{Q}(\mathbf{A}, \mathbf{B})$  is. That  $\mathbf{Q}(\mathbf{A}, \mathbf{B})$  is in fact smooth there, which is needed again below, follows from (1.18): the matrix  $\mathbf{B}^T \mathbf{A} \mathbf{A}^T \mathbf{B} = (\mathbf{A}^T \mathbf{B})^T (\mathbf{A}^T \mathbf{B})$  is positive

definite as soon as  $\mathbf{A}^T \mathbf{B}$  is invertible, and the inverse square root is a smooth function of a positive definite matrix, the map  $\mathbf{S} \mapsto \mathbf{S}^2$  of the open cone of positive definite matrices having at  $\mathbf{S}$  the differential  $\Omega \mapsto \mathbf{S}\Omega + \Omega\mathbf{S}$ , invertible by part 1 of Lemma 16 applied to  $\mathbf{S}^{1/2}$ , so that the inverse function theorem applies. Hence  $\mathcal{N}$  is open, and so is  $\mathcal{D}$ .

For the smoothness, let  $((\mathbf{A}, s), (\mathbf{B}, t), (\mathbf{Y}, \beta))$  range over  $\mathcal{N}$  and put  $\mathbf{z} := d\pi_{(\mathbf{B}, t)}((\mathbf{Y}, \beta))$ , so that  $([\mathbf{A}, s], [\mathbf{B}, t], \mathbf{z})$  ranges over  $\mathcal{D}$ . The right action of  $\mathbf{Q} := \mathbf{Q}(\mathbf{A}, \mathbf{B}) \in O(q)$  carries  $(\mathbf{B}, t)$  to the aligned representative  $(\mathbf{B}\mathbf{Q}, t)$  and is linear, so it carries  $(\mathbf{Y}, \beta)$  to  $(\mathbf{Y}\mathbf{Q}, \beta)$ ; since  $\pi$  is invariant under that action,  $d\pi_{(\mathbf{B}\mathbf{Q}, t)}((\mathbf{Y}\mathbf{Q}, \beta)) = \mathbf{z}$ , and part 3 of Lemma 16 identifies the horizontal lift (2.29) of  $\mathbf{z}$  as

$$\iota_{(\mathbf{B}\mathbf{Q}, t)}(\mathbf{z}) = \mathcal{H}_{(\mathbf{B}\mathbf{Q}, t)}((\mathbf{Y}\mathbf{Q}, \beta)),$$

a smooth function on  $\mathcal{N}$  by (2.7) and the smoothness of  $\mathbf{Q}(\mathbf{A}, \mathbf{B})$  just established. So are  $\mathbf{A}$  and  $\mathbf{V} = \mathbf{B}\mathbf{Q} - \mathbf{A}$ , and therefore, by the smoothness recorded in the second step, so is the value at  $\tau = 0$  of the solution of (2.30). Composing with the smooth map  $d\pi$  and reading (2.31) backwards,

$$((\mathbf{A}, s), (\mathbf{B}, t), (\mathbf{Y}, \beta)) \mapsto d\pi_{(\mathbf{A}, s)}((\mathbf{X}(0), \beta)) = \Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}(\mathbf{z})$$

is a smooth map from  $\mathcal{N}$  to  $TM$ . Its value depends on  $([\mathbf{A}, s], [\mathbf{B}, t], \mathbf{z})$  alone, because (2.28) is well posed; that is, it is constant on the fibres of  $\pi \times d\pi$ , and it therefore descends to a smooth map on  $\mathcal{D}$ , which is (2.33).  $\square$

Three features of (2.31) deserve to be isolated. The scalar component is transported trivially,  $\dot{\beta} = 0$ : the factor of  $\Theta$  carrying  $\sigma^2$  is a flat direction on which  $O(q)$  acts trivially, and it never mixes with the loading matrix. When  $q = 1$  the matrix  $\mathbf{X}(\tau)^T \mathbf{V} - \mathbf{V}^T \mathbf{X}(\tau)$  is skew-symmetric of size  $1 \times 1$ , hence zero, so (2.30) reads  $\dot{\mathbf{X}} = \mathbf{0}$  and  $\Pi_{[\mathbf{B}, t]}^{[\mathbf{A}, s]}$  is the identity read in the horizontal lifts; this is as it must be, since for  $q = 1$  the group  $O(1)$  is discrete,  $\pi$  is a local isometry and  $M$  is flat. For  $q \geq 2$  there is no closed form, and what stands in the way is the curvature of  $(M, g)$ : writing  $\mathcal{K}_\tau$  for the linear operator  $\mathbf{X} \mapsto \mathbf{A}_\tau \mathcal{L}_\tau(\mathbf{X}^T \mathbf{V} - \mathbf{V}^T \mathbf{X})$  of  $\mathbb{R}^{p \times q}$ , the terminal value problem (2.30) is equivalent to

$$\mathbf{X}(0) = \mathbf{Y} - \int_0^1 \mathcal{K}_\sigma(\mathbf{X}(\sigma)) d\sigma, \quad \|\mathcal{K}_\sigma\| \leq \frac{\|\mathbf{A}_\sigma\| \|\mathbf{V}\|_F}{\sigma_{\min}(\mathbf{A}_\sigma)^2}, \quad (2.35)$$

the operator norm being the one induced by  $\|\cdot\|_F$ , and the bound following from  $\|\mathbf{X}^T \mathbf{V} - \mathbf{V}^T \mathbf{X}\|_F \leq 2\|\mathbf{X}\|_F \|\mathbf{V}\|_F$  together with the observation that, in an orthonormal eigenbasis of  $\mathbf{A}_\sigma^T \mathbf{A}_\sigma$ , solving (2.6) divides the  $(i, j)$  entry by  $p_i + p_j \geq 2\sigma_{\min}(\mathbf{A}_\sigma)^2$ . So  $\mathbf{X}(0)$  differs from  $\mathbf{Y}$  by a quantity of the order of  $\|\mathbf{V}\|_F \leq d_M([\mathbf{A}, s], [\mathbf{B}, t])$  by (1.20): parallel transport departs from the naive identification of the two horizontal spaces by a term of the order of the distance travelled. Controlling that term is the entire business of the next subsection.

**Definition 20.** For  $\mathbf{u} \in B_\rho$  we write

$$\Gamma_{\mathbf{u}} := \Pi_{\text{Exp}_{\theta_0}(\mathbf{u})}^{\theta_0} : T_{\text{Exp}_{\theta_0}(\mathbf{u})} M \longrightarrow T_{\theta_0} M \quad (2.36)$$

for the map (2.28) of Proposition 19, applied with  $[\mathbf{A}, s] = \theta_0$  and  $[\mathbf{B}, t] = \text{Exp}_{\theta_0}(\mathbf{u})$ . By part 6 of Proposition 6 the curve along which the transport is taken is the radial geodesic  $\tau \mapsto \text{Exp}_{\theta_0}(\tau \mathbf{u})$  of the normal chart, and it is the unique minimizing one; the subscript  $\mathbf{u}$  therefore records the second point through that chart rather than directly, which is the form the estimates below need. This is the map written  $\Pi_{\psi(\theta_{n,h})}^{\psi(\theta)}$  in Definition III.5 of Sun et al. (2026), in the case  $\psi = \text{id}$ . Concretely, in the notation (2.3), formula (2.31) reads  $\Gamma_{\mathbf{u}}(\mathbf{z}) = d\pi_{(\mathbf{W}_0, \sigma_0^2)}((\mathbf{X}(0), \beta))$ , where  $(\mathbf{Y}, \beta) = \iota_{(\mathbf{W}_0 + \mathbf{U}_u, \sigma_0^2 + b_u)}(\mathbf{z})$  and  $\mathbf{X}$  solves (2.30) with  $\mathbf{A}_\tau = \mathbf{W}_0 + \tau \mathbf{U}_u$  and  $\mathbf{V} = \mathbf{U}_u$ . By Proposition 19,  $\Gamma_{\mathbf{u}}$  is a linear isometry and  $\Gamma_0 = \text{id}$ ; and, after both tangent spaces are identified with subspaces of  $\mathbb{R}^{p \times q} \times \mathbb{R}$  through the horizontal lifts  $\iota$  and  $\iota_{(\mathbf{W}_0 + \mathbf{U}_u, \sigma_0^2 + b_u)}$ , the map  $(\mathbf{u}, \mathbf{z}) \mapsto \Gamma_{\mathbf{u}}(\mathbf{z})$  is smooth on  $B_\rho \times (\mathbb{R}^{p \times q} \times \mathbb{R})$ , because  $\mathbf{u} \mapsto (\mathbf{U}_u, b_u)$  is linear and  $(\mathbf{A}, \mathbf{V}, \mathbf{Y}, \beta) \mapsto (\mathbf{X}(0), \beta)$  is smooth by that proposition; equivalently, it is (2.33) at  $[\mathbf{A}, s] = \theta_0$ , composed with the normal chart  $\mathbf{u} \mapsto \text{Exp}_{\theta_0}(\mathbf{u})$ .

## The transport comparison

Comparing errors measured at two different base points presupposes that the logarithm may be taken at a *moving* base point, and this is the standard fact we record first: it holds provided both points stay in a small enough neighbourhood of  $\theta_0$ .

**Lemma 21** (A uniformly normal neighbourhood). There exists  $r \in (0, \frac{\rho}{4})$  such that, writing

$$\mathcal{V} := \text{Exp}_{\theta_0}(B_r) \subseteq \mathcal{V}^\sharp := \text{Exp}_{\theta_0}(B_{2r}) \subseteq \mathcal{U}, \quad (2.37)$$

the following hold.

1. For all  $x, y \in \mathcal{V}^\sharp$  there is exactly one  $\mathbf{v} \in \mathcal{E}_x$  with  $\text{Exp}_x(\mathbf{v}) = y$ ; we denote it by  $\text{Exp}_x^{-1}(y)$ , and  $\text{Exp}_x^{-1}(x) = \mathbf{0}$ .
2. The map  $(\mathbf{u}, \mathbf{w}) \mapsto \text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u})}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w}))$  is smooth on  $B_{2r} \times B_{2r}$ , read through the horizontal lift  $\iota_{(\mathbf{w}_0 + \mathbf{U}_u, \sigma_0^2 + b_u)}$  as a map into  $\mathbb{R}^{p \times q} \times \mathbb{R}$ .

*Proof.* Let  $\mathcal{E} \subseteq TM$  be the set of pairs  $(x, \mathbf{v})$  with  $\mathbf{v} \in \mathcal{E}_x$ , the ball of Proposition 6. Its radius at  $x = [\mathbf{A}, s]$  is  $\min(\sigma_{\min}(\mathbf{A}), s)$ , which is a well defined function on  $M$  because the singular values of  $\mathbf{A}$  are unchanged by  $\mathbf{A} \mapsto \mathbf{A}\mathbf{Q}$  with  $\mathbf{Q} \in O(q)$ , and is continuous because its composition with the submersion  $\pi$  is. Hence  $\mathcal{E}$  is open in  $TM$  and contains the zero section, and  $\text{Exp} : \mathcal{E} \rightarrow M$  is smooth, being affine in the horizontal lift by (1.16) and part 3 of Lemma 16. Consider

$$E : \mathcal{E} \longrightarrow M \times M, \quad E(x, \mathbf{v}) := (x, \text{Exp}_x(\mathbf{v})).$$

Its differential at  $(\theta_0, \mathbf{0})$  is invertible: in the splitting  $T_{(\theta_0, \mathbf{0})}TM \cong T_{\theta_0}M \oplus T_{\theta_0}M$  it is  $(\mathbf{a}, \mathbf{b}) \mapsto (\mathbf{a}, \mathbf{a} + \mathbf{b})$ , because  $d(\text{Exp}_{\theta_0})_{\mathbf{0}} = \text{id}$ . The inverse function theorem therefore gives open sets  $\mathcal{N} \ni (\theta_0, \mathbf{0})$  in  $\mathcal{E}$  and  $\mathcal{O} \ni (\theta_0, \theta_0)$  in  $M \times M$  such that  $E : \mathcal{N} \rightarrow \mathcal{O}$  is a diffeomorphism. This is the usual construction of uniformly normal neighbourhoods (Lee, 2018, Theorem 5.14 and its proof); note that completeness of  $M$ , which fails here, is nowhere used, only the openness of the domain  $\mathcal{E}$  of  $\text{Exp}$ . Writing  $E^{-1}(x, y) = (x, \Xi(x, y))$ , the map  $\Xi$  is smooth on  $\mathcal{O}$  and satisfies  $\text{Exp}_x(\Xi(x, y)) = y$  and  $\Xi(x, x) = \mathbf{0}$ . Since  $\text{Exp}_{\theta_0}$  is continuous and  $(\theta_0, \theta_0) \in \mathcal{O}$ , we may choose  $r \in (0, \frac{\rho}{4})$  so small that  $\mathcal{V}^\sharp \times \mathcal{V}^\sharp \subseteq \mathcal{O}$ .

For part 1, existence is  $\text{Exp}_x(\Xi(x, y)) = y$  together with  $\Xi(x, y) \in \mathcal{E}_x$ , and uniqueness is the injectivity of  $\text{Exp}_x$  on  $\mathcal{E}_x$ , which is part 2 of Proposition 6; so  $\text{Exp}_x^{-1} = \Xi(x, \cdot)$  unambiguously, and  $\text{Exp}_x^{-1}(x) = \mathbf{0}$  because  $\Xi(x, x) = \mathbf{0}$ . Part 2 follows by composing  $\Xi$  with the smooth map  $(\mathbf{u}, \mathbf{w}) \mapsto (\text{Exp}_{\theta_0}(\mathbf{u}), \text{Exp}_{\theta_0}(\mathbf{w}))$  and with the horizontal lift, which is smooth in its base point by part 3 of Lemma 16.  $\square$

We fix such an  $r$  once and for all, together with  $\mathcal{V}$  and  $\mathcal{V}^\sharp$  as in (2.37). On  $\mathcal{V}^\sharp$  the map  $\text{Exp}_{\theta_0}^{-1}$  in the sense of part 1 of Lemma 21 agrees with the logarithmic map (1.17) of Proposition 6, both being inverses of the injective map  $\text{Exp}_{\theta_0}$  on  $B_\rho$ ; no ambiguity can therefore arise from the notation. Part 1 of the lemma says exactly that  $y \in U_x$  whenever  $x, y \in \mathcal{V}^\sharp$ , so part 6 of Proposition 6 and Proposition 19 apply to every such pair: any two points of  $\mathcal{V}^\sharp$  are joined by a unique minimizing geodesic, and the transport  $\Pi_y^x$  of (2.28) is defined throughout  $\mathcal{V}^\sharp \times \mathcal{V}^\sharp$ .

Definition III.5 of Sun et al. (2026) compares  $\text{Exp}_{\theta_n, h}^{-1}(\hat{\theta}_n)$ , transported back to  $T_{\theta_0}M$  by  $\Pi_{\theta_n, h}^{\theta_0}$ , with  $\text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n)$ . On a flat space the two would differ by exactly the vector joining the two base points; on  $M$  they differ by that vector plus a curvature correction, and the whole content of this subsection is that the correction is quadratically small. Both base points are read through the normal chart at  $\theta_0$ , which turns the comparison into a function of two chart vectors. Define, using part 1 of Lemma 21 and Definition 20,

$$F : B_{2r} \times B_{2r} \longrightarrow T_{\theta_0}M, \quad F(\mathbf{u}, \mathbf{w}) := \Gamma_{\mathbf{u}} \left( \text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u})}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w})) \right). \quad (2.38)$$

That  $F$  is smooth is a triviality, and it is worth saying once where the smoothness comes from, because none of it is particular to  $(M, g)$ : each of the two constituents of  $F$  is an instance of the smooth dependence of the

solutions of an ordinary differential equation on their time, their initial data and their parameters (Lee, 2013, Theorem D.1 and Theorem D.6).

The inner constituent is the two point logarithm  $(\mathbf{u}, \mathbf{w}) \mapsto \text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u})}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w}))$ . On any smooth Riemannian manifold the exponential map is the time one map of the geodesic flow, hence is smooth on its open domain in  $TM$ , and the inverse function theorem applied to  $(x, \mathbf{v}) \mapsto (x, \text{Exp}_x(\mathbf{v}))$  makes the two point logarithm smooth on a neighbourhood of the diagonal; this is the standard construction of uniformly normal neighbourhoods (Lee, 2018, Chapter 5 and Theorem 5.14; Lee, 2013, Theorem D.1 and Theorem D.6). Lemma 21 is that construction and nothing more, run here explicitly only because  $M$  is not complete, so that the domain of  $\text{Exp}$  has to be exhibited as an open subset of  $TM$  by hand.

The outer constituent is  $\Gamma_{\mathbf{u}}$ . On any manifold and for any connection, parallel transport along a smooth family of curves is smooth in the parameters of the family and linear in the transported vector: read in a local frame it is the flow of a linear ordinary differential equation whose coefficients are the connection coefficients evaluated along the curve, and these are smooth in every datum (Lee, 2018, Theorem 4.31 and Theorem 4.32; Lee, 2013, Theorem D.1 and Theorem D.6; see also Michor, 2008). The family here is the family  $\tau \mapsto \text{Exp}_{\theta_0}(\tau \mathbf{u})$  of radial geodesics, smooth in  $(\tau, \mathbf{u})$  by the previous paragraph, so  $(\mathbf{u}, \mathbf{z}) \mapsto \Gamma_{\mathbf{u}}(\mathbf{z})$  is smooth, which is the assertion recorded in Definition 20. Proposition 19 proves the same thing, in the stronger form (2.33) in which both base points vary, by writing that equation out in the horizontal lifts, where it becomes (2.26); but the smoothness owes nothing to the explicit form, and would hold verbatim for a metric admitting no closed form at all.

One point in the statement is not generic, and it is a matter of bookkeeping rather than of analysis. The maps  $\Gamma_{\mathbf{u}}$  have domains  $T_{\text{Exp}_{\theta_0}(\mathbf{u})}M$  that move with  $\mathbf{u}$ , so smoothness in  $(\mathbf{u}, \mathbf{z})$  asserts nothing until those spaces have been trivialized, which in general requires a choice of local frame. Here no choice is called for: all the horizontal spaces are subspaces of the single vector space  $\mathbb{R}^{p \times q} \times \mathbb{R}$ , the horizontal lifts  $\iota_{(A,s)}$  therefore trivialize globally, and part 3 of Lemma 16 is exactly the statement that they do so smoothly. That lemma is the only ingredient below which is specific to the model. Composing the two constituents in this trivialization,  $F$  is smooth on  $B_{2r} \times B_{2r}$ .

**Lemma 22** (Transport comparison). There is a constant  $C < \infty$ , depending only on  $\theta_0$  and  $r$ , such that

$$|F(\mathbf{u}, \mathbf{w}) - (\mathbf{w} - \mathbf{u})| \leq C |\mathbf{u}| |\mathbf{w} - \mathbf{u}| \quad \text{for all } \mathbf{u}, \mathbf{w} \in B_r. \quad (2.39)$$

*Proof.* By part 2 of Lemma 21, Definition 20 and the paragraphs preceding the statement,  $F$  is smooth on  $B_{2r} \times B_{2r}$ ; put

$$R(\mathbf{u}, \mathbf{w}) := F(\mathbf{u}, \mathbf{w}) - (\mathbf{w} - \mathbf{u}),$$

a smooth  $T_{\theta_0}M$ -valued function on  $B_{2r} \times B_{2r}$ . It vanishes on two slices:

- $R(\mathbf{u}, \mathbf{u}) = \mathbf{0}$  for every  $\mathbf{u} \in B_{2r}$ , because  $\text{Exp}_x^{-1}(x) = \mathbf{0}$  and  $\Gamma_{\mathbf{u}}$  is linear, so  $F(\mathbf{u}, \mathbf{u}) = \mathbf{0} = \mathbf{u} - \mathbf{u}$ ;
- $R(\mathbf{0}, \mathbf{w}) = \mathbf{0}$  for every  $\mathbf{w} \in B_{2r}$ , because  $\Gamma_{\mathbf{0}} = \text{id}$  and  $\text{Exp}_{\theta_0}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w})) = \mathbf{w}$ , so  $F(\mathbf{0}, \mathbf{w}) = \mathbf{w} = \mathbf{w} - \mathbf{0}$ .

Set

$$C := \sup_{(\mathbf{u}, \mathbf{w}) \in \overline{B_r} \times \overline{B_r}} \|\partial_1 \partial_2 R(\mathbf{u}, \mathbf{w})\| < \infty,$$

the norm being the one induced on bilinear maps  $T_{\theta_0}M \times T_{\theta_0}M \rightarrow T_{\theta_0}M$  by  $|\cdot|$ ; the supremum is finite because  $\partial_1 \partial_2 R$  is continuous and  $\overline{B_r} \times \overline{B_r}$  is a compact subset of the open set  $B_{2r} \times B_{2r}$  on which  $R$  is smooth.

Let  $\mathbf{u}, \mathbf{w} \in B_r$ . The ball  $B_r$  being convex, the segment from  $\mathbf{u}$  to  $\mathbf{w}$  lies in it, and the fundamental theorem of calculus in the second variable, together with  $R(\mathbf{u}, \mathbf{u}) = \mathbf{0}$ , gives

$$R(\mathbf{u}, \mathbf{w}) = \int_0^1 \partial_2 R(\mathbf{u}, \mathbf{u} + t(\mathbf{w} - \mathbf{u})) [\mathbf{w} - \mathbf{u}] \, dt. \quad (2.40)$$

Fix  $z \in B_r$ . The segment from  $\mathbf{0}$  to  $\mathbf{u}$  also lies in  $B_r$ , and  $\partial_2 R(\mathbf{0}, z) = \mathbf{0}$  because  $R(\mathbf{0}, \cdot)$  vanishes identically; the same argument in the first variable therefore gives

$$\partial_2 R(\mathbf{u}, z) = \int_0^1 \partial_1 \partial_2 R(\varsigma \mathbf{u}, z)[\mathbf{u}] d\varsigma, \quad \text{whence} \quad \|\partial_2 R(\mathbf{u}, z)\| \leq C|\mathbf{u}|. \quad (2.41)$$

Substituting (2.41) into (2.40) yields (2.39).  $\square$

**Remark 23.** Lemma 22 is the only place where the curvature of  $(M, g)$  enters the asymptotic theory, and (2.39) says exactly how much it costs. In the applications below  $\mathbf{u}$  is a local perturbation of size  $n^{-\frac{1}{2}}$  and  $\mathbf{w} - \mathbf{u}$  is an estimation error of size  $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ , so the right hand side is  $O_{\mathbb{P}}(n^{-1})$ ; after the multiplication by  $\sqrt{n}$  that all these statements carry, it is  $O_{\mathbb{P}}(n^{-\frac{1}{2}})$  and disappears. The two vanishing slices used in the proof are the two trivial cases of the comparison, namely no displacement of the base point and no error to compare, and it is their combination, not either alone, that produces the product  $|\mathbf{u}||\mathbf{w} - \mathbf{u}|$ .

## The estimation error in the chart

Proposition 15 speaks of  $\text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n)$ , which is defined only when  $\hat{\theta}_n$  lies in the normal neighbourhood. The next proposition is the bookkeeping that makes this a random vector defined everywhere, on an event whose probability tends to one, and identifies its horizontal lift.

**Proposition 24** (The error vector). Define

$$\mathcal{A}_n := \left\{ \hat{\theta}_n \in \mathcal{V} \right\}, \quad \hat{\mathbf{u}}_n := \begin{cases} \text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n) & \text{on } \mathcal{A}_n, \\ \mathbf{0} & \text{otherwise.} \end{cases} \quad (2.42)$$

Then, almost surely,  $\mathcal{A}_n$  holds for all  $n$  large enough; in particular  $\mathbb{P}_{\theta_0}^n(\mathcal{A}_n) \rightarrow 1$ , and the same holds with  $\mathcal{V}$  replaced by  $\text{Exp}_{\theta_0}(B_\epsilon)$  for any  $\epsilon \in (0, r]$ . On  $\mathcal{A}_n$  the horizontal lift of  $\hat{\mathbf{u}}_n$  is the *aligned error*

$$\hat{\mathbf{e}}_n := \iota(\hat{\mathbf{u}}_n) = \left( \hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2 \right), \quad \mathbf{Q}_n := \mathbf{Q}(\mathbf{W}_0, \hat{\mathbf{W}}_n) \in O(q), \quad (2.43)$$

with  $\mathbf{Q}(\cdot, \cdot)$  the orthogonal matrix (1.18). Consequently Proposition 15 may be read off the representatives: with  $\mathbf{Z} \sim N(\mathbf{0}, G^{-1})$  on  $T_{\theta_0} M$ ,

$$\sqrt{n} \left( \hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2 \right) \rightsquigarrow \iota(\mathbf{Z}) \quad \text{in } H(\mathbf{W}_0, \sigma_0^2), \quad (2.44)$$

the left hand side being  $\sqrt{n} \hat{\mathbf{e}}_n$  on  $\mathcal{A}_n$ .

*Proof.*  $\text{Exp}_{\theta_0}(B_\epsilon)$  is an open neighbourhood of  $\theta_0$  in  $M$  for every  $\epsilon \in (0, \rho]$ , by part 2 of Proposition 6; since  $\hat{\theta}_n \rightarrow \theta_0$  almost surely by Theorem 11, almost every realization eventually enters it. The identity (2.43) is (1.17) applied with  $[\mathbf{A}, s] = \theta_0$  and  $[\mathbf{B}, t] = \hat{\theta}_n$ , together with the fact that  $d\pi_{(\mathbf{W}_0, \sigma_0^2)}$  and  $\iota$  are mutually inverse on horizontal vectors. Finally, Proposition 15 gives  $\sqrt{n} \text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n) \rightsquigarrow \mathbf{Z}$ , the convention on  $\mathcal{A}_n^c$  being harmless because two random vectors that agree on events of probability tending to one have the same weak limits; applying the continuous mapping theorem to the linear isometry  $\iota$  and using (2.43) gives (2.44).  $\square$

**Remark 25.** Three comments on Proposition 15 orient everything that follows.

**1. It is an intrinsic statement.** The quantity that converges is the point  $\hat{\theta}_n$  of  $M$ , read through the logarithm at  $\theta_0$ . The orthogonal alignment  $\mathbf{Q}_n$  appearing in (2.44) is not an ad hoc Procrustes device introduced to make a limit exist: by Proposition 24 it is the logarithmic map of the first chapter, written in a representative.

**2. It concerns one law.** The expansion (1.45) and the ensuing weak convergence hold under the single law  $\mathbb{P}_{\theta_0}^n$ , and Remark 30 will show that the local regularity proved below does not follow from them, not even if they are assumed at every other point of  $M$ .

**3. No assumption is made on the multiplicities of the  $\sigma_j(\mathbf{W}_0)$ .** The result is valid when some, or all, of  $\sigma_1(\mathbf{W}_0), \dots, \sigma_q(\mathbf{W}_0)$  coincide, a case in which the individual eigenvectors of  $\mathbf{S}_n$  are *not* asymptotically normal. There is no contradiction: an individual eigenvector is not a function of the point  $\hat{\theta}_n \in M$ , whereas everything in Proposition 15 is. Section 2.3 makes this quantitative by exhibiting coordinates in which the limiting covariance is computed without ever dividing by a difference of singular values.

## Coordinates, and the local experiments

We keep the  $g_{\theta_0}$ -orthonormal basis  $e_1, \dots, e_d$  of  $T_{\theta_0}M$ , the identification  $T_{\theta_0}M \cong \mathbb{R}^d$  it provides, the matrix  $\mathbf{G}$  with entries  $\mathbf{G}_{jk} = G(e_j, e_k)$  and the coordinate vector  $\nabla l(\mathbf{x}; \theta_0) \in \mathbb{R}^d$  of the score, with entries  $s(\mathbf{x}; \theta_0)(e_j)$ , all fixed in Proposition 15 and its proof. Under this identification the Riemannian norm  $|\cdot|$  becomes the Euclidean norm, and bilinearity gives  $G(\mathbf{h}_1, \mathbf{h}_2) = \mathbf{h}_1^T \mathbf{G} \mathbf{h}_2$  and  $s(\mathbf{x}; \theta_0)(\mathbf{h}) = \nabla l(\mathbf{x}; \theta_0)^T \mathbf{h}$ ; the matrix  $\mathbf{G}$  is positive definite by Theorem 9. We abbreviate the normalized score

$$\Delta_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n \nabla l(\mathbf{X}_i; \theta_0) \in \mathbb{R}^d, \quad (2.45)$$

so that (1.46) reads

$$\sqrt{n} \hat{\mathbf{u}}_n = \mathbf{G}^{-1} \Delta_n + o_{\mathbb{P}}(1) \quad \text{under } \mathbb{P}_{\theta_0}^n. \quad (2.46)$$

By Theorem 9 the vectors  $\nabla l(\mathbf{X}_1; \theta_0), \nabla l(\mathbf{X}_2; \theta_0), \dots$  are independent, identically distributed, centred and square integrable with covariance matrix  $\mathbf{G}$ , so the multivariate central limit theorem gives

$$\Delta_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}) \quad \text{under } \mathbb{P}_{\theta_0}^n. \quad (2.47)$$

Finally, for  $\mathbf{h} \in T_{\theta_0}M$  and  $n$  large enough that  $\frac{\mathbf{h}}{\sqrt{n}} \in B_r$  we write

$$\theta_{n,\mathbf{h}} := \text{Exp}_{\theta_0} \left( \frac{\mathbf{h}}{\sqrt{n}} \right) \in \mathcal{V}, \quad \theta_{n,\mathbf{0}} = \theta_0, \quad (2.48)$$

for the *geodesic local alternative* in the direction  $\mathbf{h}$ . In this notation the local asymptotic normality expansion (1.36) of Proposition 10 reads

$$\Lambda_n(\mathbf{h}) := \sum_{i=1}^n \log \frac{p(\mathbf{X}_i; \theta_{n,\mathbf{h}})}{p(\mathbf{X}_i; \theta_0)} = \Delta_n^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h} + o_{\mathbb{P}}(1), \quad (2.49)$$

because the score is a linear functional, so that  $\sum_{i=1}^n s(\mathbf{X}_i; \theta_0) \left( \frac{\mathbf{h}}{\sqrt{n}} \right) = \Delta_n^T \mathbf{h}$ .

## Local regularity

We can now state the definition. Given an  $M$ -valued estimator sequence  $T_n$  we put, with  $\theta_{n,\mathbf{h}}$  the geodesic local alternative (2.48),

$$\hat{\mathbf{u}}_n[T_n] := \begin{cases} \text{Exp}_{\theta_0}^{-1}(T_n) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] := \begin{cases} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left( \text{Exp}_{\theta_{n,\mathbf{h}}}^{-1}(T_n) \right) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad (2.50)$$

both taking values in the single vector space  $T_{\theta_0}M$ ; the second is well defined as soon as  $\frac{\mathbf{h}}{\sqrt{n}} \in B_r$ , by part 1 of Lemma 21. For the maximum likelihood estimator the first of these is exactly the vector  $\hat{\mathbf{u}}_n$  of (2.42), and we keep that notation for it. Following Sun et al. (2026) we call an estimator sequence  $T_n$  *localizable at  $\theta_0$*  when  $\mathbb{P}_{\theta_0}^n(T_n \in \mathcal{V}) \rightarrow 1$ , which by Proposition 24 the maximum likelihood estimator is; only for such sequences do the quantities (2.50) carry any information in the limit.

**Definition 26.** A localizable estimator sequence  $T_n$  is *locally regular at  $\theta_0$  with limit law  $L$* , where  $L$  is a Borel probability measure on  $T_{\theta_0}M$ , if for every  $\mathbf{h} \in T_{\theta_0}M$

$$\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] \rightsquigarrow L \quad \text{under } \mathbb{P}_{\theta_n, \mathbf{h}}^n, \quad (2.51)$$

the limit  $L$  being the same for every  $\mathbf{h}$ . This is Definition III.5 of [Sun et al. \(2026\)](#) with  $\psi = \text{id}$  and  $\Psi = M$ : by (2.36) the map  $\Gamma_{\mathbf{h}/\sqrt{n}}$  of (2.50) is their  $\Pi_{\theta_n, \mathbf{h}}^{\theta_0}$ , so (2.51) is their display (16) verbatim.

**Lemma 27** (Contiguity). For every  $\mathbf{h} \in T_{\theta_0}M$  the sequences  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$  and  $\mathbb{P}_{\theta_0}^n$  are mutually contiguous. Consequently, if a sequence of random vectors is  $o_{\mathbb{P}}(1)$ , respectively  $O_{\mathbb{P}}(1)$ , under  $\mathbb{P}_{\theta_0}^n$ , then it is so under  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$  as well.

*Proof.* By (1.37) of Proposition 10 the log-likelihood ratio  $\Lambda_n(\mathbf{h})$  of (2.49) converges in distribution under  $\mathbb{P}_{\theta_0}^n$  to  $Z \sim N(-\frac{1}{2}G(\mathbf{h}, \mathbf{h}), G(\mathbf{h}, \mathbf{h}))$ , and  $\mathbb{E}[e^Z] = e^{-\frac{1}{2}G(\mathbf{h}, \mathbf{h}) + \frac{1}{2}G(\mathbf{h}, \mathbf{h})} = 1$ , so Le Cam's first lemma in the form of Example 6.5 of [van der Vaart \(1998\)](#) gives mutual contiguity. The transfer of  $o_{\mathbb{P}}(1)$  is immediate from the definition of contiguity. For  $O_{\mathbb{P}}(1)$ , suppose  $\mathbf{Z}_n$  were not bounded in probability under  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$ : there would be  $\varepsilon > 0$ , a subsequence  $n_k$  and constants  $M_k \rightarrow \infty$  with  $\mathbb{P}_{\theta_{n_k}, \mathbf{h}}^{n_k}(|\mathbf{Z}_{n_k}| > M_k) > \varepsilon$  for all  $k$ , whereas  $\mathbb{P}_{\theta_0}^{n_k}(|\mathbf{Z}_{n_k}| > M_k) \rightarrow 0$  because  $\mathbf{Z}_n = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_0}^n$  and  $M_k \rightarrow \infty$ ; contiguity makes these two incompatible.  $\square$

The next lemma is the bridge: on this manifold, regularity in the intrinsic sense of Definition 26 is the same thing as regularity of the ordinary Euclidean kind, read in the normal chart at  $\theta_0$ . It is what allows the classical Euclidean theory to be applied verbatim here.

**Lemma 28** (Chart form of regularity). Let  $T_n$  be a localizable estimator sequence with  $\sqrt{n} \hat{\mathbf{u}}_n[T_n] = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_0}^n$ . Then, for every  $\mathbf{h} \in T_{\theta_0}M$ ,

$$\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] = \sqrt{n} \hat{\mathbf{u}}_n[T_n] - \mathbf{h} + o_{\mathbb{P}}(1) \quad \text{under } \mathbb{P}_{\theta_n, \mathbf{h}}^n. \quad (2.52)$$

Consequently  $T_n$  is locally regular at  $\theta_0$  with limit law  $L$  if and only if

$$\sqrt{n} \hat{\mathbf{u}}_n[T_n] - \mathbf{h} \rightsquigarrow L \quad \text{under } \mathbb{P}_{\theta_n, \mathbf{h}}^n \text{ for every } \mathbf{h} \in T_{\theta_0}M. \quad (2.53)$$

Moreover the boundedness hypothesis is automatic: every locally regular sequence satisfies it.

*Proof.* Fix  $\mathbf{h}$  and abbreviate  $\hat{\mathbf{u}}_n := \hat{\mathbf{u}}_n[T_n]$ ,  $\mathbf{u}_n := \frac{\mathbf{h}}{\sqrt{n}}$  and  $\mathcal{B}_n := \{T_n \in \mathcal{V}\}$ . By hypothesis  $\mathbb{P}_{\theta_0}^n(\mathcal{B}_n) \rightarrow 1$ , hence  $\mathbb{P}_{\theta_n, \mathbf{h}}^n(\mathcal{B}_n) \rightarrow 1$  by Lemma 27, and by the same lemma  $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$  as well. On  $\mathcal{B}_n$ , and for  $n$  large enough that  $\mathbf{u}_n \in B_r$ , we have  $T_n = \text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n)$  with  $\hat{\mathbf{u}}_n \in B_r$ , so (2.38) and (2.50) give

$$\hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] = \Gamma_{\mathbf{u}_n} \left( \text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u}_n)}^{-1} (\text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n)) \right) = F(\mathbf{u}_n, \hat{\mathbf{u}}_n).$$

Lemma 22 therefore gives, on  $\mathcal{B}_n$ ,

$$|\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] - (\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h})| = \sqrt{n} |F(\mathbf{u}_n, \hat{\mathbf{u}}_n) - (\hat{\mathbf{u}}_n - \mathbf{u}_n)| \leq \sqrt{n} C \frac{|\mathbf{h}|}{\sqrt{n}} \left| \hat{\mathbf{u}}_n - \frac{\mathbf{h}}{\sqrt{n}} \right| = \frac{C|\mathbf{h}|}{\sqrt{n}} |\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}|. \quad (2.54)$$

The right hand side is a product of two factors of different natures. The first,  $\frac{C|\mathbf{h}|}{\sqrt{n}}$ , is *deterministic* and tends to zero: the constant  $C$  of Lemma 22 depends only on  $\theta_0$  and  $r$ , and  $\mathbf{h}$  is fixed throughout the proof, so neither varies with  $n$ . The second,  $|\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}|$ , is bounded in probability under  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$ , and this is where the hypothesis of the lemma and the contiguity are both spent: by the triangle inequality  $|\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}| \leq |\sqrt{n} \hat{\mathbf{u}}_n| + |\mathbf{h}|$ , in which  $|\mathbf{h}|$  is a constant and  $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_0}^n$  by hypothesis, hence  $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$  as well, by Lemma 27 as recorded at the beginning of this proof. The transfer is not a formality: (2.54) is applied under the alternative, whereas Lemma 22 is a deterministic inequality about  $(M, g)$  in which no law appears, so the only probabilistic input on its right hand side is the size of  $\hat{\mathbf{u}}_n$  under the law actually governing the data.

A sequence bounded in probability multiplied by a deterministic sequence tending to zero is  $o_{\mathbb{P}}(1)$ , of the order of that deterministic sequence. Explicitly, let  $\varepsilon, \eta > 0$ : boundedness in probability supplies  $K < \infty$  and  $N_1$  with  $\mathbb{P}_{\theta_{n,h}}^n(|\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h}| > K) < \eta$  for every  $n \geq N_1$ , and  $\frac{C|\mathbf{h}|K}{\sqrt{n}} < \varepsilon$  for every  $n$  beyond some  $N_2$  depending on  $\varepsilon$  alone, so that for  $n \geq \max(N_1, N_2)$  the right hand side of (2.54) exceeds  $\varepsilon$  with probability less than  $\eta$ . This is the assertion that it is  $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ , and a fortiori  $o_{\mathbb{P}}(1)$ , under  $\mathbb{P}_{\theta_{n,h}}^n$ . The surviving power  $n^{-\frac{1}{2}}$  is the one gained from Lemma 22: that estimate contributed the two factors  $|\mathbf{u}_n| = \frac{|\mathbf{h}|}{\sqrt{n}}$ , the size of the displacement of the base point, and  $|\hat{\mathbf{u}}_n - \mathbf{u}_n|$ , the estimation error, and the multiplication by  $\sqrt{n}$  carried by (2.54) consumed the second of them only; this is the accounting announced in Remark 23.

Finally, (2.54) was derived only on  $\mathcal{B}_n$ , and only for  $n$  large enough that  $\mathbf{u}_n \in B_r$ , whereas (2.52) is an unconditional statement; the passage from one to the other is  $\mathbb{P}_{\theta_{n,h}}^n(\mathcal{B}_n^c) \rightarrow 0$ , established at the beginning of the proof, which makes the event carrying no bound negligible in the limit. Together these prove (2.52). The equivalence of (2.51) and (2.53) is then Slutsky's lemma.

For the last assertion, let  $T_n$  be locally regular with limit law  $L$  and take  $\mathbf{h} = \mathbf{0}$  in (2.51): since  $\theta_{n,0} = \theta_0$  and  $\Gamma_0 = \text{id}$ , this reads  $\sqrt{n}\hat{\mathbf{u}}_n[T_n] \rightsquigarrow L$  under  $\mathbb{P}_{\theta_0}^n$ , and a weakly convergent sequence is bounded in probability by Prohorov's theorem.  $\square$

**Theorem 29** (Local regularity of the maximum likelihood estimator). *Let  $1 \leq q < p$  and let  $\theta_0 = [\mathbf{W}_0, \sigma_0^2] \in M$ . Then the maximum likelihood estimator  $\hat{\theta}_n$  of Section 1.4 is locally regular at  $\theta_0$  in the sense of Definition 26, with limit law*

$$L = N(\mathbf{0}, G^{-1}) \quad \text{on } T_{\theta_0}M, \quad (2.55)$$

which does not depend on  $\mathbf{h}$ . Explicitly, for every  $\mathbf{h} \in T_{\theta_0}M$ ,

$$\sqrt{n} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left( \text{Exp}_{\theta_{n,h}}^{-1} \left( \hat{\theta}_n \right) \right) \rightsquigarrow N(\mathbf{0}, G^{-1}) \quad \text{under } \mathbb{P}_{\theta_{n,h}}^n. \quad (2.56)$$

*Proof.* We work in the coordinates (2.45) and fix  $\mathbf{h} \in T_{\theta_0}M$ .

**Step 1: a single linear statistic drives everything.** The two expansions (2.46) and (2.49), both valid under  $\mathbb{P}_{\theta_0}^n$ , are linear in the *same* vector  $\Delta_n$  of (2.45), so no separate joint central limit theorem is needed: by (2.47),  $\Delta_n \rightsquigarrow N(\mathbf{0}, G)$ , and the continuous mapping theorem applied to the linear map  $\delta \mapsto (G^{-1}\delta, \delta^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T G \mathbf{h})$ , together with Slutsky's lemma, yields

$$(\sqrt{n}\hat{\mathbf{u}}_n, \Lambda_n(\mathbf{h})) \rightsquigarrow N \left( \begin{pmatrix} \mathbf{0} \\ -\frac{1}{2} \mathbf{h}^T G \mathbf{h} \end{pmatrix}, \begin{pmatrix} G^{-1} & \mathbf{h} \\ \mathbf{h}^T & \mathbf{h}^T G \mathbf{h} \end{pmatrix} \right) \quad \text{under } \mathbb{P}_{\theta_0}^n, \quad (2.57)$$

the off-diagonal block being  $\text{Cov}(G^{-1}\Delta, \Delta^T \mathbf{h}) = G^{-1}G\mathbf{h} = \mathbf{h}$  for  $\Delta \sim N(\mathbf{0}, G)$ .

**Step 2: Le Cam's third lemma.** The limit in (2.57) has exactly the form required by Example 6.7 of van der Vaart (1998), the Gaussian case of Le Cam's third lemma (Theorem 6.6 there): the last coordinate is the log-likelihood ratio and is  $N(-\frac{1}{2} \mathbf{h}^T G \mathbf{h}, \mathbf{h}^T G \mathbf{h})$ , with  $\mathbf{h}^T G \mathbf{h} = G(\mathbf{h}, \mathbf{h})$ . The cited result therefore gives

$$\sqrt{n} \hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{h}, G^{-1}) \quad \text{under } \mathbb{P}_{\theta_{n,h}}^n : \quad (2.58)$$

the limit under the contiguous alternative is the limit under the null shifted by the covariance  $\mathbf{h}$  between the statistic and the log-likelihood ratio. Equivalently  $\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, G^{-1})$  under  $\mathbb{P}_{\theta_{n,h}}^n$ , a law free of  $\mathbf{h}$ .

**Step 3: from the chart to the transported error.** The hypotheses of Lemma 28 hold for  $T_n = \hat{\theta}_n$ : it is localizable by Proposition 24, and  $\sqrt{n}\hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_0}^n$  by Proposition 15. The equivalence (2.53) of that lemma therefore turns (2.58) into (2.56), with limit law  $N(\mathbf{0}, G^{-1})$ , which in the  $g_{\theta_0}$ -orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  represents  $N(\mathbf{0}, G^{-1})$ .  $\square$

**Remark 30.** It is worth being explicit about what (2.56) excludes, since Proposition 15 alone excludes nothing of the kind. Fix a point  $\theta_* \in M$ , known to the statistician, and let  $\hat{\theta}_n^H$  be the Hodges-type modification of  $\hat{\theta}_n$

that replaces it by  $\theta_*$  whenever  $\hat{\theta}_n \in \text{Exp}_{\theta_*}(B)$ , with  $B$  the ball of radius  $n^{-\frac{1}{4}}$  in  $T_{\theta_*}M$ , and leaves it unchanged otherwise.

For any fixed true parameter  $\theta_0 \neq \theta_*$  the modification occurs on an event of probability tending to zero, by Theorem 11, so  $\hat{\theta}_n^H$  satisfies the exact analogue of Proposition 15 there. For  $\theta_0 = \theta_*$  it does better: since  $|\hat{\mathbf{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) \ll n^{-\frac{1}{4}}$  by Proposition 15, we have  $\hat{\theta}_n^H = \theta_*$  with probability tending to one, so that  $\sqrt{n}\hat{\mathbf{u}}_n \left[\hat{\theta}_n^H\right] \rightsquigarrow \delta_0$ , a limit strictly more concentrated than  $N(0, G^{-1})$ . This is the superefficiency phenomenon, and it shows that no bound whatsoever can be extracted from pointwise statements of the type of Proposition 15 alone.

And yet, at  $\theta_0 = \theta_*$  and  $\mathbf{h} \neq 0$ , the same computation gives  $\sqrt{n}\hat{\mathbf{u}}_n \left[\hat{\theta}_n^H\right] \rightsquigarrow \delta_0$  under  $\mathbb{P}_{\theta_{n,h}}^n$  as well — the estimator is pinned to  $\theta_*$  on an event of probability tending to one, since  $|\hat{\mathbf{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) \ll n^{-\frac{1}{4}}$  under the alternative too, by Lemma 27 — so by (2.52) its transported error converges to  $\delta_{-\mathbf{h}}$ , which *does* depend on  $\mathbf{h}$ . Hence  $\hat{\theta}_n^H$  is superefficient at  $\theta_*$  and is not locally regular there. Local regularity is precisely the statement that the asymptotics of Proposition 15 do not degrade as the truth is moved at the scale  $n^{-\frac{1}{2}}$ , and it is a strictly stronger property.

## 2.2 Consequences

### What local regularity adds

Proposition 15 and Theorem 29 end in the same limit law  $N(0, G^{-1})$ , and the second was deduced from the first in three short steps; it is therefore worth saying plainly, before anything is derived from it, what the second asserts that the first does not. The difference is not in the limit. It is in the *set of laws under which the limit is asserted*: Proposition 15 is a statement about the single sequence  $\mathbb{P}_{\theta_0}^n$ , whereas Theorem 29 is a statement about the whole shrinking family  $\left\{\mathbb{P}_{\theta_{n,h}}^n : \mathbf{h} \in T_{\theta_0}M\right\}$  of laws that no experiment of size  $n$  can tell apart from it. Everything in this section is a consequence of that one difference.

**Why the difference is not a technicality.** Lemma 27 says that  $\mathbb{P}_{\theta_{n,h}}^n$  and  $\mathbb{P}_{\theta_0}^n$  are mutually contiguous: for any fixed  $\mathbf{h}$ , no sequence of tests whose level under  $\mathbb{P}_{\theta_0}^n$  tends to zero can have power against  $\theta_{n,h}$  tending to one, and symmetrically with the two parameters exchanged. The qualification about the level is not decorative — the test that always rejects has power 1 — but it is exactly the qualification under which a testing problem is nontrivial. The statistician who has observed  $\mathbf{X}_1, \dots, \mathbf{X}_n$  is therefore never in a position to assert that the truth is  $\theta_0$  rather than some  $\theta_{n,h}$ ; the best that can ever be known is that the truth lies within  $O\left(n^{-\frac{1}{2}}\right)$  of the estimate, which is exactly the resolution at which the family  $\{\theta_{n,h}\}$  lives. An asymptotic property that holds at  $\theta_0$  but fails at  $\theta_{n,h}$  is thus *empirically unverifiable*: it is attached to a point that the data cannot single out. Proposition 15 is such a statement as it stands, and Remark 30 shows that it is not vacuously stable — the Hodges-type estimator constructed there satisfies the exact analogue of Proposition 15 at every point of  $M$  *other than*  $\theta_*$ , does strictly better than it at  $\theta_*$  itself, and is nevertheless worthless near  $\theta_*$ , because the point at which its behaviour is anomalous moves with the sample size in precisely the way the statistician cannot track. Local regularity is the repair: it upgrades a claim indexed by a point to a claim indexed by a neighbourhood of the resolution scale, and it is exactly the strength needed, no more. We now list what this buys.

**1. There is a lower bound at all.** This is the largest single gain, and it is a gain of *kind*, not of degree. From Proposition 15 alone *no* optimality statement about  $\hat{\theta}_n$  can be extracted, at  $\theta_0$  or anywhere else: Remark 30 exhibits a competitor whose limit law at  $\theta_*$  is  $\delta_0$ , infinitely more concentrated than  $N(0, G^{-1})$  for every reasonable measure of concentration, and the same device can be run at any finite set of points at once. The class of all estimator sequences carries no nontrivial asymptotic lower bound whatsoever. Once attention is restricted to the locally regular ones, the Hájek–Le Cam convolution theorem in the manifold form of Theorem III.1 of Sun et al. (2026) applies to the present model — its hypotheses are Theorem 9 and Definition 26, both verified here — and yields, for every locally regular  $T_n$  with limit law  $L$ ,

$$L = N(\mathbf{0}, G^{-1}) * \mu \quad \text{for some Borel probability measure } \mu \text{ on } T_{\theta_0}M. \quad (2.59)$$

Nothing below depends on the manifold form of the theorem being available. Lemma 28 turns Definition 26 into ordinary Euclidean regularity read in the normal chart, where (2.49) exhibits a locally asymptotically normal sequence of experiments with nonsingular information matrix  $\mathbf{G}$ ; the classical convolution theorem, Chapter 8 of van der Vaart (1998), applies to that sequence and returns (2.59) directly. Either way, every locally regular limit law is the law of  $\mathbf{Z} + \mathbf{N}$  with  $\mathbf{Z} \sim N(\mathbf{0}, G^{-1})$  independent of  $\mathbf{N} \sim \mu$ : relative to the limit of Theorem 29, an independent noise term has been *added*, never subtracted. Two consequences of (2.59) deserve to be named separately, because they are the statements a practitioner actually uses; a third item records, for contrast and for orientation only, the intrinsic Cramér–Rao inequality that the convolution theorem replaces here.

- *Efficiency.* Theorem 29 exhibits  $\hat{\theta}_n$  as the case  $\mu = \delta_0$  of (2.59), so the maximum likelihood estimator is asymptotically efficient in the class of locally regular estimators of  $\theta_0 \in M$ . This is Theorem V.1(i) of Sun et al. (2026), whose proof consists of exactly the verification carried out in Theorem 29.
- *A risk bound for every bowl-shaped loss.* Call  $\ell : T_{\theta_0}M \rightarrow [0, \infty)$  bowl-shaped if its sublevel sets  $\{\ell \leq c\}$  are closed, convex and symmetric about  $\mathbf{0}$ ; equivalently, if  $\ell$  is nonnegative, lower semicontinuous and quasiconvex with  $\ell(-\mathbf{v}) = \ell(\mathbf{v})$ . Anderson’s lemma applied to (2.59) gives  $\mathbb{E}[\ell(\mathbf{Z} + \mathbf{N})] \geq \mathbb{E}[\ell(\mathbf{Z})]$ , and the portmanteau theorem for nonnegative lower semicontinuous functions — this is where the sublevel sets are required to be *closed* — turns the convergence  $\sqrt{n}\hat{\mathbf{u}}_n[T_n] \rightsquigarrow L$  into

$$\liminf_n \mathbb{E}_{\mathbb{P}_{\theta_0}^n} [\ell(\sqrt{n}\hat{\mathbf{u}}_n[T_n])] \geq \mathbb{E}[\ell(\mathbf{Z})] \quad \text{for every locally regular } T_n, \quad (2.60)$$

and equally with  $\mathbb{P}_{\theta_0}^n$  replaced by any  $\mathbb{P}_{\theta_{n,h}}^n$ , the limit law being the same under all of them. See Chapter 8 of van der Vaart (1998) for the Euclidean argument, which Lemma 28 imports verbatim. Squared error is the case  $\ell = |\cdot|^2$ , but so is  $\ell = \mathbf{1}_{A^c}$  for a fixed closed convex symmetric  $A$ , that is the probability of missing a fixed region, which is what a confidence statement is about.

- *The asymptotic Cramér–Rao bound.* The intrinsic Cramér–Rao inequality on a parameter manifold, Lemma III.2 and Corollary III.1 of Sun et al. (2026), bounds below the covariance of the residual of an *unbiased* estimator in the sense of their Definition III.3, and carries at finite  $n$  two extra terms built from the curvature tensor of  $(M, g)$  which have no Euclidean counterpart; by the linearity of the curvature operator recorded there, those terms are  $O(n^{-1})$  while the main term  $G^{-1}$  is  $O(1)$ , so that the bound reduces asymptotically to the curvature-free  $G^{-1}$ . We use it nowhere. Converting it into a statement about  $\hat{\theta}_n$  would require an unbiasedness hypothesis, which nothing here gives any reason to expect of the maximum likelihood estimator at finite  $n$ , and the convergence  $\text{Cov}(\sqrt{n}\hat{\mathbf{u}}_n) \rightarrow \mathbf{G}^{-1}$ , which does *not* follow from Proposition 15: weak convergence to  $N(\mathbf{0}, \mathbf{G}^{-1})$  controls second moments only through the  $\liminf$  inequality (2.60), and upgrading that to convergence needs the uniform integrability of  $|\sqrt{n}\hat{\mathbf{u}}_n|^2$ , which is nowhere established here and is not innocuous,  $\hat{\mathbf{u}}_n$  being defined by truncation to  $\mathbf{0}$  off  $\mathcal{A}_n$  in (2.42). The comparison we do make is the one the convolution theorem supplies, namely (2.59) and (2.60), which are statements about laws and need no moment hypothesis at all.

None of this is available to someone holding Proposition 15 alone, for the simple reason that (2.59) is false without the word *regular*: the Hodges sequence has  $L = \delta_0$  at  $\theta_*$ , and  $\delta_0$  is not of the form  $N(\mathbf{0}, G^{-1}) * \mu$ , such a convolution having a nonsingular Gaussian factor and hence a density.

**2. The influence operator is pinned down by a pointwise identity.** Asymptotic linearity says that the estimator is, to first order, an average of a fixed function of the data; it places no constraint at all on *which* function. Local regularity places exactly one constraint, and it is a constraint that can be checked at a single point of the model, with no asymptotics at all. Proposition 35 below states it: an asymptotically linear sequence is locally regular *if and only if* its influence operator  $\text{IF}'$  satisfies

$$\mathbb{E}[s(\mathbf{X}; \theta_0)(z) \text{IF}'(\mathbf{X})] = z \quad \text{for every } z \in T_{\theta_0}M, \quad (2.61)$$

that is, if and only if the cross-covariance between the influence operator and the score is the identity operator of  $T_{\theta_0}M$ . This is the specialization to  $\psi = \text{id}$  of the calculus of influence functions of Lemma III.3 of [Sun et al. \(2026\)](#); the proposition below adds the converse, which is what makes (2.61) usable as a criterion. The gain is concrete. A statistician who has computed an influence operator for a competing estimator, by whatever means, may decide whether that estimator is locally regular by evaluating one expectation under the true law, and may then read off its asymptotic covariance and compare it with  $G^{-1}$  — and, by part 2 of that proposition, the comparison always goes the same way. Without local regularity the influence operator is merely *some* centred square integrable function subject to no normalization, and no comparison between two estimators is possible at all.

**3. The estimator becomes equivariant in the limit experiment.** The expansion (2.49) of Proposition 10 says that the local experiments  $(\mathbb{P}_{\theta_n, h}^n : h \in T_{\theta_0}M)$  converge, in Le Cam's sense, to the Gaussian shift experiment  $(N(h, G^{-1}) : h \in \mathbb{R}^d)$ , in which one observes a single Gaussian vector whose mean is the unknown local parameter and whose covariance is known ([van der Vaart, 1998](#), Chapter 9). Proposition 15 identifies the behaviour of  $\sqrt{n}\hat{u}_n$  at the *single* point  $h = 0$  of that limit experiment, which is no more informative than knowing the value of a function at one point of its domain. Equation (2.58) identifies it at every  $h$  at once, and the answer,  $N(h, G^{-1})$ , is exactly the law of the identity estimator  $y \mapsto y$  in the limit experiment: in the limit,  $\sqrt{n}\hat{u}_n$  is the natural shift-equivariant estimator of the local parameter. Local regularity is the assertion that the map sending  $h$  to the law of  $\sqrt{n}\hat{u}_n$  under  $\mathbb{P}_{\theta_n, h}^n$  converges to an equivariant family, and equivariance is the property that transfers the optimality theory of the limit experiment — where it is a finite dimensional Gaussian computation — back to the sequence. This is the structural reason behind item 1: (2.59) is the image, under the convergence of experiments, of the elementary fact that an equivariant estimator of a Gaussian mean cannot improve on the observation itself.

**4. The property survives smooth transformation.** Let  $\varphi$  be a smooth map from a neighbourhood of  $\theta_0$  in  $M$  into a fixed finite dimensional vector space; shrinking the radius  $r$  of (2.37) if necessary, we may assume that  $\varphi$  is defined on all of  $\mathcal{V}$ . As in (2.50), the random vector  $\varphi(T_n)$  is to be read with the convention that it is set to a fixed value, say  $\varphi(\theta_0)$ , off the event  $\{T_n \in \mathcal{V}\}$ ; the convention is harmless because that event has probability tending to one under every  $\mathbb{P}_{\theta_n, h}^n$ , by localizability and Lemma 27, and two random vectors agreeing on events of probability tending to one have the same weak limits. With that convention, if  $T_n$  is locally regular at  $\theta_0$  with limit law  $L$ , then  $\varphi(T_n)$  satisfies

$$\sqrt{n}(\varphi(T_n) - \varphi(\theta_{n, h})) \rightsquigarrow (d\varphi_{\theta_0})_{\#} L \quad \text{under } \mathbb{P}_{\theta_n, h}^n, \quad (2.62)$$

with a limit free of  $h$ . This is the delta method applied in the normal chart to (2.53). Indeed, put  $\tilde{\varphi} := \varphi \circ \text{Exp}_{\theta_0}$ , smooth on  $B_r$  and with differential  $d\varphi_{\theta_0}$  at  $0$  because  $d(\text{Exp}_{\theta_0})_0 = \text{id}$ ; Taylor's theorem, with the second derivative of  $\tilde{\varphi}$  bounded on the compact set  $\overline{B_{r/2}}$ , supplies a constant  $c < \infty$  with

$$|\tilde{\varphi}(a) - \tilde{\varphi}(b) - d\varphi_{\theta_0}(a - b)| \leq c(|a|^2 + |b|^2) \quad \text{for } a, b \in \overline{B_{r/2}}.$$

Apply this with  $a = \hat{u}_n[T_n]$  and  $b = \frac{h}{\sqrt{n}}$ , both in  $\overline{B_{r/2}}$  with probability tending to one, and multiply by  $\sqrt{n}$ : the left hand side of (2.62) equals  $d\varphi_{\theta_0}(\sqrt{n}\hat{u}_n[T_n] - h)$  up to a remainder bounded by  $c n^{-\frac{1}{2}}(|\sqrt{n}\hat{u}_n[T_n]|^2 + |h|^2)$ , which is  $O_{\mathbb{P}}(n^{-\frac{1}{2}})$  because  $\sqrt{n}\hat{u}_n[T_n] = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_n, h}^n$  by Lemmas 27 and 28. Part 3 of Corollary 45 below is the instance  $\varphi = \Sigma$ , carried out in full and with an exact remainder, and Remark 46 records that part 3 is the only part of that corollary requiring anything from Section 2.1. The point is that (2.62) has no analogue starting from Proposition 15: the delta method applied to a pointwise central limit theorem returns a pointwise central limit theorem, and a family of pointwise statements indexed by  $\theta$  carries no information about how they fit together, as Remark 30 demonstrates. Local regularity is stable under the operations one performs on estimators in practice; pointwise asymptotic normality is not.

**5. Confidence sets that cover what they claim to cover.** This is the operational payoff. Corollary 43 constructs the Wald region  $\mathcal{C}_{n,1-\alpha} \subseteq M$  and shows that its coverage tends to  $1 - \alpha$  along every sequence of local alternatives, not merely at  $\theta_0$ . Remark 44 shows what the failure looks like when regularity is absent: the Hodges region covers  $\theta_{n,h}$  with limiting probability 1 for small  $h$  and 0 for large  $h$ , and  $1 - \alpha$  for no  $h$  whatsoever,  $h = 0$  included. Since the statistician does not know  $h$  and, by contiguity, cannot come to know it, a region whose coverage depends on  $h$  reports a confidence level that is not the level it achieves.

The rest of the section carries this out. The next subsection collects the three elementary facts that the arguments need. The two subsections after it prove the characterization announced in item 2 and answer the question raised by item 3, namely in what sense  $\hat{\theta}_n$  is asymptotically normal *on the manifold*. The last two contain the consequences themselves: the confidence sets of item 5, and the instance of item 4 at the level of the fitted covariance matrix.

### Three preliminaries

The first preliminary is that the covariance matrix is a *quadratic* function of the local parameter, exactly and with no remainder, which is what makes every delta method argument below an identity rather than an approximation. Throughout the section we write  $\Sigma_u := \Sigma(\text{Exp}_{\theta_0}(u))$  for the covariance matrix at the point of  $M$  with normal coordinate  $u \in B_\rho$ .

**Lemma 31.** Let  $\mathcal{D}_0 : T_{\theta_0}M \rightarrow \text{Sym}(p)$  be the linear map

$$\mathcal{D}_0(h) := D_h \Sigma(\theta_0) = U_h W_0^T + W_0 U_h^T + b_h I_p \quad (2.63)$$

of (1.26). Then, for every  $u \in B_\rho$ , one has the *exact* identity

$$\Sigma_u = \Sigma_0 + \mathcal{D}_0(u) + U_u U_u^T, \quad (2.64)$$

with no remainder. Moreover, with  $r$  the radius fixed in (2.37), there are constants  $0 < c_1 \leq c_2 < \infty$ , depending only on  $\theta_0$  and  $r$ , such that

$$c_1 I_p \preceq \Sigma_u \preceq c_2 I_p \quad \text{for every } u \in \overline{B_r}. \quad (2.65)$$

In particular  $u \mapsto \Sigma_u^{-1}$  is smooth and bounded on  $\overline{B_r}$ .

*Proof.* By (1.16) and (1.6),

$$\Sigma_u = (W_0 + U_u)(W_0 + U_u)^T + (\sigma_0^2 + b_u) I_p = \Sigma_0 + U_u W_0^T + W_0 U_u^T + b_u I_p + U_u U_u^T,$$

which is (2.64). For the bounds, the first summand of the middle expression is positive semidefinite, and  $|b_u| \leq |u| \leq r < \rho \leq \sigma_0^2$  by (2.4) and (2.5); hence  $\Sigma_u \succeq (\sigma_0^2 - r) I_p$  and we may take  $c_1 := \sigma_0^2 - r > 0$ . Similarly  $\|U_u\| \leq \|U_u\|_F \leq r$ , so that  $\|\Sigma_u\| \leq (\|W_0\| + r)^2 + \sigma_0^2 + r =: c_2$ .  $\square$

The second preliminary is the general notion of asymptotic linearity, of which (1.45) is the instance proved in Section 1.4. Naming it lets us record that the influence operator of a sequence is determined by the sequence, so that (1.44) is *the* influence operator of  $\hat{\theta}_n$  and not merely one of them.

**Definition 32** (Asymptotic linearity). A localizable estimator sequence  $T_n$  is *asymptotically linear* at  $\theta_0$  with influence operator  $\text{IF}' : \mathbb{R}^p \rightarrow T_{\theta_0}M$  if  $\text{IF}'$  is Borel measurable,  $\text{IF}'(X_1)$  is centred and square integrable, and

$$\sqrt{n} \hat{u}_n[T_n] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}'(X_i) + o_{\mathbb{P}}(1) \quad \text{under } \mathbb{P}_{\theta_0}^n, \quad (2.66)$$

with  $\hat{u}_n[T_n]$  the chart error of (2.50). Proposition 15 says that  $\hat{\theta}_n$  is asymptotically linear in this sense, with influence operator  $\text{IF}$  of (1.44).

**Proposition 33** (Uniqueness of the influence operator). If  $\text{IF}'$  and  $\text{IF}''$  both satisfy (2.66) for the same estimator sequence  $T_n$ , then  $\text{IF}'(\mathbf{X}_1) = \text{IF}''(\mathbf{X}_1)$  almost surely.

*Proof.* Subtracting the two expansions (2.66) gives  $n^{-\frac{1}{2}} \sum_{i=1}^n \mathbf{D}(\mathbf{X}_i) = o_{\mathbb{P}}(1)$  with  $\mathbf{D} := \text{IF}' - \text{IF}''$ , which is centred and square integrable by Definition 32. The multivariate central limit theorem gives  $n^{-\frac{1}{2}} \sum_{i=1}^n \mathbf{D}(\mathbf{X}_i) \rightsquigarrow N(\mathbf{0}, \text{Cov}(\mathbf{D}(\mathbf{X}_1)))$ , and a sequence converging in distribution to a limit and also to  $\mathbf{0}$  in probability forces that limit to be  $\delta_0$ ; hence  $\text{Cov}(\mathbf{D}(\mathbf{X}_1)) = \mathbf{0}$  and,  $\mathbf{D}(\mathbf{X}_1)$  being centred,  $\mathbf{D}(\mathbf{X}_1) = \mathbf{0}$  almost surely.  $\square$

The third preliminary is that the Fisher information operator may be evaluated at a nearby point, and in particular at the estimator, without losing smoothness or positive definiteness. To compare the operators at two points we read them in one and the same frame, transported along the radial geodesic.

**Lemma 34.** For  $\mathbf{u} \in B_r$  let  $\mathbf{G}(\mathbf{u})$  be the  $d \times d$  matrix

$$\mathbf{G}(\mathbf{u})_{jk} := G_{\text{Exp}_{\theta_0}(\mathbf{u})}(\Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_j), \Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_k)), \quad j, k = 1, \dots, d, \quad (2.67)$$

the matrix of the Fisher information operator at  $\text{Exp}_{\theta_0}(\mathbf{u})$  in the frame obtained by transporting  $\mathbf{e}_1, \dots, \mathbf{e}_d$  along the radial geodesic. Then  $\mathbf{u} \mapsto \mathbf{G}(\mathbf{u})$  is smooth on  $B_r$ , positive definite everywhere, and  $\mathbf{G}(\mathbf{0}) = \mathbf{G}$ .

*Proof.* Write  $(\mathbf{W}_u, \sigma_u^2) := (\mathbf{W}_0 + \mathbf{U}_u, \sigma_0^2 + b_u)$  for the representative of  $\text{Exp}_{\theta_0}(\mathbf{u})$  used in (1.16), which is affine in  $\mathbf{u}$ , and let  $(\tilde{\mathbf{U}}_j(\mathbf{u}), \tilde{b}_j(\mathbf{u})) := \iota_{(\mathbf{W}_u, \sigma_u^2)}(\Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_j))$  be the horizontal lift there. By (1.27) and (1.26) applied at the point  $\text{Exp}_{\theta_0}(\mathbf{u})$ ,

$$\mathbf{G}(\mathbf{u})_{jk} = \frac{1}{2} \text{tr} \{ \Sigma_u^{-1} \mathbf{N}_j(\mathbf{u}) \Sigma_u^{-1} \mathbf{N}_k(\mathbf{u}) \}, \quad \mathbf{N}_j(\mathbf{u}) := \tilde{\mathbf{U}}_j(\mathbf{u}) \mathbf{W}_u^T + \mathbf{W}_u \tilde{\mathbf{U}}_j(\mathbf{u})^T + \tilde{b}_j(\mathbf{u}) \mathbf{I}_p.$$

Each ingredient is smooth in  $\mathbf{u}$ : the matrix  $\Sigma_u$  is polynomial in  $\mathbf{u}$  by (2.64) and is bounded below by  $c_1 \mathbf{I}_p$  by (2.65), so  $\Sigma_u^{-1}$  is smooth;  $\Gamma_{\mathbf{u}}^{-1}$  is smooth by Proposition 19 and Definition 20; and  $\iota_{(\mathbf{W}_u, \sigma_u^2)}$  is smooth in  $\mathbf{u}$  by part 3 of Lemma 16. That  $\mathbf{G}(\mathbf{0}) = \mathbf{G}$  is  $\Gamma_0 = \text{id}$ , and positive definiteness is Theorem 9 at the point  $\text{Exp}_{\theta_0}(\mathbf{u})$ , the change of frame being invertible.  $\square$

## The influence operator of a locally regular sequence

**Proposition 35** (Regularity as a normalization of the influence operator). Let  $T_n$  be a localizable estimator sequence which is asymptotically linear at  $\theta_0$  in the sense of Definition 32, with influence operator  $\text{IF}' : \mathbb{R}^p \rightarrow T_{\theta_0} M$ . In the coordinates fixed in (2.45), let  $\mathbf{V} := \text{Cov}(\text{IF}'(\mathbf{X}_1))$  and  $\mathbf{K} := \mathbb{E}[\text{IF}'(\mathbf{X}_1) \nabla l(\mathbf{X}_1; \theta_0)^T]$  be its covariance matrix and its cross-covariance matrix with the score.

1. For every  $\mathbf{h} \in T_{\theta_0} M$ ,

$$\sqrt{n} \hat{\mathbf{u}}_n[T_n] \rightsquigarrow N(\mathbf{K} \mathbf{h}, \mathbf{V}) \quad \text{under } \mathbb{P}_{\theta_n, \mathbf{h}}^n. \quad (2.68)$$

Consequently  $T_n$  is locally regular at  $\theta_0$  if and only if  $\mathbf{K} = \mathbf{I}_d$ , that is, if and only if (2.61) holds; and in that case the limit law is  $L = N(\mathbf{0}, \mathbf{V})$ .

2. If  $T_n$  is locally regular then  $\mathbf{V} \succeq \mathbf{G}^{-1}$ , with equality if and only if  $\text{IF}'(\mathbf{X}_1) = \text{IF}(\mathbf{X}_1)$  almost surely,  $\text{IF}$  being the influence function (1.44) of  $\hat{\theta}_n$ .

*Proof.* 1. Write  $\psi_i := \text{IF}'(\mathbf{X}_i) \in \mathbb{R}^d$  and  $\Psi_n := n^{-\frac{1}{2}} \sum_{i=1}^n \psi_i$ , so that Definition 32 reads  $\sqrt{n} \hat{\mathbf{u}}_n[T_n] = \Psi_n + o_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_0}^n$ . The vectors  $(\psi_i, \nabla l(\mathbf{X}_i; \theta_0)) \in \mathbb{R}^{2d}$  are independent, identically distributed, centred and square integrable, the second block by Theorem 9 and the first by Definition 32, so the multivariate central limit theorem gives

$$(\Psi_n, \Delta_n) \rightsquigarrow N\left(\mathbf{0}, \begin{pmatrix} \mathbf{V} & \mathbf{K} \\ \mathbf{K}^T & \mathbf{G} \end{pmatrix}\right) \quad \text{under } \mathbb{P}_{\theta_0}^n,$$

with  $\Delta_n$  the normalized score (2.45). Applying the continuous mapping theorem to the linear map  $(a, \delta) \mapsto (a, \delta^T h - \frac{1}{2} h^T G h)$ , and then Slutsky's lemma together with the expansion (2.49) of  $\Lambda_n(h)$ ,

$$(\Psi_n, \Lambda_n(h)) \rightsquigarrow N \left( \begin{pmatrix} \mathbf{0} \\ -\frac{1}{2} h^T G h \end{pmatrix}, \begin{pmatrix} V & K h \\ h^T K^T & h^T G h \end{pmatrix} \right) \quad \text{under } \mathbb{P}_{\theta_0}^n,$$

the off-diagonal block being  $\text{Cov}(\psi_1, \nabla l(\mathbf{X}_1; \theta_0)^T h) = K h$ . This is again the situation of Example 6.7 of van der Vaart (1998), so Le Cam's third lemma gives  $\Psi_n \rightsquigarrow N(K h, V)$  under  $\mathbb{P}_{\theta_n, h}^n$ ; and  $\sqrt{n} \hat{u}_n[T_n] - \Psi_n = o_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_n, h}^n$  as well, by Lemma 27. This proves (2.68).

In particular  $\sqrt{n} \hat{u}_n[T_n] = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_0}^n$ , so Lemma 28 applies and makes local regularity equivalent to the requirement that  $\sqrt{n} \hat{u}_n[T_n] - h$  have a limit law free of  $h$  under  $\mathbb{P}_{\theta_n, h}^n$ . By (2.68) that limit is  $N((K - I_d)h, V)$ , whose mean must therefore not depend on  $h$ ; taking  $h = 0$  shows that the constant value is  $0$ , so  $(K - I_d)h = 0$  for every  $h$  and  $K = I_d$ . Conversely  $K = I_d$  makes that limit  $N(0, V)$  for every  $h$ . Finally, for  $z \in T_{\theta_0} M$  the  $j$ -th coordinate of  $\mathbb{E}[s(\mathbf{X}; \theta_0)(z) \text{IF}'(\mathbf{X})]$  is  $\mathbb{E}[(\nabla l(\mathbf{X}; \theta_0)^T z) \psi_{1j}] = (K z)_j$ , so  $K = I_d$  and (2.61) are the same statement.

2. Assume  $K = I_d$  and put  $D := \psi_1 - G^{-1} \nabla l(\mathbf{X}_1; \theta_0)$ , which is centred and square integrable. Then  $\mathbb{E}[D \nabla l(\mathbf{X}_1; \theta_0)^T] = K - G^{-1} G = 0$ , so  $D$  is uncorrelated with  $G^{-1} \nabla l(\mathbf{X}_1; \theta_0)$  and

$$V = \text{Cov}(G^{-1} \nabla l(\mathbf{X}_1; \theta_0)) + \text{Cov}(D) = G^{-1} + \text{Cov}(D) \succeq G^{-1},$$

using  $\text{Cov}(G^{-1} \nabla l(\mathbf{X}_1; \theta_0)) = G^{-1} G G^{-1} = G^{-1}$ . Equality forces  $\text{Cov}(D) = 0$  and hence,  $D$  being centred,  $D = 0$  almost surely, which by (1.44) and the identification made at the end of the proof of Proposition 15 says  $\text{IF}'(\mathbf{X}_1) = \text{IF}(\mathbf{X}_1)$  almost surely.  $\square$

**Remark 36.** Part 2 is the convolution theorem (2.59) restricted to the asymptotically linear class, proved from scratch in five lines rather than cited: the decomposition  $\psi_1 = G^{-1} \nabla l(\mathbf{X}_1; \theta_0) + D$  with  $D$  uncorrelated with the score is exactly the decomposition  $L = N(0, G^{-1}) * \mu$ , read at the level of influence operators rather than of limit laws. Part 1 should be read alongside Proposition 33: taken together they say that an asymptotically linear sequence has exactly one influence operator, that local regularity is the single linear condition (2.61) on it, and that among the influence operators satisfying that condition the score-proportional one (1.44) is the unique minimizer of the asymptotic covariance.

Note also where local regularity is *not* needed. Part 1 shows that (2.68) holds for *any* asymptotically linear sequence, so the behaviour under local alternatives is always computable, from  $K$  alone; what regularity adds is not the computability but the answer  $K = I_d$ . The Hodges sequence of Remark 30 is the natural illustration. At  $\theta_0 = \theta_*$  it is localizable, being equal to  $\theta_* \in \mathcal{V}$  with probability tending to one, and on that same event  $\hat{u}_n[\hat{\theta}_n^H] = \text{Exp}_{\theta_*}^{-1}(\theta_*) = 0$ , so that  $\sqrt{n} \hat{u}_n[\hat{\theta}_n^H] = o_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_*}^n$ . It is therefore asymptotically linear at  $\theta_*$  with influence operator  $\text{IF}' \equiv 0$ , which is centred and square integrable; hence  $K = 0$  and  $V = 0$ , and part 1 predicts  $\sqrt{n} \hat{u}_n[\hat{\theta}_n^H] \rightsquigarrow N(K h, V) = N(0, 0) = \delta_0$  under  $\mathbb{P}_{\theta_n, h}^n$  for every  $h$ , which is exactly what Remark 30 computes by hand. It violates (2.61) as flagrantly as possible,  $K = 0$  being as far from  $I_d$  as a matrix can be. So the Hodges sequence is no counterexample to (2.61), which it simply violates while failing to be locally regular; it is a counterexample to dispensing with the hypothesis of local regularity in (2.59). Part 2 is untouched by it, since part 2 assumes  $K = I_d$  and it is precisely that assumption which fails here; no contradiction arises from  $V = 0 \prec G^{-1}$ .

## Asymptotic normality as a random point of the manifold

Proposition 15 and Theorem 29 are statements about a random *vector*,  $\sqrt{n} \text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n)$ , living in the fixed vector space  $T_{\theta_0} M$ . It is natural to ask what they say about the random *point*  $\hat{\theta}_n \in M$  itself, in the spirit of Example III.1 of Sun et al. (2026), where a Gaussian law on a manifold is a law whose density is proportional to  $\exp\left(-\frac{d(x, \mu)^2}{2\tau^2}\right)$

and is centred at a point  $\mu$  of the manifold. The obstruction to asking the question naively is that  $M$  carries no scalar multiplication, so the rescaling by  $\sqrt{n}$  that all these statements involve cannot be performed on  $M$ ; and the dilation  $\theta \mapsto \text{Exp}_{\theta_0}(\sqrt{n} \text{Exp}_{\theta_0}^{-1}(\theta))$ , which one might substitute for it, does not map  $M$  into itself, the domain  $B_\rho$  of Proposition 6 being a bounded ball. The rescaling must therefore be moved onto the *metric*: one asks not that the law of  $\hat{\theta}_n$  converge — it converges to  $\delta_{\theta_0}$ , which says only that  $\hat{\theta}_n$  is consistent — but that it be indistinguishable, at the resolution  $n^{-\frac{1}{2}}$ , from an explicit Gaussian-shaped law on  $M$  centred at the true parameter. That is the formulation below, and it turns out to be exactly local regularity.

The formulation uses the Riemannian distance of  $(M, g)$ , so we first record that this distance is finite — a fact needed nowhere earlier and therefore not yet stated.

**Lemma 37** ( $M$  is connected). The manifold  $M$  is connected; consequently its Riemannian distance  $d_M$  is finite on  $M \times M$ .

*Proof.* Write  $\text{St} := \{U \in \mathbb{R}^{p \times q} : U^T U = I_q\}$  and let  $\text{Sym}^{++}(q)$  be the set of positive definite  $q \times q$  symmetric matrices, which is convex and hence connected. The continuous map  $(U, P) \mapsto UP$  from  $\text{St} \times \text{Sym}^{++}(q)$  into  $\mathbb{R}^{p \times q}$  has as image exactly the set of rank  $q$  matrices: any such  $A$  is  $UP$  with  $P := (A^T A)^{\frac{1}{2}}$ , which is positive definite, and  $U := AP^{-1}$ , which satisfies  $U^T U = P^{-1} A^T A P^{-1} = I_q$ . The set  $\text{St}$  is itself connected, being the image of the connected group  $SO(p)$  under  $R \mapsto RE$ , where  $E \in \mathbb{R}^{p \times q}$  consists of the first  $q$  columns of  $I_p$ : given  $U \in \text{St}$ , complete it to  $[U \ U^\perp] \in O(p)$  and, if the determinant is  $-1$ , change the sign of one column of  $U^\perp$ , which is possible precisely because  $q < p$ . So the set of rank  $q$  matrices is connected, hence so is  $\Theta$ , the product of that set with  $(0, \infty)$ , and hence so is its continuous image  $M = \pi(\Theta)$ . A connected Riemannian manifold is path connected, and the infimum of the lengths of the joining curves is finite, so  $d_M < \infty$  throughout.  $\square$

**Definition 38** (Wrapped normal laws and the rescaled metric). Let  $d_M$  denote the Riemannian distance of the connected Riemannian manifold  $(M, g)$ , and write  $\text{law}_{\mathbb{P}}(Y)$  for the law of a random element  $Y$  under  $\mathbb{P}$ , the subscript being omitted when the law is clear.

1. For  $\theta \in M$ , a centred Gaussian law  $\nu$  on  $T_\theta M$  and  $n \geq 1$ , let  $\nu^{(n)}$  be the law of  $n^{-\frac{1}{2}}Z$  with  $Z \sim \nu$ , conditioned on the event  $\{n^{-\frac{1}{2}}Z \in \mathcal{E}_\theta\}$ , which has positive probability because  $\mathcal{E}_\theta$  is an open ball about the origin; and put

$$\mathcal{N}_n(\theta, \nu) := (\text{Exp}_\theta)_\# \nu^{(n)}, \quad (2.69)$$

the push-forward of  $\nu^{(n)}$  by the exponential map at  $\theta$ : the *wrapped normal law* on  $M$  with centre  $\theta$ , dispersion  $n^{-1}$  and shape  $\nu$ . We abbreviate  $\mathcal{N}_n(\theta) := \mathcal{N}_n(\theta, N(\mathbf{0}, G_\theta^{-1}))$ , the shape being given by the inverse Fisher information operator at  $\theta$  itself.

2. For Borel probability measures  $P, Q$  on  $M$  put

$$\text{BL}_n(P, Q) := \sup \left\{ \left| \int_M f dP - \int_M f dQ \right| : \|f\|_\infty \leq 1, |f(x) - f(y)| \leq \sqrt{n} d_M(x, y) \ \forall x, y \in M \right\}, \quad (2.70)$$

the bounded Lipschitz distance for the rescaled metric  $\sqrt{n} d_M$ , a genuine metric by Lemma 37; and write BL for the ordinary bounded Lipschitz distance on  $T_{\theta_0} M \cong \mathbb{R}^d$ , which metrizes weak convergence there.

Two comments on the definition. First, the conditioning in part 1 is a formality for every  $\theta$  in the compact set  $\mathcal{K} := \text{Exp}_{\theta_0}(\overline{B_{r/2}}) \subseteq \text{Exp}_{\theta_0}(B_r)$ , and this is all we shall need, since the centres occurring below are  $\theta_0$  and  $\theta_{n,h}$ , for which  $|\mathbf{h}|/\sqrt{n} \leq \frac{r}{2}$  once  $n$  is large. For  $\theta = [\mathbf{A}, s]$  ranging in  $\mathcal{K}$  the radius  $\min(\sigma_{\min}(\mathbf{A}), s)$  of  $\mathcal{E}_\theta$  is a positive continuous function of  $\theta$ , as observed in the proof of Lemma 21, hence is bounded below by some  $\rho' > 0$ ; and the

operator norm of  $G_\theta^{-1}$  equals that of  $G(\text{Exp}_{\theta_0}^{-1}(\theta))^{-1}$ , the frame of (2.67) being orthonormal, which is bounded above on  $\overline{B_{r/2}}$  by Lemma 34. The Gaussian tail bound then gives  $\mathbb{P}\left(n^{-\frac{1}{2}}\mathbf{Z} \notin \mathcal{E}_\theta\right) \leq \mathbb{P}(|\mathbf{Z}| > \sqrt{n}\rho') = O(e^{-cn})$  for some  $c > 0$ , uniformly over  $\theta \in \mathcal{K}$ , so the conditioning perturbs the law by  $O(e^{-cn})$  in total variation and is invisible in everything that follows. The same bound holds, a fortiori, whenever the shape  $\nu$  has a covariance whose operator norm is bounded along the sequence, which is the only property of it used below. Second,  $\text{BL}_n$  is nondecreasing in  $n$ , the class of competing  $f$  growing with  $n$ , and  $\text{BL}_n(P, Q) \rightarrow 0$  is a strictly stronger requirement than weak convergence: it says that  $P$  and  $Q$  cannot be told apart by any bounded test function varying on the scale  $n^{-\frac{1}{2}}$ , which is the scale on which  $\hat{\theta}_n$  and  $\theta_0$  differ. Two laws both collapsing to  $\delta_{\theta_0}$  are automatically close in the weak topology and the closeness says nothing;  $\text{BL}_n$  is what removes that triviality.

**Lemma 39** (The normal chart is bi-Lipschitz). There exist  $\varepsilon \in (0, r]$  and  $L \geq 1$ , depending only on  $\theta_0$ , such that

$$L^{-1}|\mathbf{u} - \mathbf{w}| \leq d_M(\text{Exp}_{\theta_0}(\mathbf{u}), \text{Exp}_{\theta_0}(\mathbf{w})) \leq L|\mathbf{u} - \mathbf{w}| \quad \text{for all } \mathbf{u}, \mathbf{w} \in B_\varepsilon. \quad (2.71)$$

Moreover  $d_M(\theta_0, \text{Exp}_{\theta_0}(\mathbf{u})) = |\mathbf{u}|$  for every  $\mathbf{u} \in B_\rho$ .

*Proof.* Fix  $\varepsilon_0 \in (0, r]$ . By part 2 of Proposition 6 the map  $\text{Exp}_{\theta_0}$  is a diffeomorphism of  $B_\rho$  onto the open set  $\mathcal{U}$ , so  $d(\text{Exp}_{\theta_0})_{\mathbf{u}}$  is a linear isomorphism for every  $\mathbf{u} \in B_\rho$ ; by continuity and compactness of  $\overline{B_{\varepsilon_0}} \subseteq B_\rho$  there is  $L_0 \geq 1$  with

$$L_0^{-1}|\boldsymbol{\xi}| \leq |d(\text{Exp}_{\theta_0})_{\mathbf{u}}(\boldsymbol{\xi})|_g \leq L_0|\boldsymbol{\xi}| \quad \text{for all } \mathbf{u} \in \overline{B_{\varepsilon_0}} \text{ and } \boldsymbol{\xi} \in T_{\theta_0}M, \quad (2.72)$$

where  $|\cdot|_g$  is the norm of  $g$  at the relevant point.

The upper bound in (2.71) follows with  $L = L_0$  for all  $\mathbf{u}, \mathbf{w} \in B_{\varepsilon_0}$ : the segment  $t \mapsto \mathbf{u} + t(\mathbf{w} - \mathbf{u})$  stays in the convex set  $B_{\varepsilon_0}$ , its image under  $\text{Exp}_{\theta_0}$  joins the two points and has, by (2.72),  $g$ -length at most  $L_0|\mathbf{u} - \mathbf{w}|$ , and  $d_M$  is the infimum of the lengths of such curves.

For the lower bound put  $\varepsilon := \frac{\varepsilon_0}{4}$  and take  $\mathbf{u}, \mathbf{w} \in B_\varepsilon$ , so that  $|\mathbf{u} - \mathbf{w}| < \frac{\varepsilon_0}{2}$ . Let  $c$  be any piecewise smooth curve in  $M$  from  $\text{Exp}_{\theta_0}(\mathbf{u})$  to  $\text{Exp}_{\theta_0}(\mathbf{w})$ . If  $c$  stays inside  $\text{Exp}_{\theta_0}\left(B_{\frac{\varepsilon_0}{2}}\right)$ , then  $\text{Exp}_{\theta_0}^{-1} \circ c$  is a curve in  $B_{\frac{\varepsilon_0}{2}}$  from  $\mathbf{u}$  to  $\mathbf{w}$ , of Euclidean length at least  $|\mathbf{u} - \mathbf{w}|$ , so (2.72) gives  $\text{length}(c) \geq L_0^{-1}|\mathbf{u} - \mathbf{w}|$ . Otherwise, let  $t_*$  be the first time at which  $c$  leaves  $\text{Exp}_{\theta_0}\left(B_{\frac{\varepsilon_0}{2}}\right)$ ; the restriction of  $\text{Exp}_{\theta_0}^{-1} \circ c$  to  $[0, t_*]$  joins  $\mathbf{u}$ , of norm less than  $\frac{\varepsilon_0}{4}$ , to a point of norm  $\frac{\varepsilon_0}{2}$ , hence has Euclidean length at least  $\frac{\varepsilon_0}{4} > \frac{1}{2}|\mathbf{u} - \mathbf{w}|$ , and  $\text{length}(c) \geq \frac{1}{2}L_0^{-1}|\mathbf{u} - \mathbf{w}|$ . Taking the infimum over  $c$  and setting  $L := 2L_0$  gives (2.71).

The last assertion is the Gauss lemma. Since  $\text{Exp}_{\theta_0}$  is a diffeomorphism of the ball  $B_\rho \subseteq T_{\theta_0}M$  onto  $\mathcal{U}$ , the set  $\mathcal{U}$  is a geodesic ball centred at  $\theta_0$ , and inside a geodesic ball the radial geodesic  $t \mapsto \text{Exp}_{\theta_0}(t\mathbf{u})$ ,  $t \in [0, 1]$ , is the unique minimizing curve in  $M$  from  $\theta_0$  to  $\text{Exp}_{\theta_0}(\mathbf{u})$  (Lee, 2018, Chapter 6). Its length is  $|\mathbf{u}|$ , geodesics having constant speed and the initial speed being  $|\mathbf{u}|$ .  $\square$

We fix  $\varepsilon$  and  $L$  as in Lemma 39 for the rest of this subsection and write  $\mathcal{V}_\varepsilon := \text{Exp}_{\theta_0}(B_\varepsilon) \subseteq \mathcal{V}$ . The next lemma is the translation device: it says that comparing two  $M$ -valued sequences in  $\text{BL}_n$  is the same as comparing their rescaled chart images in the ordinary bounded Lipschitz distance, so that every statement about random points of  $M$  made below reduces to a statement about random vectors of  $T_{\theta_0}M$ , where the preceding results live.

**Lemma 40** (Rescaled comparison). For each  $n$  let  $Y_n$  and  $Y'_n$  be  $M$ -valued random elements, possibly defined under different laws, with  $\mathbb{P}(Y_n \in \mathcal{V}_\varepsilon) \rightarrow 1$  and  $\mathbb{P}(Y'_n \in \mathcal{V}_\varepsilon) \rightarrow 1$ , and put

$$\mathbf{y}_n := \begin{cases} \sqrt{n} \text{Exp}_{\theta_0}^{-1}(Y_n) & \text{on } \{Y_n \in \mathcal{V}_\varepsilon\}, \\ \mathbf{0} & \text{otherwise,} \end{cases}$$

and likewise  $\mathbf{y}'_n$  from  $Y'_n$ . Then, with  $P_n := \text{law}(Y_n)$  and  $P'_n := \text{law}(Y'_n)$  on  $M$ ,

$$\text{BL}_n(P_n, P'_n) \longrightarrow 0 \quad \Longleftrightarrow \quad \text{BL}(\text{law}(\mathbf{y}_n), \text{law}(\mathbf{y}'_n)) \longrightarrow 0. \quad (2.73)$$

In particular, if  $\mathbf{y}_n$  and  $\mathbf{y}'_n$  converge in distribution to the same limit, then  $\text{BL}_n(P_n, P'_n) \rightarrow 0$ .

*Proof.* Abbreviate  $\eta_n := \mathbb{P}(Y_n \notin \mathcal{V}_\varepsilon) + \mathbb{P}(Y'_n \notin \mathcal{V}_\varepsilon) \rightarrow 0$ .

( $\Leftarrow$ ) Let  $f : M \rightarrow \mathbb{R}$  satisfy  $\|f\|_\infty \leq 1$  and  $|f(x) - f(y)| \leq \sqrt{n} d_M(x, y)$ , and define  $g$  on  $\sqrt{n}B_\varepsilon$  by  $g(\mathbf{z}) := f(\text{Exp}_{\theta_0}(n^{-\frac{1}{2}}\mathbf{z}))$ . By the upper bound in (2.71),  $|g(\mathbf{z}) - g(\mathbf{z}')| \leq \sqrt{n} L n^{-\frac{1}{2}} |\mathbf{z} - \mathbf{z}'| = L |\mathbf{z} - \mathbf{z}'|$ , and  $\|g\|_\infty \leq 1$ . Extend  $g$  to all of  $T_{\theta_0}M$  by the McShane formula  $\tilde{g}(\mathbf{z}) := \inf_{\mathbf{z}' \in \sqrt{n}B_\varepsilon} (g(\mathbf{z}') + L|\mathbf{z} - \mathbf{z}'|)$ , truncated to  $[-1, 1]$ ; both bounds are preserved and  $\tilde{g} = g$  on  $\sqrt{n}B_\varepsilon$ . On  $\{Y_n \in \mathcal{V}_\varepsilon\}$  we have  $f(Y_n) = \tilde{g}(\mathbf{y}_n)$ , and both are bounded by 1 everywhere, so

$$\left| \int_M f dP_n - \int_M f dP'_n \right| \leq |\mathbb{E}[\tilde{g}(\mathbf{y}_n)] - \mathbb{E}[\tilde{g}(\mathbf{y}'_n)]| + 2\eta_n \leq L \text{BL}(\text{law}(\mathbf{y}_n), \text{law}(\mathbf{y}'_n)) + 2\eta_n,$$

the last step because  $L^{-1}\tilde{g}$  is bounded by 1 and 1-Lipschitz,  $L$  being at least 1. Taking the supremum over  $f$  gives  $\text{BL}_n(P_n, P'_n) \leq L \text{BL} + 2\eta_n$ .

( $\Rightarrow$ ) Symmetrically, let  $g : T_{\theta_0}M \rightarrow \mathbb{R}$  satisfy  $\|g\|_\infty \leq 1$  and  $\text{Lip}(g) \leq 1$ , and define  $f$  on  $\mathcal{V}_\varepsilon$  by  $f(x) := g(\sqrt{n} \text{Exp}_{\theta_0}^{-1}(x))$ . For  $x, y \in \mathcal{V}_\varepsilon$  the lower bound in (2.71) gives  $|f(x) - f(y)| \leq \sqrt{n} |\text{Exp}_{\theta_0}^{-1}(x) - \text{Exp}_{\theta_0}^{-1}(y)| \leq L \sqrt{n} d_M(x, y)$ , so  $f$  is  $L$ -Lipschitz for the metric  $\sqrt{n}d_M$  on  $\mathcal{V}_\varepsilon$ ; extend it to  $M$  by the McShane formula for that metric — finite by Lemma 37, which is what makes the formula meaningful — again truncated to  $[-1, 1]$ . The same two displays, read in the other direction, give  $\text{BL}(\text{law}(\mathbf{y}_n), \text{law}(\mathbf{y}'_n)) \leq L \text{BL}_n(P_n, P'_n) + 2\eta_n$ .

The final assertion holds because BL metrizes weak convergence on  $T_{\theta_0}M \cong \mathbb{R}^d$ .  $\square$

**Theorem 41** (The maximum likelihood estimator as an asymptotically wrapped normal point of  $M$ ). *Let  $1 \leq q < p$  and  $\theta_0 = [\mathbf{W}_0, \sigma_0^2] \in M$ , and let  $\mathcal{N}_n(\cdot)$  and  $\text{BL}_n$  be as in Definition 38.*

1. (Fixed centre.)

$$\text{BL}_n(\text{law}_{\mathbb{P}_{\theta_0}^n}(\hat{\theta}_n), \mathcal{N}_n(\theta_0)) \rightarrow 0. \quad (2.74)$$

*That is, at the resolution  $n^{-\frac{1}{2}}$  the maximum likelihood estimator is a wrapped normal random point of  $M$ , centred at  $[\mathbf{W}_0, \sigma_0^2]$ , with dispersion  $n^{-1}$  and with shape the inverse Fisher information operator.*

2. (Moving centre.) For every  $\mathbf{h} \in T_{\theta_0}M$ ,

$$\text{BL}_n(\text{law}_{\mathbb{P}_{\theta_n, \mathbf{h}}^n}(\hat{\theta}_n), \mathcal{N}_n(\theta_n, \mathbf{h})) \rightarrow 0, \quad (2.75)$$

*the approximating law being the wrapped normal centred at the true parameter  $\theta_n, \mathbf{h}$ , with shape the inverse Fisher information operator at that same point.*

3. (The moving centre statement is local regularity.) Let  $T_n$  be a localizable estimator sequence with  $\sqrt{n}\hat{\mathbf{u}}_n[T_n] = O_{\mathbb{P}}(1)$  under  $\mathbb{P}_{\theta_0}^n$ , let  $\nu$  be a centred Gaussian law on  $T_{\theta_0}M$ , and for  $\mathbf{u} \in B_\rho$  let  $\nu_{\mathbf{u}}$  denote its transport to  $T_{\text{Exp}_{\theta_0}(\mathbf{u})}M$  by the linear isometry  $\Gamma_{\mathbf{u}}^{-1}$  of Definition 20. Then  $T_n$  is locally regular at  $\theta_0$  with limit law  $\nu$ , in the sense of Definition 26, if and only if

$$\text{BL}_n(\text{law}_{\mathbb{P}_{\theta_n, \mathbf{h}}^n}(T_n), \mathcal{N}_n(\theta_n, \mathbf{h}, \nu_{\mathbf{h}/\sqrt{n}})) \rightarrow 0 \quad \text{for every } \mathbf{h} \in T_{\theta_0}M. \quad (2.76)$$

*Proof.* Fix  $\mathbf{h} \in T_{\theta_0}M$  and abbreviate  $\mathbf{u}_n := \frac{\mathbf{h}}{\sqrt{n}}$  and  $\theta_n := \theta_{n, \mathbf{h}} = \text{Exp}_{\theta_0}(\mathbf{u}_n)$ . Parts 1 and 2 are proved together, part 1 being the case  $\mathbf{h} = 0$ , in which  $\mathbf{u}_n = 0$ ,  $\theta_n = \theta_0$  and  $\Gamma_0 = \text{id}$ ; part 3 is then read off the same computation. Throughout,  $\mathbf{G}(\cdot)$  is the transported information matrix (2.67) of Lemma 34.

**Step 1: the estimator side.** By Proposition 24 and Lemma 27,  $\mathbb{P}_{\theta_n}^n(\hat{\theta}_n \in \mathcal{V}_\varepsilon) \rightarrow 1$ , and (2.58) gives

$$\sqrt{n} \hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{h}, \mathbf{G}^{-1}) \quad \text{under } \mathbb{P}_{\theta_n}^n. \quad (2.77)$$

For  $\mathbf{h} = 0$  this is Proposition 15 itself, and neither Lemma 27 nor Theorem 29 is used.

**Step 2: the wrapped normal side.** Let  $\Theta_n \sim \mathcal{N}_n(\theta_n, \lambda_n)$ , where  $\lambda_n := N(0, G_{\theta_n}^{-1})$  for parts 1 and 2 and  $\lambda_n := \nu_{\mathbf{u}_n}$  for part 3. In both cases  $\lambda_n$  is a centred Gaussian law on  $T_{\theta_n}M$ , and we compute its covariance matrix

in the frame  $\Gamma_{\mathbf{u}_n}^{-1}(\mathbf{e}_1), \dots, \Gamma_{\mathbf{u}_n}^{-1}(\mathbf{e}_d)$  of  $T_{\theta_n}M$ , which is  $g_{\theta_n}$ -orthonormal because  $\mathbf{e}_1, \dots, \mathbf{e}_d$  is  $g_{\theta_0}$ -orthonormal and  $\Gamma_{\mathbf{u}_n}$  is a linear isometry. The coordinates of a vector  $\mathbf{v} \in T_{\theta_n}M$  in that frame are by construction the  $\mathbf{e}$ -coordinates of  $\Gamma_{\mathbf{u}_n}(\mathbf{v}) \in T_{\theta_0}M$ . For parts 1 and 2 the matrix of the bilinear form  $G_{\theta_n}$  in that frame is  $\mathbf{G}(\mathbf{u}_n)$ , by the very definition (2.67), so the covariance matrix of  $\lambda_n$  is  $\mathbf{G}(\mathbf{u}_n)^{-1}$ ; for part 3 it is the fixed covariance matrix  $\mathbf{C}_\nu$  of  $\nu$  in the basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$ , transport being an isometry. Either way the covariance stays bounded, so the conditioning built into  $\lambda_n^{(n)}$  costs  $O(e^{-cn})$  in total variation, as recorded after Definition 38, and may be discarded. Writing  $\Theta_n = \text{Exp}_{\theta_n}(\mathbf{v}_n)$  with  $\mathbf{v}_n \sim \lambda_n^{(n)}$ , the vector  $\sqrt{n} \Gamma_{\mathbf{u}_n}(\mathbf{v}_n)$  therefore has, in the basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$ , the law  $N(0, \mathbf{G}(\mathbf{u}_n)^{-1})$ , respectively  $N(0, \mathbf{C}_\nu)$ ; in particular  $\sqrt{n}|\mathbf{v}_n| = \sqrt{n}|\Gamma_{\mathbf{u}_n}(\mathbf{v}_n)| = O_{\mathbb{P}}(1)$ .

We check next that  $\Theta_n \in \mathcal{V}_\varepsilon$  with probability tending to one. The set of pairs  $(x, \mathbf{v})$  in  $TM$  with  $\mathbf{v} \in \mathcal{E}_x$  and  $\text{Exp}_x(\mathbf{v}) \in \mathcal{V}_\varepsilon$  is open, the domain of  $\text{Exp}$  being open in  $TM$  and  $\text{Exp}$  continuous on it as established in the proof of Lemma 21, and it contains  $(\theta_0, 0)$ ; hence there is  $\delta > 0$  such that  $\text{Exp}_x(\mathbf{v}) \in \mathcal{V}_\varepsilon$  whenever  $x \in \text{Exp}_{\theta_0}(B_\delta)$  and  $|\mathbf{v}| < \delta$ . Since  $|\mathbf{u}_n| \rightarrow 0$  and  $\mathbb{P}(|\mathbf{v}_n| < \delta) \rightarrow 1$ , the claim follows. Put  $\mathbf{w}_n := \text{Exp}_{\theta_0}^{-1}(\Theta_n)$  on that event and  $\mathbf{w}_n := 0$  off it.

On that event  $\mathbf{u}_n$  and  $\mathbf{w}_n$  lie in  $B_r$ , so  $\theta_n$  and  $\Theta_n$  lie in  $\mathcal{V}^\#$ ; since moreover  $\mathbf{v}_n \in \mathcal{E}_{\theta_n}$  and  $\text{Exp}_{\theta_n}(\mathbf{v}_n) = \Theta_n$ , the uniqueness clause of part 1 of Lemma 21 identifies  $\mathbf{v}_n$  with  $\text{Exp}_{\theta_n}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w}_n))$ , and the definition (2.38) of  $F$  then reads  $\Gamma_{\mathbf{u}_n}(\mathbf{v}_n) = F(\mathbf{u}_n, \mathbf{w}_n)$ . Lemma 22 gives

$$|F(\mathbf{u}_n, \mathbf{w}_n) - (\mathbf{w}_n - \mathbf{u}_n)| \leq C|\mathbf{u}_n||\mathbf{w}_n - \mathbf{u}_n|, \quad (2.78)$$

whence first, by the triangle inequality and because  $\Gamma_{\mathbf{u}_n}$  is an isometry,  $|\mathbf{w}_n - \mathbf{u}_n| \leq \frac{|\mathbf{v}_n|}{1-C|\mathbf{u}_n|}$  for  $n$  large enough that  $C|\mathbf{u}_n| < 1$ , so that  $\sqrt{n}|\mathbf{w}_n - \mathbf{u}_n| = O_{\mathbb{P}}(1)$ ; and then, feeding this back into (2.78) multiplied by  $\sqrt{n}$ ,

$$\sqrt{n} \mathbf{w}_n = \mathbf{h} + \sqrt{n} \Gamma_{\mathbf{u}_n}(\mathbf{v}_n) + O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right).$$

By Lemma 34,  $\mathbf{G}(\mathbf{u}_n) \rightarrow \mathbf{G}(0) = \mathbf{G}$ , so  $N(0, \mathbf{G}(\mathbf{u}_n)^{-1}) \rightsquigarrow N(0, \mathbf{G}^{-1})$ ; Slutsky's lemma therefore gives

$$\sqrt{n} \mathbf{w}_n \rightsquigarrow N(\mathbf{h}, \mathbf{G}^{-1}) \quad \text{in parts 1 and 2,} \quad \sqrt{n} \mathbf{w}_n \rightsquigarrow N(\mathbf{h}, \mathbf{C}_\nu) \quad \text{in part 3.} \quad (2.79)$$

**Step 3: conclusion.** Parts 1 and 2 follow from Lemma 40 applied with  $Y_n := \hat{\theta}_n$  under  $\mathbb{P}_{\theta_n}^n$  and  $Y'_n := \Theta_n$ : the localization hypotheses were verified in Steps 1 and 2, the chart error of Lemma 40 agrees with  $\sqrt{n}\hat{\mathbf{u}}_n$  on  $\{\hat{\theta}_n \in \mathcal{V}_\varepsilon\}$ , an event of probability tending to one, and the two rescaled chart images have the same weak limit  $N(\mathbf{h}, \mathbf{G}^{-1})$  by (2.77) and (2.79).

For part 3, apply the equivalence (2.73) of the same lemma with  $Y_n := T_n$  under  $\mathbb{P}_{\theta_n}^n$  and  $Y'_n := \Theta_n$ ; the hypothesis  $\sqrt{n}\hat{\mathbf{u}}_n[T_n] = O_{\mathbb{P}}(1)$  together with localizability and Lemma 27 gives  $\mathbb{P}_{\theta_n}^n(T_n \in \mathcal{V}_\varepsilon) \rightarrow 1$  and makes the two versions of the chart error agree there. The left hand side of (2.73) is then (2.76) for the given  $\mathbf{h}$ , and by (2.79) and the fact that BL metrizes weak convergence its right hand side holds if and only if  $\sqrt{n}\hat{\mathbf{u}}_n[T_n] \rightsquigarrow N(\mathbf{h}, \mathbf{C}_\nu)$  under  $\mathbb{P}_{\theta_n}^n$ , that is, if and only if  $\sqrt{n}\hat{\mathbf{u}}_n[T_n] - \mathbf{h} \rightsquigarrow \nu$  there. Requiring this for every  $\mathbf{h}$  is exactly condition (2.53) of Lemma 28, which that lemma shows to be equivalent to local regularity with limit law  $\nu$ ; and its last assertion makes the boundedness hypothesis automatic on the regularity side.  $\square$

**Remark 42 (Reading the theorem). 1. What each part costs.** Part 1 uses Proposition 15 and the chart estimate Lemma 39, and nothing else; in particular it does *not* use local regularity. It does not really use the transport comparison Lemma 22 either: at  $\mathbf{h} = 0$  one has  $\mathbf{u}_n = 0$ ,  $\Gamma_0 = \text{id}$  and therefore  $\mathbf{w}_n = \mathbf{v}_n$  exactly, so that (2.78) is invoked only in the trivial slice  $R(0, \cdot) = 0$  already recorded in the proof of that lemma. Corollary 43 below is different in this respect: it needs the transport comparison even at  $\mathbf{h} = 0$ , because there the moving base point is  $\hat{\theta}_n$  and not  $\theta_{n,\mathbf{h}}$ . Part 2 is part 1 with the centre moved, and the only ingredient that changes is (2.77), which for  $\mathbf{h} \neq 0$  is precisely (2.58), the substance of Theorem 29. Part 3 says that this is not an artefact of the proof: for a localizable sequence with the boundedness property, the moving centre statement and local regularity are the *same* statement. So the answer to the question whether local regularity is needed for asymptotic normality at the level of the manifold is: not for the version centred at a fixed, known  $\theta_0$ , which is a restatement of Proposition 15; and for the version centred at the true parameter, whatever it is, local regularity is not merely sufficient but necessary.

**2. The shape may be perturbed.** Step 2 uses about the shape only that its covariance, read in the transported frame, is bounded and converges to  $G^{-1}$ . Part 2 therefore also holds with  $\mathcal{N}_n(\theta_{n,h})$  replaced by  $\mathcal{N}_n(\theta_{n,h}, (N(\mathbf{0}, G_{\theta_0}^{-1}))_{h/\sqrt{n}})$ , the transport of the information at  $\theta_0$  itself: at the resolution  $n^{-\frac{1}{2}}$  the shape need not be re-evaluated at the moving centre. What may *not* be moved is the centre, which is the content of the next comment.

**3. Why the moving centre version is the one that means something.** The centre of the approximating law in (2.74) is a point the statistician must be told. The centre in (2.75) is the true parameter, whichever member of the contiguous family it is. By Lemma 27 no procedure can distinguish these members, so only the second version supports the sentence one actually wants to say, namely that  $\theta_n$  is distributed around the truth like a wrapped normal law with dispersion  $n^{-1}$  and shape  $G^{-1}$ ; and it is that sentence which Corollary 43 below converts into a confidence region. The Hodges-type sequence of Remark 30 satisfies the analogue of part 1 at every  $\theta_0 \neq \theta_*$ , and at  $\theta_*$  satisfies a degenerate analogue with  $\mathcal{N}_n$  replaced by  $\delta_{\theta_*}$ ; it satisfies no version of part 2 at  $\theta_*$ , since by (2.52) its transported error converges to  $\delta_{-h}$ , which depends on  $h$ . Part 3 is the formal statement that this is the only way things can go wrong.

**4. The link with Example III.1 of Sun et al. (2026).** On  $\text{Exp}_\theta(\mathcal{E}_\theta)$  the wrapped law  $\mathcal{N}_n(\theta)$  has a density with respect to the Riemannian volume measure of  $(M, g)$  proportional to

$$\exp\left(-\frac{n}{2} G_\theta(\text{Exp}_\theta^{-1}(x), \text{Exp}_\theta^{-1}(x))\right) \cdot J_\theta(x)^{-1}, \quad J_\theta(x) := \left| \det d(\text{Exp}_\theta)_{\text{Exp}_\theta^{-1}(x)} \right|, \quad (2.80)$$

the Jacobian factor converting the Lebesgue measure of the tangent space into the volume measure. Two observations. First, the exponent is the Riemannian Gaussian exponent of Example III.1 of Sun et al. (2026), with the quadratic form  $G_\theta$  in place of  $\tau^{-2}g_\theta$ : if the information operator happened to be isotropic,  $G_\theta = \tau^{-2}g_\theta$ , then by the last assertion of Lemma 39 the exponent would be  $-\frac{n}{2\tau^2} d_M(\theta, x)^2$ , and (2.80) would be, up to  $J_\theta$ , the density displayed in that example with variance parameter  $\frac{\tau^2}{n}$ . The Fisher information of the probabilistic principal component analysis model is not isotropic — Section 2.3 computes it, and (2.87) already shows that different directions of  $T_{\theta_0}M$  are treated very differently — so the honest limiting shape is the anisotropic wrapped law rather than the isotropic Riemannian Gaussian. Second, the Jacobian is invisible at the relevant scale: on the support of  $\mathcal{N}_n(\theta)$  one has  $|\text{Exp}_\theta^{-1}(x)| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$ , and the standard expansion of the volume density in normal coordinates gives  $J_\theta(x) = 1 + O(|\text{Exp}_\theta^{-1}(x)|^2) = 1 + O_{\mathbb{P}}(n^{-1})$ . At the resolution  $n^{-\frac{1}{2}}$  at which  $\text{BL}_n$  measures, the distinction between the wrapped normal law, the intrinsic Riemannian Gaussian density and the Gaussian read in the normal chart is therefore immaterial, and the theorem is insensitive to which of the three is chosen; we have chosen the wrapped law because it requires no curvature computation. Nothing in this observation is used anywhere below; it is recorded only to say what (2.75) amounts to.

**5. What is not claimed.** The metric  $\text{BL}_n$  is a weak metric, and nothing above asserts convergence in total variation, nor any control of densities; that would be a genuinely stronger statement, and none of the consequences drawn below needs it. Nor is any uniformity in  $h$  asserted: the statements hold along each fixed sequence of local alternatives, which is the exact strength of Definition 26, and the genuinely uniform version would require the local asymptotic normality expansion along arbitrary bounded sequences  $h_n$ , which Proposition 10 as stated does not provide. Remark 44 says the same thing where it matters.

## Locally uniform confidence sets

The first consequence proper is the inferential payoff, and it is the one that distinguishes local regularity from Proposition 15 in practice.

**Corollary 43** (Locally uniform Wald confidence sets). Let  $\alpha \in (0, 1)$ , let  $\chi_{d,1-\alpha}^2$  be the  $(1 - \alpha)$  quantile of the chi-squared law with  $d$  degrees of freedom, and define the random subset of  $M$

$$\mathcal{C}_{n,1-\alpha} := \text{Exp}_{\hat{\theta}_n}(\{\mathbf{y} \in T_{\hat{\theta}_n}M : n G_{\hat{\theta}_n}(\mathbf{y}, \mathbf{y}) \leq \chi_{d,1-\alpha}^2\}), \quad (2.81)$$

which is well defined on an event of probability tending to one. Then, for every  $\mathbf{h} \in T_{\theta_0}M$ ,

$$\mathbb{P}_{\theta_n, \mathbf{h}}^n (\theta_n, \mathbf{h} \in \mathcal{C}_{n, 1-\alpha}) \longrightarrow 1 - \alpha. \quad (2.82)$$

In particular the coverage is asymptotically correct not only at  $\theta_0$ , which is the case  $\mathbf{h} = \mathbf{0}$  and needs no more probabilistic input than Proposition 15, but along every sequence of local alternatives at the estimation scale  $n^{-\frac{1}{2}}$ .

*Proof.* Fix  $\mathbf{h}$  and work under  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$ , which by Lemma 27 is contiguous to  $\mathbb{P}_{\theta_0}^n$ ; all the  $o_{\mathbb{P}}(1)$  and  $O_{\mathbb{P}}(1)$  statements of Proposition 15 and Proposition 24 therefore remain valid. In particular  $\mathbb{P}_{\theta_n, \mathbf{h}}^n(\mathcal{A}_n) \rightarrow 1$ ,  $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$  in probability and  $\sqrt{n}\hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$ .

*Well definedness.* By Lemma 34 and the continuous mapping theorem,  $\mathbf{G}(\hat{\mathbf{u}}_n) \rightarrow \mathbf{G}$  in probability, so that  $\lambda_{\min}(\mathbf{G}(\hat{\mathbf{u}}_n))$  is bounded away from zero in probability. For any  $\mathbf{y} \in T_{\hat{\theta}_n}M$ , the vector  $\mathbf{z} := \Gamma_{\hat{\mathbf{u}}_n}(\mathbf{y})$  has the same norm as  $\mathbf{y}$ ,  $\Gamma_{\hat{\mathbf{u}}_n}$  being a linear isometry, and its coordinates in  $\mathbf{e}_1, \dots, \mathbf{e}_d$  are the coordinates of  $\mathbf{y}$  in the frame transported along the radial geodesic; hence, by (2.67),

$$G_{\hat{\theta}_n}(\mathbf{y}, \mathbf{y}) = \mathbf{z}^T \mathbf{G}(\hat{\mathbf{u}}_n) \mathbf{z} \geq \lambda_{\min}(\mathbf{G}(\hat{\mathbf{u}}_n)) |\mathbf{y}|^2. \quad (2.83)$$

The set inside (2.81) is therefore contained in a ball of radius  $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$  about  $\mathbf{0}$ . Meanwhile the radius of the domain  $\mathcal{E}_{\hat{\theta}_n}$  of  $\text{Exp}_{\hat{\theta}_n}$ , namely  $\min\left(\sigma_{\min}(\hat{\mathbf{W}}_n), \hat{\sigma}_n^2\right)$ , is a continuous function of the point  $\hat{\theta}_n \in M$ , as observed in the proof of Lemma 21, and therefore converges in probability to  $\rho > 0$  by Theorem 11. Hence, with probability tending to one, that set is a subset of  $\mathcal{E}_{\hat{\theta}_n}$  and  $\mathcal{C}_{n, 1-\alpha}$  is defined.

*Reduction to a quadratic form.* Both  $\hat{\theta}_n$  and  $\theta_n, \mathbf{h}$  lie in  $\mathcal{V}$  with probability tending to one, so part 1 of Lemma 21 makes  $\mathbf{y}_n := \text{Exp}_{\hat{\theta}_n}^{-1}(\theta_n, \mathbf{h})$  well defined there, and the injectivity of  $\text{Exp}_{\hat{\theta}_n}$  on  $\mathcal{E}_{\hat{\theta}_n}$  shows that, on an event of probability tending to one, membership  $\theta_n, \mathbf{h} \in \mathcal{C}_{n, 1-\alpha}$  is equivalent to

$$n G_{\hat{\theta}_n}(\mathbf{y}_n, \mathbf{y}_n) \leq \chi_{d, 1-\alpha}^2. \quad (2.84)$$

Put  $\mathbf{z}_n := \Gamma_{\hat{\mathbf{u}}_n}(\mathbf{y}_n) \in T_{\theta_0}M$ , so that the first equality of (2.83), applied with  $\mathbf{y} = \mathbf{y}_n$ , reads  $G_{\hat{\theta}_n}(\mathbf{y}_n, \mathbf{y}_n) = \mathbf{z}_n^T \mathbf{G}(\hat{\mathbf{u}}_n) \mathbf{z}_n$ .

*The limit.* Apply Lemma 22 with  $\mathbf{u} = \hat{\mathbf{u}}_n$  and  $\mathbf{w} = \frac{\mathbf{h}}{\sqrt{n}}$ , both in  $B_r$  with probability tending to one, noting that  $F\left(\hat{\mathbf{u}}_n, \frac{\mathbf{h}}{\sqrt{n}}\right) = \mathbf{z}_n$  by (2.38):

$$|\sqrt{n}\mathbf{z}_n - (\mathbf{h} - \sqrt{n}\hat{\mathbf{u}}_n)| \leq \sqrt{n} C |\hat{\mathbf{u}}_n| \left| \frac{\mathbf{h}}{\sqrt{n}} - \hat{\mathbf{u}}_n \right| = \frac{C}{\sqrt{n}} |\sqrt{n}\hat{\mathbf{u}}_n| |\mathbf{h} - \sqrt{n}\hat{\mathbf{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right),$$

so that  $\sqrt{n}\mathbf{z}_n = -(\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h}) + o_{\mathbb{P}}(1)$ . By (2.58),  $\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$  under  $\mathbb{P}_{\theta_n, \mathbf{h}}^n$ , and this law is symmetric, so  $\sqrt{n}\mathbf{z}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ . Combining with  $\mathbf{G}(\hat{\mathbf{u}}_n) \rightarrow \mathbf{G}$  in probability, Slutsky's lemma and the continuous mapping theorem applied to  $(\mathbf{z}, \mathbf{A}) \mapsto \mathbf{z}^T \mathbf{A} \mathbf{z}$  give, through the first equality of (2.83),

$$n G_{\hat{\theta}_n}(\mathbf{y}_n, \mathbf{y}_n) = (\sqrt{n}\mathbf{z}_n)^T \mathbf{G}(\hat{\mathbf{u}}_n) (\sqrt{n}\mathbf{z}_n) \rightsquigarrow \mathbf{Z}^T \mathbf{G} \mathbf{Z}, \quad \mathbf{Z} \sim N(\mathbf{0}, \mathbf{G}^{-1}),$$

and  $\mathbf{Z}^T \mathbf{G} \mathbf{Z} = \|\mathbf{G}^{\frac{1}{2}} \mathbf{Z}\|^2 \sim \chi_d^2$  because  $\mathbf{G}^{\frac{1}{2}} \mathbf{Z} \sim N(\mathbf{0}, \mathbf{I}_d)$ . The distribution function of  $\chi_d^2$  being continuous, the probability of (2.84) converges to  $\mathbb{P}(\chi_d^2 \leq \chi_{d, 1-\alpha}^2) = 1 - \alpha$ , which is (2.82).  $\square$

**Remark 44.** Three comments.

**1. The region is a statistic.** Nothing in (2.81) refers to  $\theta_0$ : the exponential map and the information operator are evaluated at  $\hat{\theta}_n$ , and Section 2.3 gives both in closed form, so  $\mathcal{C}_{n, 1-\alpha}$  can be computed from the data alone.

**2. Only local regularity is used.** The proof uses about  $\hat{\theta}_n$  only that it is locally regular at  $\theta_0$  with limit law  $N(\mathbf{0}, \mathbf{G}^{-1})$ . Localizability is part of Definition 26, and consistency in probability follows, since local regularity forces  $\sqrt{n}\hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$  and hence  $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$  in probability. Corollary 43 therefore holds verbatim for any locally

regular sequence with that limit law,  $\hat{\theta}_n$  being replaced by it in (2.81) and  $\mathcal{A}_n$  by the corresponding localization event; by (2.59) these are exactly the locally regular sequences that attain the convolution bound.

**3. What “locally uniform” does and does not mean.** Statement (2.82) is made along each fixed sequence of local alternatives, which is the exact strength of Definition 26 and is already enough to rule out the failure exhibited in Remark 30. Indeed, at  $\theta_0 = \theta_*$  the Hodges-type estimator is pinned to  $\theta_*$  with  $\mathbb{P}_{\theta_{n,h}}^n$ -probability tending to one, so that on that event the analogous region is the *deterministic* set  $\text{Exp}_{\theta_*}(\{\mathbf{v} : n G_{\theta_*}(\mathbf{v}, \mathbf{v}) \leq \chi_{d,1-\alpha}^2\})$ , which by the injectivity of  $\text{Exp}_{\theta_*}$  contains  $\theta_{n,h} = \text{Exp}_{\theta_*}\left(\frac{\mathbf{h}}{\sqrt{n}}\right)$  precisely when  $G(\mathbf{h}, \mathbf{h}) \leq \chi_{d,1-\alpha}^2$ , a condition free of  $n$ . Its coverage of  $\theta_{n,h}$  therefore tends to 1 when  $G(\mathbf{h}, \mathbf{h}) \leq \chi_{d,1-\alpha}^2$  and to 0 otherwise, and equals  $1 - \alpha$  in the limit for no  $\mathbf{h}$  at all,  $\mathbf{h} = \mathbf{0}$  included. The genuinely uniform statement,  $\sup_{|\mathbf{h}| \leq c} \left| \mathbb{P}_{\theta_{n,h}}^n(\theta_{n,h} \in \mathcal{C}_{n,1-\alpha}) - (1 - \alpha) \right| \rightarrow 0$ , would require the expansion (2.49) along arbitrary bounded sequences  $\mathbf{h}_n$  rather than along fixed  $\mathbf{h}$ , which Proposition 10 as stated does not provide; we do not need it anywhere.

## The fitted covariance matrix

The second consequence records what all of this says about the object the model is really about, namely the fitted covariance matrix  $\hat{\Sigma}_n$  of (1.43), which, unlike  $\hat{\theta}_n$  itself, is an honest element of the fixed vector space  $\text{Sym}(p)$ . It is the instance  $\varphi = \Sigma$  of item 4 above, and the linear map  $\mathcal{D}_0$  of (2.63) plays the role of  $d\varphi_{\theta_0}$ : it is the differential  $d\Sigma_{\theta_0}$  of the map  $\Sigma$  of (1.6), read through the exponential chart.

**Corollary 45** (Covariance level central limit theorem). Let  $1 \leq q < p$  and  $\theta_0 \in M$ .

1. Under  $\mathbb{P}_{\theta_0}^n$  the fitted covariance matrix is asymptotically linear in  $\text{Sym}(p)$  with influence function  $\mathcal{D}_0 \circ \text{IF}$ ,

$$\sqrt{n} \left( \hat{\Sigma}_n - \Sigma_0 \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathcal{D}_0(\text{IF}(\mathbf{X}_i)) + o_{\mathbb{P}}(1), \quad (2.85)$$

and consequently

$$\sqrt{n} \left( \hat{\Sigma}_n - \Sigma_0 \right) \rightsquigarrow \mathcal{D}_0(\mathbf{Z}), \quad \mathbf{Z} \sim N(\mathbf{0}, G^{-1}) \text{ on } T_{\theta_0}M. \quad (2.86)$$

2. The map  $\mathcal{D}_0$  is injective, and, in the notation of Lemma 47, its image is

$$\mathcal{D}_0(T_{\theta_0}M) = \left\{ \mathbf{J}_0 \begin{pmatrix} \mathbf{M}_{11} & \mathbf{M}_{21}^T \\ \mathbf{M}_{21} & \beta \mathbf{I}_{p-q} \end{pmatrix} \mathbf{J}_0^T : \mathbf{M}_{11} \in \text{Sym}(q), \mathbf{M}_{21} \in \mathbb{R}^{(p-q) \times q}, \beta \in \mathbb{R} \right\}, \quad (2.87)$$

where  $\mathbf{J}_0 := [\mathbf{U}_0 \ \mathbf{U}_0^\perp] \in O(p)$  as in (2.94) below. Hence the limit in (2.86) is a centred Gaussian law carried by a  $d$ -dimensional subspace of  $\text{Sym}(p)$ , namely the tangent space at  $\Sigma_0$  to the immersed submanifold  $\Sigma(M)$  of Proposition 1; it is degenerate in  $\text{Sym}(p)$  exactly when  $p - q \geq 2$ , the codimension being

$$\frac{p(p+1)}{2} - d = \frac{(p-q)(p-q+1)}{2} - 1 = \frac{(p-q+2)(p-q-1)}{2}. \quad (2.88)$$

3. The convergence (2.86) is locally regular: for every  $\mathbf{h} \in T_{\theta_0}M$ ,

$$\sqrt{n} \left( \hat{\Sigma}_n - \Sigma(\theta_{n,h}) \right) \rightsquigarrow \mathcal{D}_0(\mathbf{Z}) \quad \text{under } \mathbb{P}_{\theta_{n,h}}^n, \quad (2.89)$$

with the same limit law, free of  $\mathbf{h}$ .

*Proof.* **1.** On  $\mathcal{A}_n$  we have  $\hat{\theta}_n = \text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n)$ , so (2.64) at  $\mathbf{u} = \hat{\mathbf{u}}_n$  is the exact identity

$$\hat{\Sigma}_n - \Sigma_0 = \mathcal{D}_0(\hat{\mathbf{u}}_n) + \mathbf{U}_{\hat{\mathbf{u}}_n} \mathbf{U}_{\hat{\mathbf{u}}_n}^T \quad \text{on } \mathcal{A}_n, \quad (2.90)$$

with no remainder, because  $\text{Exp}_{\theta_0}$  is affine in the horizontal lift. Since  $\iota$  is a linear isometry,  $\|U_{\hat{u}_n} U_{\hat{u}_n}^T\|_F \leq \|U_{\hat{u}_n}\| \|U_{\hat{u}_n}\|_F \leq \|U_{\hat{u}_n}\|_F^2 \leq |\hat{u}_n|^2$ , and  $|\hat{u}_n| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$  by Proposition 15; hence  $\sqrt{n} \|U_{\hat{u}_n} U_{\hat{u}_n}^T\|_F = O_{\mathbb{P}}(n^{-\frac{1}{2}}) = o_{\mathbb{P}}(1)$ . Multiplying (2.90) by  $\sqrt{n}$  therefore gives

$$\sqrt{n} (\hat{\Sigma}_n - \Sigma_0) = \mathcal{D}_0(\sqrt{n} \hat{u}_n) + o_{\mathbb{P}}(1) \quad \text{on } \mathcal{A}_n, \quad (2.91)$$

and the restriction to  $\mathcal{A}_n$  is harmless: the two sides of (2.91) differ only on  $\mathcal{A}_n^c$ , whose probability tends to zero by Proposition 24, so their difference is  $o_{\mathbb{P}}(1)$  and they have the same weak limits. Now (2.85) follows from (1.45) and the linearity of  $\mathcal{D}_0$ , and (2.86) from Proposition 15, Slutsky's lemma and the continuous mapping theorem applied to the linear, hence continuous, map  $\mathcal{D}_0$ .

**2.** If  $\mathcal{D}_0(\mathbf{h}) = \mathbf{0}$  then  $G(\mathbf{h}, \mathbf{h}) = 0$  by (1.27), whence  $\mathbf{h} = \mathbf{0}$  by Theorem 9; so  $\mathcal{D}_0$  is injective and its image has dimension  $d$ . That image is the one described in (2.87) by (2.97) of Lemma 48 below, together with part 3 of that lemma and Lemma 49: as  $\mathbf{h}$  ranges over  $T_{\theta_0}M$ , the triple  $(2\Psi + b_{\mathbf{h}}\mathbf{I}_q, \mathbf{C}\mathbf{D}_0, b_{\mathbf{h}})$  ranges over all of  $\text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}$ , the matrix  $\mathbf{D}_0$  being invertible. The image is that of  $d\Sigma_{\theta_0}$  because  $\mathcal{D}_0(\mathbf{h}) = \frac{d}{dt}|_{t=0} \Sigma(\text{Exp}_{\theta_0}(t\mathbf{h}))$  by (1.26) and  $t \mapsto \text{Exp}_{\theta_0}(t\mathbf{h})$  has velocity  $\mathbf{h}$  at  $t = 0$ . Reading (2.87) off, the only constraint that  $\mathcal{D}_0(T_{\theta_0}M)$  imposes on a symmetric matrix is that its trailing  $(p-q) \times (p-q)$  block, in the basis  $\mathbf{J}_0$ , be a multiple of the identity; the codimension is therefore  $\dim \text{Sym}(p-q) - 1 = \frac{(p-q)(p-q+1)}{2} - 1$ . Writing  $m := p-q$ , this is  $\frac{m^2+m-2}{2} = \frac{(m+2)(m-1)}{2}$ , and it vanishes if and only if  $m = 1$ , since  $m+2 \geq 3$ . The same number is  $\frac{p(p+1)}{2} - d$ : by (1.2),

$$\frac{p(p+1)}{2} - d = \frac{p(p+1)}{2} - pq - 1 + \frac{q(q-1)}{2} = \frac{p^2 - 2pq + q^2 + p - q - 2}{2} = \frac{m^2 + m - 2}{2}.$$

**3.** Fix  $\mathbf{h}$  and work under  $\mathbb{P}_{\theta_{n,h}}^n$ , which is contiguous to  $\mathbb{P}_{\theta_0}^n$  by Lemma 27; in particular  $\mathbb{P}_{\theta_{n,h}}^n(\mathcal{A}_n) \rightarrow 1$  and the  $O_{\mathbb{P}}$  statements above persist, as in the proof of Corollary 43. Applying (2.64) at  $\mathbf{u} = \frac{\mathbf{h}}{\sqrt{n}}$  gives the second exact identity

$$\Sigma(\theta_{n,h}) - \Sigma_0 = \frac{1}{\sqrt{n}} \mathcal{D}_0(\mathbf{h}) + \frac{1}{n} \mathbf{U}_{\mathbf{h}} \mathbf{U}_{\mathbf{h}}^T,$$

so subtracting it from (2.90) and multiplying by  $\sqrt{n}$ ,

$$\sqrt{n} (\hat{\Sigma}_n - \Sigma(\theta_{n,h})) = \mathcal{D}_0(\sqrt{n} \hat{u}_n - \mathbf{h}) + \sqrt{n} \mathbf{U}_{\hat{u}_n} \mathbf{U}_{\hat{u}_n}^T - \frac{1}{\sqrt{n}} \mathbf{U}_{\mathbf{h}} \mathbf{U}_{\mathbf{h}}^T \quad \text{on } \mathcal{A}_n.$$

The last term is deterministic and  $O(n^{-\frac{1}{2}})$ . By (2.58),  $\sqrt{n} \hat{u}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$  under  $\mathbb{P}_{\theta_{n,h}}^n$ ; in particular  $\sqrt{n} \hat{u}_n = O_{\mathbb{P}}(1)$  there, so the middle term is again  $O_{\mathbb{P}}(n^{-\frac{1}{2}})$  by the bound of part 1. The continuous mapping theorem applied to  $\mathcal{D}_0$  now gives (2.89), the limit being the image of  $N(\mathbf{0}, \mathbf{G}^{-1})$ , that is  $\mathcal{D}_0(\mathbf{Z})$ , and free of  $\mathbf{h}$ .  $\square$

**Remark 46** (What is used where). Part 1 of Corollary 45 uses nothing from Section 2.1 beyond the localization of Proposition 24, and even there any normal neighbourhood of  $\theta_0$  would serve in place of  $\mathcal{V}$ . Its proof invokes Proposition 15, the exact expansion (2.64) and the continuous mapping theorem, and no more; neither the parallel transport of Proposition 19, nor the transport comparison of Lemma 22, nor the contiguity of Lemma 27, nor the local regularity of Theorem 29 appears anywhere in it. It could have been stated at the end of Section 1.4; it is placed here only because part 3 belongs here, and because it is the covariance level reading of the whole chapter.

The reason no transport is needed is the converse of the phenomenon that made Section 2.1 necessary in the first place. The intrinsic error  $\text{Exp}_{\theta_{n,h}}^{-1}(\hat{\theta}_n)$  of Definition 26 lives in  $T_{\theta_{n,h}}M$ , a vector space that *moves with  $n$* , which is exactly why  $\Gamma_{\mathbf{h}/\sqrt{n}}$  has to be inserted before any limit can be taken. The covariance level error  $\hat{\Sigma}_n - \Sigma(\theta)$ , by contrast, is a difference of two elements of the single fixed vector space  $\text{Sym}(p)$ , for every  $\theta$  at once; there is nothing to transport, the delta method applies verbatim, and curvature is invisible because  $\Sigma(M) \subseteq \text{Sym}(p)$  is

being viewed extrinsically. That the delta method is here an identity rather than an approximation is (2.64): the quadratic remainder is written out and cancelled, so (2.89) is a sharper form of the general (2.62).

What part 3 costs is correspondingly small: given (2.58), which is the substance of Theorem 29 and does use Lemma 27 and Le Cam's third lemma, the passage to (2.89) is three lines of exact algebra. One should resist reading more into part 3 than it says: it asserts that the *maximum likelihood* fitted covariance is locally regular as an estimator of  $\Sigma(\theta)$ , and nothing about other estimators.

This is the place to close the account opened at the start of the section. Theorem 29 is a statement about *other parameters* than  $\theta_0$ , and none of the geometry it rests on — Lemma 16, Proposition 19 and Lemmas 21 and 22, which are results about  $(M, g)$  in their own right, independent of any statistics — was needed by Proposition 15, which lives entirely inside the single vector space  $T_{\theta_0}M$ . They are what the two-base-point comparison costs, and Lemma 27 is the one probabilistic ingredient added alongside them. In the terminology of Sun et al. (2026), Theorem 29 verifies their Definition III.5 for the probabilistic principal component analysis model. We have proved it through the chart reduction of Lemma 28 and the classical Euclidean results of van der Vaart (1998), rather than by citing the manifold version, because Lemma 22 makes that reduction exact, and because of the localization of the curvature recorded in Remark 23.

## 2.3 The Fisher Information Operator in Closed Form

Proposition 15 identifies the limiting covariance of the maximum likelihood estimator as  $G_{\theta_0}^{-1}$ , and Theorem 29 shows that this limit is not an accident of the point  $\theta_0$ ; but nothing so far says what  $G_{\theta_0}^{-1}$  is. This section computes it. The only inputs are (1.26), (1.25) and (1.27) of Section 1.3, the spectral decomposition (2.92) of Lemma 47 and the horizontal lift (2.2); no asymptotics are used until Theorem 52, which merely reads Proposition 15 in the coordinates constructed here.

Two features of the answer deserve to be announced. First, the whole calculation is a  $2 \times 2$  block computation in the singular basis of  $\mathbf{W}_0$ , and it produces a set of coordinates in which  $G_{\theta_0}$  is block diagonal with blocks of size 1, 1 and  $q + 1$ ; the only genuinely coupled block is the one that mixes the  $q$  leading directions with the noise variance, and it is inverted by a one-line Schur complement. Second, no assumption whatsoever is made on the multiplicities of the singular values of  $\mathbf{W}_0$ : the formulas below are valid when  $\sigma_1(\mathbf{W}_0) = \dots = \sigma_q(\mathbf{W}_0)$ , a case in which individual eigenvectors of  $\mathbf{S}_n$  are not asymptotically normal at all. This is the quantitative form of the third comment of Remark 25.

### Block coordinates at the true parameter

Everything rests on the spectral decomposition of  $\Sigma_0$ , which the first chapter computes eigenvalue by eigenvalue in (1.39) but does not assemble into a basis. We assemble it now.

**Lemma 47** (Spectral decomposition of the true covariance). Fix a singular value decomposition  $\mathbf{W}_0 = \mathbf{U}_0 \mathbf{D}_0 \mathbf{V}_0^T$ , in which  $\mathbf{U}_0 \in \mathbb{R}^{p \times q}$  has orthonormal columns,  $\mathbf{V}_0 \in O(q)$ , and  $\mathbf{D}_0 = \text{diag}(\sigma_1(\mathbf{W}_0), \dots, \sigma_q(\mathbf{W}_0))$  has strictly positive diagonal because  $\text{rank}(\mathbf{W}_0) = q$ . Put  $\mathbf{P}_0 := \mathbf{U}_0 \mathbf{U}_0^T$ , the orthogonal projection onto the column space of  $\mathbf{W}_0$ . Then

$$\Sigma_0 = \mathbf{U}_0 (\mathbf{D}_0^2 + \sigma_0^2 \mathbf{I}_q) \mathbf{U}_0^T + \sigma_0^2 (\mathbf{I}_p - \mathbf{P}_0), \quad (2.92)$$

the two summands acting on orthogonal complementary subspaces; this is the spectral decomposition realizing the eigenvalues (1.39), and the gap separating the signal eigenvalues from the noise eigenvalue is

$$\lambda_q(\Sigma_0) - \lambda_{q+1}(\Sigma_0) = \sigma_q(\mathbf{W}_0)^2 > 0. \quad (2.93)$$

Passing to another representative  $(\mathbf{W}_0 \mathbf{Q}, \sigma_0^2)$  of  $\theta_0$ , with  $\mathbf{Q} \in O(q)$ , leaves  $\mathbf{U}_0$ ,  $\mathbf{D}_0$  and  $\mathbf{P}_0$  unchanged and replaces  $\mathbf{V}_0$  by  $\mathbf{Q}^T \mathbf{V}_0$ .

*Proof.* We have  $\mathbf{W}_0 \mathbf{W}_0^T = \mathbf{U}_0 \mathbf{D}_0^2 \mathbf{U}_0^T$  and  $\mathbf{I}_p = \mathbf{P}_0 + (\mathbf{I}_p - \mathbf{P}_0)$ , whence (2.92); the ranges of  $\mathbf{P}_0$  and  $\mathbf{I}_p - \mathbf{P}_0$  are orthogonal complements. Equation (2.93) follows from (1.39) because  $\sigma_q(\mathbf{W}_0) = \sigma_{\min}(\mathbf{W}_0) > 0$ . The last sentence is immediate from  $\mathbf{W}_0 \mathbf{Q} = \mathbf{U}_0 \mathbf{D}_0 (\mathbf{Q}^T \mathbf{V}_0)^T$ .  $\square$

We fix the singular value decomposition  $\mathbf{W}_0 = \mathbf{U}_0 \mathbf{D}_0 \mathbf{V}_0^T$  of Lemma 47 once and for all, and we complete  $\mathbf{U}_0$  to an orthogonal matrix: we let  $\mathbf{U}_0^\perp \in \mathbb{R}^{p \times (p-q)}$  have orthonormal columns spanning the orthogonal complement of the column space of  $\mathbf{W}_0$ , so that

$$\mathbf{J}_0 := [\mathbf{U}_0 \ \mathbf{U}_0^\perp] \in O(p), \quad \mathbf{I}_p - \mathbf{P}_0 = \mathbf{U}_0^\perp (\mathbf{U}_0^\perp)^T. \quad (2.94)$$

We abbreviate the signal eigenvalues of  $\Sigma_0$  supplied by Lemma 47 as

$$\tau_j := \lambda_j(\Sigma_0) = \sigma_j(\mathbf{W}_0)^2 + \sigma_0^2 > \sigma_0^2 \quad (j = 1, \dots, q), \quad \mathbf{T}_0 := \mathbf{D}_0^2 + \sigma_0^2 \mathbf{I}_q = \text{diag}(\tau_1, \dots, \tau_q). \quad (2.95)$$

The letters  $\mathbf{B}, \mathbf{C}, \Psi, \mathbf{y}, \mathbf{z}, \mathbf{J}_0, \mathbf{T}_0, \tau_j, \kappa, \mathbf{K}$  and  $\mathcal{G}$  are local to this section.

**Lemma 48** (Block form of the differential). Let  $\mathbf{h} \in T_{\theta_0} M$  have horizontal lift  $\iota(\mathbf{h}) = (\mathbf{U}_h, b_h)$  as in (2.3), and set

$$\mathbf{B} := \mathbf{U}_0^T \mathbf{U}_h \mathbf{V}_0 \in \mathbb{R}^{q \times q}, \quad \mathbf{C} := (\mathbf{U}_0^\perp)^T \mathbf{U}_h \mathbf{V}_0 \in \mathbb{R}^{(p-q) \times q}, \quad \Psi := \frac{1}{2} (\mathbf{B} \mathbf{D}_0 + \mathbf{D}_0 \mathbf{B}^T) \in \text{Sym}(q). \quad (2.96)$$

Then:

1.  $\mathbf{J}_0^T \Sigma_0 \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0 & \mathbf{0} \\ \mathbf{0} & \sigma_0^2 \mathbf{I}_{p-q} \end{pmatrix}$  and  $\mathbf{J}_0^T \Sigma_0^{-1} \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0^{-1} & \mathbf{0} \\ \mathbf{0} & \sigma_0^{-2} \mathbf{I}_{p-q} \end{pmatrix}$ .
2.  $\mathbf{U}_h = \mathbf{U}_0 \mathbf{B} \mathbf{V}_0^T + \mathbf{U}_0^\perp \mathbf{C} \mathbf{V}_0^T$  and  $|\mathbf{h}|^2 = \|\mathbf{B}\|_F^2 + \|\mathbf{C}\|_F^2 + b_h^2$ .
3.  $\mathbf{D}_0 \mathbf{B} \in \text{Sym}(q)$ ; conversely every triple  $(\mathbf{B}, \mathbf{C}, b)$  with  $\mathbf{D}_0 \mathbf{B}$  symmetric arises from exactly one  $\mathbf{h} \in T_{\theta_0} M$ .
4. In the basis (2.94),

$$\mathbf{J}_0^T \mathcal{D}_0(\mathbf{h}) \mathbf{J}_0 = \begin{pmatrix} 2\Psi + b_h \mathbf{I}_q & \mathbf{D}_0 \mathbf{C}^T \\ \mathbf{C} \mathbf{D}_0 & b_h \mathbf{I}_{p-q} \end{pmatrix}, \quad (2.97)$$

with  $\mathcal{D}_0(\mathbf{h}) = D_h \Sigma(\theta_0)$  as in (2.63).

*Proof.* 1. Substitute  $\mathbf{P}_0 = \mathbf{U}_0 \mathbf{U}_0^T$  and  $\mathbf{I}_p - \mathbf{P}_0 = \mathbf{U}_0^\perp (\mathbf{U}_0^\perp)^T$  into (2.92) and use  $\mathbf{U}_0^T \mathbf{U}_0 = \mathbf{I}_q$ ,  $(\mathbf{U}_0^\perp)^T \mathbf{U}_0^\perp = \mathbf{I}_{p-q}$  and  $\mathbf{U}_0^T \mathbf{U}_0^\perp = \mathbf{0}$ .

2. Since  $\mathbf{J}_0 \in O(p)$  and  $\mathbf{V}_0 \in O(q)$ , the matrix  $\mathbf{J}_0^T \mathbf{U}_h \mathbf{V}_0$  has  $\mathbf{B}$  and  $\mathbf{C}$  stacked as its blocks, which is the first identity; the Frobenius norm is invariant under multiplication by orthogonal matrices on either side, so  $\|\mathbf{U}_h\|_F^2 = \|\mathbf{B}\|_F^2 + \|\mathbf{C}\|_F^2$ , and the second identity is (2.4).

3. By part 3 of Proposition 1 the pair  $(\mathbf{U}_h, b_h)$  is horizontal at  $(\mathbf{W}_0, \sigma_0^2)$  if and only if  $\mathbf{U}_h^T \mathbf{W}_0 = \mathbf{W}_0^T \mathbf{U}_h$ . Using part 2 and  $\mathbf{W}_0 = \mathbf{U}_0 \mathbf{D}_0 \mathbf{V}_0^T$ ,

$$\mathbf{W}_0^T \mathbf{U}_h = \mathbf{V}_0 \mathbf{D}_0 \mathbf{U}_0^T (\mathbf{U}_0 \mathbf{B} \mathbf{V}_0^T + \mathbf{U}_0^\perp \mathbf{C} \mathbf{V}_0^T) = \mathbf{V}_0 (\mathbf{D}_0 \mathbf{B}) \mathbf{V}_0^T,$$

which is symmetric if and only if  $\mathbf{D}_0 \mathbf{B}$  is. Conversely, given such a triple, part 2 defines  $\mathbf{U}_h$ , the displayed identity makes  $(\mathbf{U}_h, b)$  horizontal, and  $\mathbf{h} := d\pi_{(\mathbf{W}_0, \sigma_0^2)}((\mathbf{U}_h, b))$  is the unique preimage because  $\iota$  is a bijection onto  $H_{(\mathbf{W}_0, \sigma_0^2)}$ .

4. By (2.63),  $\mathcal{D}_0(\mathbf{h}) = \mathbf{U}_h \mathbf{W}_0^T + \mathbf{W}_0 \mathbf{U}_h^T + b_h \mathbf{I}_p$ , and by part 2,

$$U_h W_0^T = (U_0 B V_0^T + U_0^\perp C V_0^T) V_0 D_0 U_0^T = U_0 B D_0 U_0^T + U_0^\perp C D_0 U_0^T,$$

so that  $J_0^T U_h W_0^T J_0 = \begin{pmatrix} B D_0 & 0 \\ C D_0 & 0 \end{pmatrix}$ . Adding the transpose of this matrix and  $b_h I_p$ , and using  $B D_0 + (B D_0)^T = 2\Psi$ , gives (2.97).  $\square$

The three blocks have direct meanings, and it is worth recording them before any computation. By (2.97) the matrix  $2\Psi + b_h I_q$  is the first order change of  $\Sigma$  inside the signal block, the matrix  $C D_0$  is its change in the signal-noise block, and  $b_h I_{p-q}$  is its change in the noise block — where the model forces the change to be a *multiple of the identity*, which is the whole source of the efficiency gain computed below, and which is the only constraint, as (2.87) recorded. Note also that  $\Psi$ , not  $B$ , is the natural unknown: the map  $B \mapsto \Psi$  is exactly the operation that quotients out the  $\frac{q(q-1)}{2}$  directions along which  $\Sigma$  does not move.

**Lemma 49** (Linear coordinates on  $T_{\theta_0} M$ ). The map

$$\kappa : T_{\theta_0} M \longrightarrow \text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}, \quad \kappa(\mathbf{h}) := (\Psi, \mathbf{C}, b_h) \quad (2.98)$$

of (2.96) is a linear isomorphism.

*Proof.* Linearity is clear:  $\iota$  is linear by (2.2), and  $U_h \mapsto (B, C) \mapsto \Psi$  is a composition of linear maps.

For injectivity, suppose  $\kappa(\mathbf{h}) = \mathbf{0}$ . Then  $b_h = 0$  and  $\mathbf{C} = \mathbf{0}$ , while  $\Psi = \mathbf{0}$  reads  $B D_0 = -D_0 B^T$ . Since  $\text{rank}(\mathbf{W}_0) = q$ , all  $\sigma_j(\mathbf{W}_0)$  are strictly positive and  $D_0$  is invertible, so part 3 of Lemma 48, that is  $D_0 B = B^T D_0$ , gives  $B^T = D_0 B D_0^{-1}$ . Substituting this into  $B D_0 = -D_0 B^T$  yields  $B D_0 = -D_0^2 B D_0^{-1}$ , and multiplying on the right by  $D_0$ ,

$$B D_0^2 + D_0^2 B = \mathbf{0}.$$

Both  $D_0^2$  factors are diagonal, so the  $(i, j)$  entry of the left hand side is  $(\sigma_i(\mathbf{W}_0)^2 + \sigma_j(\mathbf{W}_0)^2) B_{ij}$ , and the bracket is strictly positive; hence  $B = \mathbf{0}$ . By part 2 of Lemma 48 this forces  $U_h = \mathbf{0}$ , so  $\iota(\mathbf{h}) = (\mathbf{0}, 0)$  and  $\mathbf{h} = \mathbf{0}$ .

For surjectivity it suffices to compare dimensions,  $\kappa$  being injective and linear:

$$\dim(\text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}) = \frac{q(q+1)}{2} + (p-q)q + 1 = pq + 1 - \frac{q(q-1)}{2} = d$$

by (1.2).  $\square$

Lemma 49 lets us treat  $(\Psi, \mathbf{C}, b)$  as coordinates on  $T_{\theta_0} M$ . We use throughout the *scalar* coordinates

$$(\Psi_{ij})_{1 \leq i \leq j \leq q}, \quad (C_{ki})_{1 \leq k \leq p-q, 1 \leq i \leq q}, \quad b, \quad (2.99)$$

which identify the target of  $\kappa$  with  $\mathbb{R}^d$ , and we let  $\mathcal{G}$  denote the  $d \times d$  symmetric matrix representing the Fisher form in them, that is

$$G_{\theta_0}(\mathbf{h}_1, \mathbf{h}_2) = \kappa(\mathbf{h}_1)^T \mathcal{G} \kappa(\mathbf{h}_2) \quad \text{for all } \mathbf{h}_1, \mathbf{h}_2 \in T_{\theta_0} M. \quad (2.100)$$

Note that  $\kappa$  is *not* a  $g_{\theta_0}$ -isometry, so  $\mathcal{G}$  is not the matrix  $\mathbf{G}$  of (2.45); the relation between the two is recorded in the proof of Theorem 52 and is the only place where it is needed.

## The Fisher form and its inverse

**Proposition 50** (Explicit Fisher form). For every  $\mathbf{h} \in T_{\theta_0}M$ , in the notation (2.96),

$$G_{\theta_0}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr} \left[ \left( \mathbf{T}_0^{-1} (2\Psi + b_{\mathbf{h}} \mathbf{I}_q) \right)^2 \right] + \frac{1}{\sigma_0^2} \operatorname{tr} \left( \mathbf{T}_0^{-1} \mathbf{D}_0^2 \mathbf{C}^T \mathbf{C} \right) + \frac{(p-q)b_{\mathbf{h}}^2}{2\sigma_0^4}, \quad (2.101)$$

and, in the scalar coordinates (2.99),

$$G_{\theta_0}(\mathbf{h}, \mathbf{h}) = \underbrace{\sum_{i=1}^q \frac{(2\Psi_{ii} + b_{\mathbf{h}})^2}{2\tau_i^2}}_{\text{coupled block}} + \frac{(p-q)b_{\mathbf{h}}^2}{2\sigma_0^4} + \underbrace{\sum_{1 \leq i < j \leq q} \frac{4\Psi_{ij}^2}{\tau_i \tau_j}}_{\Psi \text{ off-diagonal}} + \underbrace{\sum_{i=1}^q \sum_{k=1}^{p-q} \left( \frac{1}{\sigma_0^2} - \frac{1}{\tau_i} \right) \mathbf{C}_{ki}^2}_{\mathbf{C} \text{-block}}. \quad (2.102)$$

In particular  $\mathcal{G}$  is block diagonal with respect to the partition of (2.99) into the three groups  $\{\Psi_{ij}\}_{i < j}$ ,  $\{\mathbf{C}_{ki}\}$  and  $(\Psi_{11}, \dots, \Psi_{qq}, b)$ , with

$$\mathcal{G}_{\Psi_{ij}\Psi_{ij}} = \frac{4}{\tau_i \tau_j} \quad (i < j), \quad \mathcal{G}_{\mathbf{C}_{ki}\mathbf{C}_{ki}} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i} = \frac{\sigma_i(\mathbf{W}_0)^2}{\sigma_0^2 \tau_i}, \quad \mathcal{G}_{\mathbf{c}} = \begin{pmatrix} \Pi & \boldsymbol{\nu} \\ \boldsymbol{\nu}^T & \eta \end{pmatrix}, \quad (2.103)$$

the last being the coupled block in the ordering  $(\Psi_{11}, \dots, \Psi_{qq}, b)$ , where

$$\Pi := \operatorname{diag} (2\tau_1^{-2}, \dots, 2\tau_q^{-2}), \quad \boldsymbol{\nu} := (\tau_1^{-2}, \dots, \tau_q^{-2})^T \in \mathbb{R}^q, \quad \eta := \frac{1}{2} \sum_{i=1}^q \frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4}. \quad (2.104)$$

*Proof.* Write  $\mathbf{M} := 2\Psi + b_{\mathbf{h}} \mathbf{I}_q$  and  $b := b_{\mathbf{h}}$ . Conjugating by  $\mathbf{J}_0$  does not change traces, so by (1.27) with  $\mathbf{h}_1 = \mathbf{h}_2 = \mathbf{h}$  it suffices to compute the trace of the square of  $\mathbf{J}_0^T \Sigma_0^{-1} \mathcal{D}_0(\mathbf{h}) \mathbf{J}_0$ . Multiplying the two matrices of Lemma 48,

$$\mathbf{J}_0^T \Sigma_0^{-1} \mathcal{D}_0(\mathbf{h}) \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0^{-1} \mathbf{M} & \mathbf{T}_0^{-1} \mathbf{D}_0 \mathbf{C}^T \\ \sigma_0^{-2} \mathbf{C} \mathbf{D}_0 & \sigma_0^{-2} b \mathbf{I}_{p-q} \end{pmatrix}. \quad (2.105)$$

The two diagonal blocks of the square of (2.105) are

$$(\mathbf{T}_0^{-1} \mathbf{M})^2 + \sigma_0^{-2} \mathbf{T}_0^{-1} \mathbf{D}_0 \mathbf{C}^T \mathbf{C} \mathbf{D}_0 \quad \text{and} \quad \sigma_0^{-2} \mathbf{C} \mathbf{D}_0 \mathbf{T}_0^{-1} \mathbf{D}_0 \mathbf{C}^T + \sigma_0^{-4} b^2 \mathbf{I}_{p-q},$$

so, using the cyclic property of the trace and the fact that the diagonal matrices  $\mathbf{T}_0$  and  $\mathbf{D}_0$  commute,

$$\operatorname{tr} \left[ (\Sigma_0^{-1} \mathcal{D}_0(\mathbf{h}))^2 \right] = \operatorname{tr} \left[ (\mathbf{T}_0^{-1} \mathbf{M})^2 \right] + \frac{2}{\sigma_0^2} \operatorname{tr} (\mathbf{T}_0^{-1} \mathbf{D}_0^2 \mathbf{C}^T \mathbf{C}) + \frac{(p-q)b^2}{\sigma_0^4}.$$

Halving gives (2.101).

For (2.102), both  $\mathbf{T}_0^{-1}$  and  $\mathbf{D}_0^2$  are diagonal, so

$$\operatorname{tr} (\mathbf{T}_0^{-1} \mathbf{D}_0^2 \mathbf{C}^T \mathbf{C}) = \sum_{i=1}^q \frac{\sigma_i(\mathbf{W}_0)^2}{\tau_i} (\mathbf{C}^T \mathbf{C})_{ii} = \sum_{i=1}^q \frac{\sigma_i(\mathbf{W}_0)^2}{\tau_i} \sum_{k=1}^{p-q} \mathbf{C}_{ki}^2,$$

and  $\frac{\sigma_i(\mathbf{W}_0)^2}{\sigma_0^2 \tau_i} = \frac{\tau_i - \sigma_0^2}{\sigma_0^2 \tau_i} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i}$  by (2.95). For the first term,  $\mathbf{M}$  is symmetric, so

$$\operatorname{tr} \left[ (\mathbf{T}_0^{-1} \mathbf{M})^2 \right] = \sum_{i,j=1}^q \frac{\mathbf{M}_{ij} \mathbf{M}_{ji}}{\tau_i \tau_j} = \sum_{i=1}^q \frac{\mathbf{M}_{ii}^2}{\tau_i^2} + 2 \sum_{i < j} \frac{\mathbf{M}_{ij}^2}{\tau_i \tau_j},$$

and  $\mathbf{M}_{ii} = 2\Psi_{ii} + b$ ,  $\mathbf{M}_{ij} = 2\Psi_{ij}$  for  $i \neq j$ ; halving gives the first two groups of terms in (2.102).

Finally, (2.102) is a sum of three quadratic forms in *disjoint* sets of the scalar coordinates (2.99), which is exactly the assertion of block diagonality; reading off the coefficients gives the first two entries of (2.103). For the third, expanding the coupled part of (2.102),

$$\sum_{i=1}^q \frac{(2\Psi_{ii} + b)^2}{2\tau_i^2} + \frac{(p-q)b^2}{2\sigma_0^4} = \sum_{i=1}^q \frac{2\Psi_{ii}^2}{\tau_i^2} + 2b \sum_{i=1}^q \frac{\Psi_{ii}}{\tau_i^2} + b^2 \left( \frac{1}{2} \sum_{i=1}^q \frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4} \right),$$

whose matrix in the ordering  $(\Psi_{11}, \dots, \Psi_{qq}, b)$  is  $\mathcal{G}_c$  of (2.103): the coefficient of  $\Psi_{ii}^2$  is  $2\tau_i^{-2} = \Pi_{ii}$ , that of  $b \Psi_{ii}$  is  $2\tau_i^{-2} = 2\nu_i$ , and that of  $b^2$  is  $\eta$ .  $\square$

**Proposition 51** (Explicit inverse). The matrix  $\mathcal{G}$  of (2.100) is invertible, and  $\mathcal{G}^{-1}$  is block diagonal with respect to the same partition, with

$$(\mathcal{G}^{-1})_{\Psi_{ij}\Psi_{ij}} = \frac{\tau_i\tau_j}{4} \quad (i < j), \quad (\mathcal{G}^{-1})_{C_{ki}C_{ki}} = \frac{\sigma_0^2\tau_i}{\sigma_i(\mathbf{W}_0)^2} = \left( \frac{1}{\sigma_0^2} - \frac{1}{\tau_i} \right)^{-1}, \quad (2.106)$$

these two groups being diagonal, and, on the coupled block,

$$(\mathcal{G}^{-1})_{bb} = \frac{2\sigma_0^4}{p-q}, \quad (\mathcal{G}^{-1})_{\Psi_{ii}b} = -\frac{\sigma_0^4}{p-q}, \quad (\mathcal{G}^{-1})_{\Psi_{ii}\Psi_{jj}} = \delta_{ij} \frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}. \quad (2.107)$$

*Proof.*  $\mathcal{G}$  is positive definite because  $G_{\theta_0}$  is, by Theorem 9, and  $\kappa$  is a bijection; so it is invertible, and the inverse of a block diagonal matrix is block diagonal with the blocks inverted. The first two groups are diagonal by (2.103), and their inverses are the reciprocals, which is (2.106).

For the coupled block,  $\Pi^{-1} = \text{diag} \left( \frac{\tau_1^2}{2}, \dots, \frac{\tau_q^2}{2} \right)$  and

$$\Pi^{-1}\boldsymbol{\nu} = \left( \frac{\tau_i^2}{2} \cdot \frac{1}{\tau_i^2} \right)_{i=1}^q = \frac{1}{2}\mathbf{1}_q, \quad (2.108)$$

so the Schur complement of  $\Pi$  in  $\mathcal{G}_c$  is

$$\zeta := \eta - \boldsymbol{\nu}^T \Pi^{-1} \boldsymbol{\nu} = \eta - \frac{1}{2} \sum_{i=1}^q \frac{1}{\tau_i^2} = \frac{p-q}{2\sigma_0^4} \quad (2.109)$$

by (2.104); it is strictly positive because  $q < p$ , which is where the standing assumption  $q < p$  enters the inversion. The standard block inversion formula

$$\begin{pmatrix} \Pi & \boldsymbol{\nu} \\ \boldsymbol{\nu}^T & \eta \end{pmatrix}^{-1} = \begin{pmatrix} \Pi^{-1} + \zeta^{-1} \Pi^{-1} \boldsymbol{\nu} \boldsymbol{\nu}^T \Pi^{-1} & -\zeta^{-1} \Pi^{-1} \boldsymbol{\nu} \\ -\zeta^{-1} \boldsymbol{\nu}^T \Pi^{-1} & \zeta^{-1} \end{pmatrix}$$

together with (2.108) and (2.109) gives  $\zeta^{-1} = \frac{2\sigma_0^4}{p-q}$ , then  $-\zeta^{-1} (\Pi^{-1} \boldsymbol{\nu})_i = -\frac{2\sigma_0^4}{p-q} \cdot \frac{1}{2} = -\frac{\sigma_0^4}{p-q}$ , and finally

$$(\Pi^{-1})_{ij} + \zeta^{-1} (\Pi^{-1} \boldsymbol{\nu})_i (\Pi^{-1} \boldsymbol{\nu})_j = \delta_{ij} \frac{\tau_i^2}{2} + \frac{2\sigma_0^4}{p-q} \cdot \frac{1}{2} \cdot \frac{1}{2} = \delta_{ij} \frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)},$$

which is (2.107).  $\square$

**Theorem 52** (Theorem 15 in block coordinates). Let  $1 \leq q < p$  and  $\theta_0 = [\mathbf{W}_0, \sigma_0^2] \in M$ . Define, on the event  $\mathcal{A}_n$  of (2.42), the aligned error blocks

$$\begin{aligned} B_n &:= U_0^T \left( \hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0 \right) \mathbf{V}_0, & C_n &:= (U_0^\perp)^T \left( \hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0 \right) \mathbf{V}_0, \\ \Psi_n &:= \frac{1}{2} (B_n D_0 + D_0 B_n^T), & b_n &:= \hat{\sigma}_n^2 - \sigma_0^2, \end{aligned} \quad (2.110)$$

with  $\mathbf{Q}_n$  as in (2.43), and set  $(\Psi_n, C_n, b_n) := (\mathbf{0}, \mathbf{0}, 0)$  off  $\mathcal{A}_n$ . Then

$$\sqrt{n}(\Psi_n, \mathbf{C}_n, b_n) \rightsquigarrow N(\mathbf{0}, \mathcal{G}^{-1}) \quad \text{in } \text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}, \quad (2.111)$$

with  $\mathcal{G}^{-1}$  given by Proposition 51. Explicitly, writing  $\mathbf{Z} = (\mathbf{Z}_\Psi, \mathbf{Z}_C, Z_b)$  for the limit:

1. the  $(p-q)q$  entries of  $\mathbf{Z}_C$  are mutually independent, independent of  $(\mathbf{Z}_\Psi, Z_b)$ , and  $\text{Var}((\mathbf{Z}_C)_{ki}) = \frac{\sigma_0^2 \tau_i}{\sigma_i(\mathbf{W}_0)^2}$ ;
2. the  $\frac{q(q-1)}{2}$  strictly upper triangular entries of  $\mathbf{Z}_\Psi$  are mutually independent, independent of everything else, and  $\text{Var}((\mathbf{Z}_\Psi)_{ij}) = \frac{\tau_i \tau_j}{4}$  for  $i < j$ ;
3. the remaining  $q+1$  coordinates  $((\mathbf{Z}_\Psi)_{11}, \dots, (\mathbf{Z}_\Psi)_{qq}, Z_b)$  are jointly Gaussian with the covariances (2.107); in particular

$$\sqrt{n}(\hat{\sigma}_n^2 - \sigma_0^2) \rightsquigarrow N\left(0, \frac{2\sigma_0^4}{p-q}\right). \quad (2.112)$$

*Proof.* By (2.43) we have  $\iota(\hat{\mathbf{u}}_n) = (\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2)$  on  $\mathcal{A}_n$ , so by (2.96) the triple  $(\Psi_n, \mathbf{C}_n, b_n)$  of (2.110) is exactly  $\kappa(\hat{\mathbf{u}}_n)$  there; and both vanish off  $\mathcal{A}_n$  because  $\hat{\mathbf{u}}_n = \mathbf{0}$  there by (2.42). Hence  $(\Psi_n, \mathbf{C}_n, b_n) = \kappa(\hat{\mathbf{u}}_n)$  identically.

Let  $\mathbf{K}$  denote the matrix of  $\kappa$  with respect to the  $g_{\theta_0}$ -orthonormal basis  $\mathbf{e}_1, \dots, \mathbf{e}_d$  of (2.45) on the source and the scalar coordinates (2.99) on the target. Comparing (2.100) with  $\mathbf{G}_{jk} = G_{\theta_0}(\mathbf{e}_j, \mathbf{e}_k)$  gives  $\mathbf{G} = \mathbf{K}^T \mathcal{G} \mathbf{K}$ , so

$$\mathbf{K} \mathbf{G}^{-1} \mathbf{K}^T = \mathbf{K} (\mathbf{K}^T \mathcal{G} \mathbf{K})^{-1} \mathbf{K}^T = \mathbf{K} \mathbf{K}^{-1} \mathcal{G}^{-1} \mathbf{K}^{-T} \mathbf{K}^T = \mathcal{G}^{-1}. \quad (2.113)$$

By Proposition 15 we have  $\sqrt{n}\hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$  in the coordinates of that basis, so the continuous mapping theorem applied to the linear map  $\mathbf{K}$  and the identity (2.113) give (2.111).

The three assertions are then read off Proposition 51: a Gaussian vector whose covariance matrix is block diagonal has independent blocks, and within the first two groups the blocks are diagonal, so those coordinates are mutually independent. Statement (2.112) is the  $(b, b)$  entry of (2.107) together with  $b_n = \hat{\sigma}_n^2 - \sigma_0^2$ .  $\square$

## The score and the influence function in coordinates

The same block decomposition makes the score (1.25) and the influence operator (1.44) completely explicit. Throughout, for  $\mathbf{x} \in \mathbb{R}^p$  we write

$$\mathbf{y} := \mathbf{U}_0^T \mathbf{x} \in \mathbb{R}^q, \quad \mathbf{z} := (\mathbf{U}_0^\perp)^T \mathbf{x} \in \mathbb{R}^{p-q}, \quad (2.114)$$

so that, by part 1 of Lemma 48, when  $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma_0)$  the vectors  $\mathbf{U}_0^T \mathbf{X}$  and  $(\mathbf{U}_0^\perp)^T \mathbf{X}$  are independent, the first being  $N_q(\mathbf{0}, \mathbf{T}_0)$  and the second  $N_{p-q}(\mathbf{0}, \sigma_0^2 \mathbf{I}_{p-q})$ .

**Proposition 53** (Score and influence function in coordinates). Let  $\tilde{\mathbf{s}}(\mathbf{x}) \in \mathbb{R}^d$  be the coordinate vector of the score in the scalar coordinates (2.99), defined by  $s(\mathbf{x}; \theta_0)(\mathbf{h}) = \tilde{\mathbf{s}}(\mathbf{x})^T \kappa(\mathbf{h})$  for all  $\mathbf{h} \in T_{\theta_0} M$ ; it is related to the vector  $\mathbf{s}$  of (2.45) by  $\mathbf{s} = \mathbf{K}^T \tilde{\mathbf{s}}$ . Then

$$\tilde{\mathbf{s}}_{\Psi_{ii}} = \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2}, \quad \tilde{\mathbf{s}}_{\Psi_{ij}} = \frac{2\mathbf{y}_i \mathbf{y}_j}{\tau_i \tau_j} \quad (i < j), \quad \tilde{\mathbf{s}}_{C_{ki}} = \frac{\sigma_i(\mathbf{W}_0) \mathbf{y}_i \mathbf{z}_k}{\sigma_0^2 \tau_i}, \quad (2.115)$$

$$\tilde{\mathbf{s}}_b = \frac{1}{2} \left[ \sum_{i=1}^q \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2} + \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{\sigma_0^4} \right]. \quad (2.116)$$

Moreover  $\mathbb{E}[\tilde{\mathbf{s}}(\mathbf{X})] = \mathbf{0}$  and  $\mathbb{E}[\tilde{\mathbf{s}}(\mathbf{X}) \tilde{\mathbf{s}}(\mathbf{X})^T] = \mathcal{G}$ , the influence operator (1.44) has coordinate vector  $\kappa(\text{IF}(\mathbf{x})) = \mathcal{G}^{-1} \tilde{\mathbf{s}}(\mathbf{x})$ , and explicitly

$$\text{IF}_b(\mathbf{x}) = \frac{\mathbf{x}^T (\mathbf{I}_p - \mathbf{P}_0) \mathbf{x} - (p-q)\sigma_0^2}{p-q}, \quad \text{IF}_{C_{ki}}(\mathbf{x}) = \frac{\mathbf{y}_i \mathbf{z}_k}{\sigma_i(\mathbf{W}_0)}, \quad (2.117)$$

$$\text{IF}_{\Psi_{ij}}(\mathbf{x}) = \frac{\mathbf{y}_i \mathbf{y}_j}{2} \quad (i < j), \quad \text{IF}_{\Psi_{ii}}(\mathbf{x}) = \frac{1}{2} [\mathbf{y}_i^2 - \tau_i - \text{IF}_b(\mathbf{x})]. \quad (2.118)$$

*Proof.* The relation  $\mathbf{s} = \mathbf{K}^T \tilde{\mathbf{s}}$  is immediate from  $\mathbf{s}(\mathbf{x})^T \mathbf{h} = s(\mathbf{x}; \theta_0)(\mathbf{h}) = \tilde{\mathbf{s}}(\mathbf{x})^T \mathbf{K} \mathbf{h}$ , valid for all  $\mathbf{h}$ .

Abbreviate  $\mathbf{E} := \Sigma_0^{-1} [\mathbf{x} \mathbf{x}^T - \Sigma_0] \Sigma_0^{-1}$ , so that (1.25) reads  $s(\mathbf{x}; \theta_0)(\mathbf{h}) = \frac{1}{2} \text{tr} \{ \mathbf{E} \mathbf{D}_0(\mathbf{h}) \}$ . Since  $\mathbf{J}_0^T \mathbf{x} = (\mathbf{y}^T, \mathbf{z}^T)^T$  by (2.114), part 1 of Lemma 48 gives

$$\mathbf{J}_0^T \mathbf{E} \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0^{-1} (\mathbf{y} \mathbf{y}^T - \mathbf{T}_0) \mathbf{T}_0^{-1} & \sigma_0^{-2} \mathbf{T}_0^{-1} \mathbf{y} \mathbf{z}^T \\ \sigma_0^{-2} \mathbf{z} \mathbf{y}^T \mathbf{T}_0^{-1} & \sigma_0^{-4} (\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q}) \end{pmatrix} =: \begin{pmatrix} \mathbf{E}_{11} & \mathbf{E}_{12} \\ \mathbf{E}_{12}^T & \mathbf{E}_{22} \end{pmatrix}. \quad (2.119)$$

Pairing (2.119) with (2.97), the two off-diagonal contributions to the trace are equal, because  $\text{tr} (\mathbf{E}_{12}^T \mathbf{D}_0 \mathbf{C}^T) = \text{tr} ((\mathbf{E}_{12}^T \mathbf{D}_0 \mathbf{C}^T)^T) = \text{tr} (\mathbf{C} \mathbf{D}_0 \mathbf{E}_{12}) = \text{tr} (\mathbf{E}_{12} \mathbf{C} \mathbf{D}_0)$ , and therefore

$$2 s(\mathbf{x}; \theta_0)(\mathbf{h}) = \text{tr} (\mathbf{E}_{11} [2\Psi + b_h \mathbf{I}_q]) + \frac{2}{\sigma_0^2} \mathbf{z}^T \mathbf{C} \mathbf{D}_0 \mathbf{T}_0^{-1} \mathbf{y} + b_h \text{tr} \mathbf{E}_{22}. \quad (2.120)$$

Now read off the coefficient of each scalar coordinate. The matrix  $\Psi$  is symmetric, so  $\text{tr} (\mathbf{E}_{11} \Psi) = \sum_i (\mathbf{E}_{11})_{ii} \Psi_{ii} + 2 \sum_{i < j} (\mathbf{E}_{11})_{ij} \Psi_{ij}$ , while  $(\mathbf{E}_{11})_{ii} = \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2}$  and  $(\mathbf{E}_{11})_{ij} = \frac{\mathbf{y}_i \mathbf{y}_j}{\tau_i \tau_j}$  for  $i \neq j$ ; dividing the resulting  $2 \text{tr} (\mathbf{E}_{11} \Psi)$  by the factor 2 in (2.120) gives the first two formulas of (2.115). The second term of (2.120) equals  $\frac{2}{\sigma_0^2} \sum_{k,i} \mathbf{C}_{ki} \mathbf{z}_k \frac{\sigma_i(\mathbf{W}_0) \mathbf{y}_i}{\tau_i}$ , which after division by 2 gives the third formula of (2.115). The coefficient of  $b_h$  is  $\frac{1}{2} [\text{tr} \mathbf{E}_{11} + \text{tr} \mathbf{E}_{22}]$ , which is (2.116).

The identities  $\mathbb{E} [\tilde{\mathbf{s}}] = \mathbf{0}$  and  $\mathbb{E} [\tilde{\mathbf{s}} \tilde{\mathbf{s}}^T] = \mathcal{G}$  are Theorem 9 read through (2.100), since for all  $\mathbf{h}_1, \mathbf{h}_2 \in T_{\theta_0} M$

$$\mathbb{E} [s(\mathbf{X}; \theta_0)(\mathbf{h}_1) s(\mathbf{X}; \theta_0)(\mathbf{h}_2)] = G_{\theta_0}(\mathbf{h}_1, \mathbf{h}_2) = \kappa(\mathbf{h}_1)^T \mathcal{G} \kappa(\mathbf{h}_2)$$

and  $\kappa$  is onto. They can also be checked directly from (2.115): with  $\mathbf{y}$  and  $\mathbf{z}$  independent centred Gaussians,  $\mathbb{E} [\tilde{\mathbf{s}}_{\Psi_{ii}}^2] = \text{Var}(\mathbf{y}_i^2) / \tau_i^4 = 2 / \tau_i^2$ ,  $\mathbb{E} [\tilde{\mathbf{s}}_{\Psi_{ij}}^2] = 4 / (\tau_i \tau_j)$ ,  $\mathbb{E} [\tilde{\mathbf{s}}_{C_{ki}}^2] = \sigma_i(\mathbf{W}_0)^2 \tau_i \sigma_0^2 / (\sigma_0^4 \tau_i^2) = \sigma_i(\mathbf{W}_0)^2 / (\sigma_0^2 \tau_i)$ ,  $\mathbb{E} [\tilde{\mathbf{s}}_{\Psi_{ii}} \tilde{\mathbf{s}}_b] = 1 / \tau_i^2$  and  $\mathbb{E} [\tilde{\mathbf{s}}_b^2] = \eta$ , every product containing an odd power of a coordinate of  $\mathbf{z}$  having mean zero; these are exactly the entries (2.103).

By Definition 14 the vector  $\text{IF}(\mathbf{x})$  of (1.44) is characterized by  $G_{\theta_0}(\text{IF}(\mathbf{x}), \mathbf{h}) = s(\mathbf{x}; \theta_0)(\mathbf{h})$  for all  $\mathbf{h}$ , that is, by (2.100), by  $\kappa(\text{IF}(\mathbf{x}))^T \mathcal{G} \kappa(\mathbf{h}) = \tilde{\mathbf{s}}(\mathbf{x})^T \kappa(\mathbf{h})$  for all  $\mathbf{h}$ ; since  $\kappa$  is onto,  $\mathcal{G} \kappa(\text{IF}(\mathbf{x})) = \tilde{\mathbf{s}}(\mathbf{x})$ .

It remains to compute  $\mathcal{G}^{-1} \tilde{\mathbf{s}}$  block by block. For  $i < j$ , by (2.106),  $\text{IF}_{\Psi_{ij}} = \frac{\tau_i \tau_j}{4} \cdot \frac{2 \mathbf{y}_i \mathbf{y}_j}{\tau_i \tau_j} = \frac{\mathbf{y}_i \mathbf{y}_j}{2}$ , and  $\text{IF}_{C_{ki}} = \frac{\sigma_0^2 \tau_i}{\sigma_i(\mathbf{W}_0)^2} \cdot \frac{\sigma_i(\mathbf{W}_0) \mathbf{y}_i \mathbf{z}_k}{\sigma_0^2 \tau_i} = \frac{\mathbf{y}_i \mathbf{z}_k}{\sigma_i(\mathbf{W}_0)}$ , which are the second formula of (2.117) and the first of (2.118). On the coupled block, (2.107) and (2.116) give

$$\begin{aligned} \text{IF}_b &= \frac{2\sigma_0^4}{p-q} \tilde{\mathbf{s}}_b - \frac{\sigma_0^4}{p-q} \sum_{i=1}^q \tilde{\mathbf{s}}_{\Psi_{ii}} \\ &= \frac{\sigma_0^4}{p-q} \sum_{i=1}^q \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2} + \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{p-q} - \frac{\sigma_0^4}{p-q} \sum_{i=1}^q \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2} = \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{p-q}, \end{aligned}$$

the  $\mathbf{y}$  terms cancelling identically; since  $\|\mathbf{z}\|^2 = \mathbf{x}^T \mathbf{U}_0^\perp (\mathbf{U}_0^\perp)^T \mathbf{x} = \mathbf{x}^T (\mathbf{I}_p - \mathbf{P}_0) \mathbf{x}$  by (2.94), this is the first formula of (2.117). Similarly

$$\begin{aligned} \text{IF}_{\Psi_{ii}} &= \sum_{j=1}^q \left[ \delta_{ij} \frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)} \right] \tilde{\mathbf{s}}_{\Psi_{jj}} - \frac{\sigma_0^4}{p-q} \tilde{\mathbf{s}}_b \\ &= \frac{\mathbf{y}_i^2 - \tau_i}{2} + \frac{\sigma_0^4}{2(p-q)} \sum_{j=1}^q \frac{\mathbf{y}_j^2 - \tau_j}{\tau_j^2} - \frac{\sigma_0^4}{2(p-q)} \sum_{j=1}^q \frac{\mathbf{y}_j^2 - \tau_j}{\tau_j^2} - \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{2(p-q)}, \end{aligned}$$

which is the second formula of (2.118). □

**Corollary 54** (The noise variance). The maximum likelihood estimator of the noise variance satisfies (2.112), and it is asymptotically linear with influence function

$$\text{IF}_{\sigma^2}(\mathbf{x}) = \frac{\mathbf{x}^T (\mathbf{I}_p - \mathbf{P}_0) \mathbf{x} - (p - q)\sigma_0^2}{p - q}, \quad \mathbb{E}[\text{IF}_{\sigma^2}(\mathbf{X})] = 0, \quad \mathbb{E}[\text{IF}_{\sigma^2}(\mathbf{X})^2] = \frac{2\sigma_0^4}{p - q}. \quad (2.121)$$

*Proof.* The weak limit is (2.112). The influence function is the first formula of (2.117), because  $b_n = \hat{\sigma}_n^2 - \sigma_0^2$  is the  $b$  coordinate of  $\kappa(\hat{\mathbf{u}}_n)$  by (2.110), so that the  $b$  coordinate of (1.45) reads  $\sqrt{n}(\hat{\sigma}_n^2 - \sigma_0^2) = n^{-\frac{1}{2}} \sum_{i=1}^n \text{IF}_b(\mathbf{X}_i) + o_{\mathbb{P}}(1)$ . For the moments,  $(\mathbf{U}_0^\perp)^T \mathbf{X} \sim N_{p-q}(\mathbf{0}, \sigma_0^2 \mathbf{I}_{p-q})$  gives  $\mathbb{E}\|\mathbf{z}\|^2 = (p - q)\sigma_0^2$  and  $\text{Var}(\|\mathbf{z}\|^2) = 2(p - q)\sigma_0^4$ , whence  $\mathbb{E}[\text{IF}_{\sigma^2}^2] = \frac{2(p-q)\sigma_0^4}{(p-q)^2} = \frac{2\sigma_0^4}{p-q}$ , consistent with (2.112) as it must be, since the asymptotic covariance of an asymptotically linear sequence is the covariance of its influence function.  $\square$

**Remark 55** (Where the model helps, and where it does not). Formulas (2.117)–(2.118) answer a question the asymptotic theory alone does not: in which directions is the constrained estimator  $\hat{\Sigma}_n$  actually better than the sample covariance matrix  $\mathbf{S}_n$ ? The answer is: in exactly one block, and the reason is the computation of  $\hat{\Sigma}_n$  carried out in the proof of Theorem 11, which exhibits it as  $\mathbf{S}_n$  with the trailing  $p - q$  eigenvalues replaced by their average.

By (2.85) the influence function of  $\hat{\Sigma}_n$  is  $\mathcal{D}_0 \circ \text{IF}$ , whose blocks are, by (2.97),  $2\text{IF}_\Psi + \text{IF}_b \mathbf{I}_q$ ,  $\text{IF}_C \mathbf{D}_0$  and  $\text{IF}_b \mathbf{I}_{p-q}$ . By (2.117)–(2.118) these are

$$(2\text{IF}_\Psi + \text{IF}_b \mathbf{I}_q)_{ij} = \mathbf{y}_i \mathbf{y}_j - \delta_{ij} \tau_i, \quad (\text{IF}_C \mathbf{D}_0)_{ki} = \sigma_i(\mathbf{W}_0) \frac{\mathbf{y}_i \mathbf{z}_k}{\sigma_i(\mathbf{W}_0)} = \mathbf{z}_k \mathbf{y}_i, \quad \text{IF}_b \mathbf{I}_{p-q}, \quad (2.122)$$

the diagonal case of the first identity being exactly the cancellation of  $\text{IF}_b$  in the second formula of (2.118). Now the influence function of  $\mathbf{S}_n$ , read in the same basis (2.94), is  $\mathbf{J}_0^T (\mathbf{x} \mathbf{x}^T - \Sigma_0) \mathbf{J}_0$ , whose blocks are  $\mathbf{y} \mathbf{y}^T - \mathbf{T}_0$ ,  $\mathbf{z} \mathbf{y}^T$  and  $\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q}$ . The first two coincide with (2.122): *to first order the maximum likelihood fit reproduces the sample covariance matrix in the signal block and in the signal–noise block, and gains nothing there.* Only the trailing block differs, where  $\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q}$  is replaced by its average

$$\frac{\text{tr}(\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q})}{p - q} \mathbf{I}_{p-q} = \text{IF}_b \mathbf{I}_{p-q} :$$

the isotropy constraint of the model averages the  $p - q$  noise directions, dividing the variance by  $p - q$ . This is the infinitesimal shadow of the exact identity established in the proof of Theorem 11, which says the same thing non-asymptotically:  $\hat{\Sigma}_n$  differs from  $\mathbf{S}_n$  only through the trailing eigenvalues, which it replaces by their mean. It also explains why the leading blocks cannot be improved: by (2.87) the model imposes no constraint there at all, the image of  $\mathcal{D}_0$  being everything except for the isotropy of the trailing block.

**Remark 56** (Cross-checks against classical principal component analysis). Three consistency checks, each of which recovers a classical formula, and none of which is used elsewhere.

**1. Pooled noise eigenvalues.** The variance  $\frac{2\sigma_0^4}{p-q}$  of (2.112) is the classical asymptotic variance of the average of the  $p - q$  sample eigenvalues attached to a population eigenvalue of multiplicity  $p - q$ . Indeed  $\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j>q} \lambda_j(\mathbf{S}_n)$  exactly, by (1.42), and

$$\sum_{j>q} \lambda_j(\mathbf{S}_n) = \text{tr}(\mathbf{S}_n) - \max \{ \text{tr}(\mathbf{P} \mathbf{S}_n) : \mathbf{P} \text{ an orthogonal projection of rank } q \}$$

by Ky Fan’s variational principle. Write  $\mathbf{P}_n$  for the maximizer, which is the projection onto the span of  $\mathbf{v}_1, \dots, \mathbf{v}_q$ . Since  $\mathbf{P}_0$  is the maximizer at  $\Sigma_0$ , by (2.93), we have on the one hand  $\text{tr}(\mathbf{P}_n \mathbf{S}_n) \geq \text{tr}(\mathbf{P}_0 \mathbf{S}_n)$  and on the other

$$\text{tr}(\mathbf{P}_n \mathbf{S}_n) - \text{tr}(\mathbf{P}_0 \mathbf{S}_n) \leq \text{tr}((\mathbf{P}_n - \mathbf{P}_0)(\mathbf{S}_n - \Sigma_0)) \leq \|\mathbf{P}_n - \mathbf{P}_0\|_F \|\mathbf{S}_n - \Sigma_0\|_F,$$

because  $\text{tr}(\mathbf{P}_n \boldsymbol{\Sigma}_0) \leq \text{tr}(\mathbf{P}_0 \boldsymbol{\Sigma}_0)$ . The multivariate central limit theorem applied to the independent, identically distributed and square integrable matrices  $\mathbf{X}_i \mathbf{X}_i^T$  gives  $\|\mathbf{S}_n - \boldsymbol{\Sigma}_0\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ , and the Davis–Kahan  $\sin \Theta$  theorem, with the eigengap (2.93), gives  $\|\mathbf{P}_n - \mathbf{P}_0\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$  as well (see for example [Chen et al., 2021](#), Chapter 2, or [Kato, 1995](#), Chapter II). Hence  $\mathbf{P}_n$  may be replaced by  $\mathbf{P}_0$  at a cost of  $O_{\mathbb{P}}(n^{-1})$ , and the statistic that remains is  $\frac{1}{p-q} \text{tr}((\mathbf{I}_p - \mathbf{P}_0) \mathbf{S}_n)$ , whose centred and  $\sqrt{n}$ -scaled version is exactly the average of the influence functions (2.121).

**2. The leading block is not improved.** By (2.107), and writing  $\text{Var}(2(\mathbf{Z}_{\Psi})_{ii} + Z_b) = 4\text{Var}((\mathbf{Z}_{\Psi})_{ii}) + 4\text{Cov}((\mathbf{Z}_{\Psi})_{ii}, Z_b) + \text{Var}(Z_b)$ ,

$$\text{Var}(2(\mathbf{Z}_{\Psi})_{ii} + Z_b) = 4 \left( \frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)} \right) + 4 \left( -\frac{\sigma_0^4}{p-q} \right) + \frac{2\sigma_0^4}{p-q} = 2\tau_i^2 = \text{Var}\left((\mathbf{U}_0^T \mathbf{X})_i^2\right),$$

and likewise  $\text{Var}(2(\mathbf{Z}_{\Psi})_{ij}) = \tau_i \tau_j = \text{Var}(\mathbf{y}_i \mathbf{y}_j)$  for  $i < j$  — the sample covariance variances for those coordinates, as Remark 55 requires.

**3. Anderson’s eigenvector variances.** Suppose the  $\sigma_i(\mathbf{W}_0)$  are simple. By part 2 of Lemma 48, the component of the aligned loading error orthogonal to the column space of  $\mathbf{W}_0$  is  $\mathbf{U}_0^\perp \mathbf{C}_n \mathbf{V}_0^T$ ; since the  $i$ -th column of  $\mathbf{W}_0 \mathbf{V}_0$  is  $\sigma_i(\mathbf{W}_0)$  times the  $i$ -th column of  $\mathbf{U}_0$ , the tilt of the estimated  $i$ -th direction out of the signal subspace is  $\mathbf{U}_0^\perp$  applied to the  $i$ -th column of  $\mathbf{C}_n$ , divided by  $\sigma_i(\mathbf{W}_0)$ , with limiting per-coordinate variance

$$\frac{1}{\sigma_i(\mathbf{W}_0)^2} \cdot \frac{\sigma_0^2 \tau_i}{\sigma_i(\mathbf{W}_0)^2} = \frac{\sigma_0^2 \tau_i}{(\tau_i - \sigma_0^2)^2} = \frac{\lambda_i(\boldsymbol{\Sigma}_0) \lambda_j(\boldsymbol{\Sigma}_0)}{(\lambda_i(\boldsymbol{\Sigma}_0) - \lambda_j(\boldsymbol{\Sigma}_0))^2} \quad \text{for any } j > q,$$

which is Anderson’s formula for the noise directions. When the  $\sigma_i(\mathbf{W}_0)$  are repeated this eigenvector-by-eigenvector reading breaks down — individual eigenvectors are then not asymptotically normal — but nothing in Propositions 50 and 51 is affected: no formula in this section divides by a difference of singular values, and the only inverted quantities are  $\tau_i$ ,  $\sigma_0^2$ ,  $\sigma_i(\mathbf{W}_0)$  and  $p - q$ , all strictly positive under the standing assumptions  $\text{rank}(\mathbf{W}_0) = q$ ,  $\sigma_0^2 > 0$  and  $q < p$ . This is the promised quantitative form of the third comment of Remark 25.

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